

Design and Simulation of analog/RF ICs with Open-Source CAD Tools and PDKs: All-Region Single-Piece MOSFET Theory

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Outline

- **Introduction: ACM timeline**
- **ACM2 model**
- **Long-channel: I_D and g_m/I_D models**

ACM timeline

- **1993 ϕs -based model (SBMICRO, Campinas, Br)**
- **1995 Long-channel charge-based model (SSE, Nov.)**
- **1998 Most referenced ACM paper (JSSC, Oct.)**
- **2000 ACM model in SMASH simulator (CICC, Orlando)**

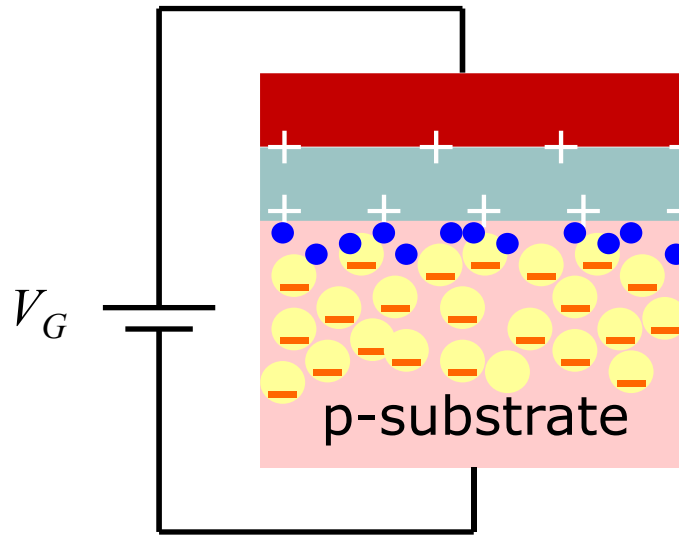


- **2018 Short course on ACM for ST organized by Prof. Sylvain Bourdel in Grenoble**
- **2021 4-parameter single-piece model (NorCAS, Oslo)**
- **2023 ACM2 in VERILOG-AMS (NEWCAS, Edinburgh)**

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The capacitive model of the MOSFET



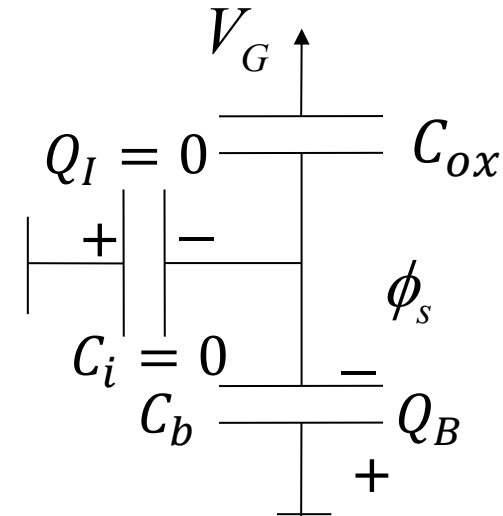
ϕ_s surface potential

C_{ox} oxide capacitance per unit area

C_b depletion capacitance per unit area

Q_I carrier charge density

V_{T0} threshold voltage



$$\frac{\Delta\phi_s}{\Delta V_G} = \frac{C_{ox}}{C_{ox} + C_b} = \frac{1}{n}$$

$$\phi_s = 2\phi_F + \frac{V_G - V_{T0}}{n} = 2\phi_F + V_P$$

ACM2 current law

From 3 approximations: normalized current vs. normalized charge densities at source and drain



$$i_D = \frac{(q_S + q_D + 2)}{1 + \zeta(q_S - q_D)} (q_S - q_D)$$

$$i_D = I_D / I_S$$

$$I_S = \frac{W}{L} \mu_s n C_{ox} \frac{\phi_t^2}{2}$$

normalization (specific) current

$$q_{S(D)} = Q_{S(D)} / (-n C_{ox} \phi_t)$$

$-n C_{ox} \phi_t$ thermal charge

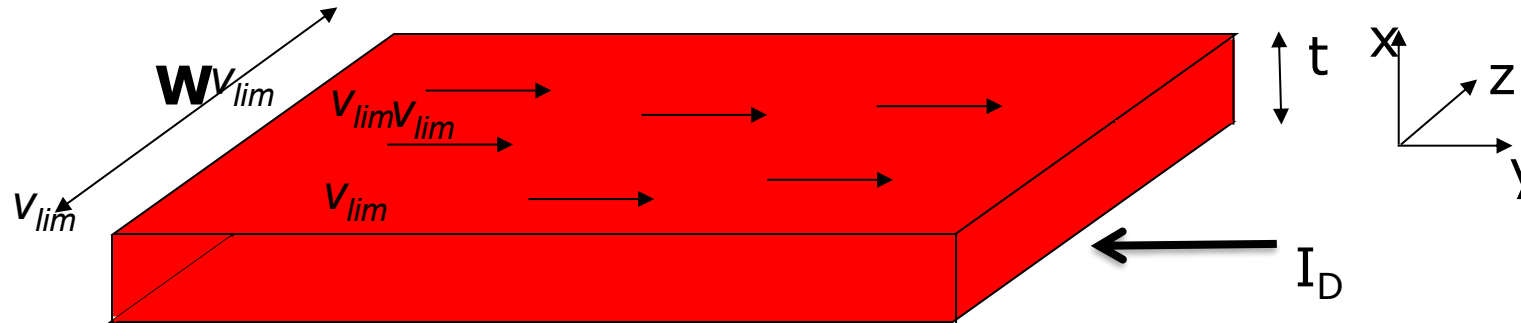
$$SI : q_{S(D)} \gg 1 \quad WI : q_{S(D)} \ll 1$$

Short-channel parameter ζ :

$$\zeta = \frac{(\mu_s \phi_t / L)}{v_{lim}}$$

ratio of diffusion-related velocity to saturation velocity

Physics-based saturation



Saturation current due to saturation velocity of the carriers

$$I_{Dsat} = -W Q_{Dsat} v_{lim}$$

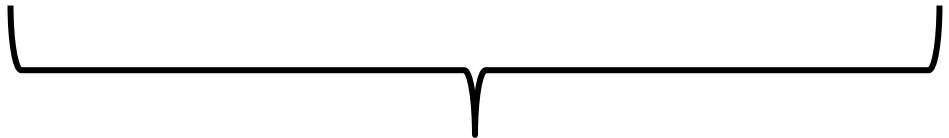
Q_{Dsat} is the saturation inversion charge per unit area, or, using normalized variables:

$$i_{Dsat} = \frac{2}{\zeta} q_{dsat}$$

"Carrier velocity approaches v_{sat} , but never reaches v_{sat} " Y.Taur TED March 2019

Physics-based saturation: design model

$$i_{Dsat} = \frac{2}{\zeta} q_{dsat} \qquad i_{Dsat} = \frac{(q_s + q_{Dsat} + 2)}{1 + \zeta(q_s - q_{Dsat})} (q_s - q_{Dsat})$$


$$q_{Dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

or equivalently

$$q_s = \sqrt{1 + \frac{2}{\zeta} q_{dsat}} - 1 + q_{dsat}$$

The Unified Charge Control Model (UCCM)

$$\frac{V_P - V_C}{\phi_t} = q_I - 1 + \ln q_I$$
$$V_S \leq V_C \leq V_D$$
$$q_S \leq q_I \leq q_D$$



$$\frac{V_{DS}}{\phi_t} = q_S - q_D + \ln \frac{q_S}{q_D}$$

The “Regional” Weak (WI) and Strong Inversion (SI) Approximations

WI

$$q_I \ll 1 \rightarrow V_P - V_C \ll -\phi_t$$

$$q_I \cong e^{\frac{V_P - V_C + \phi_t}{\phi_t}}$$

Error <10% for $q_I < 0.22$

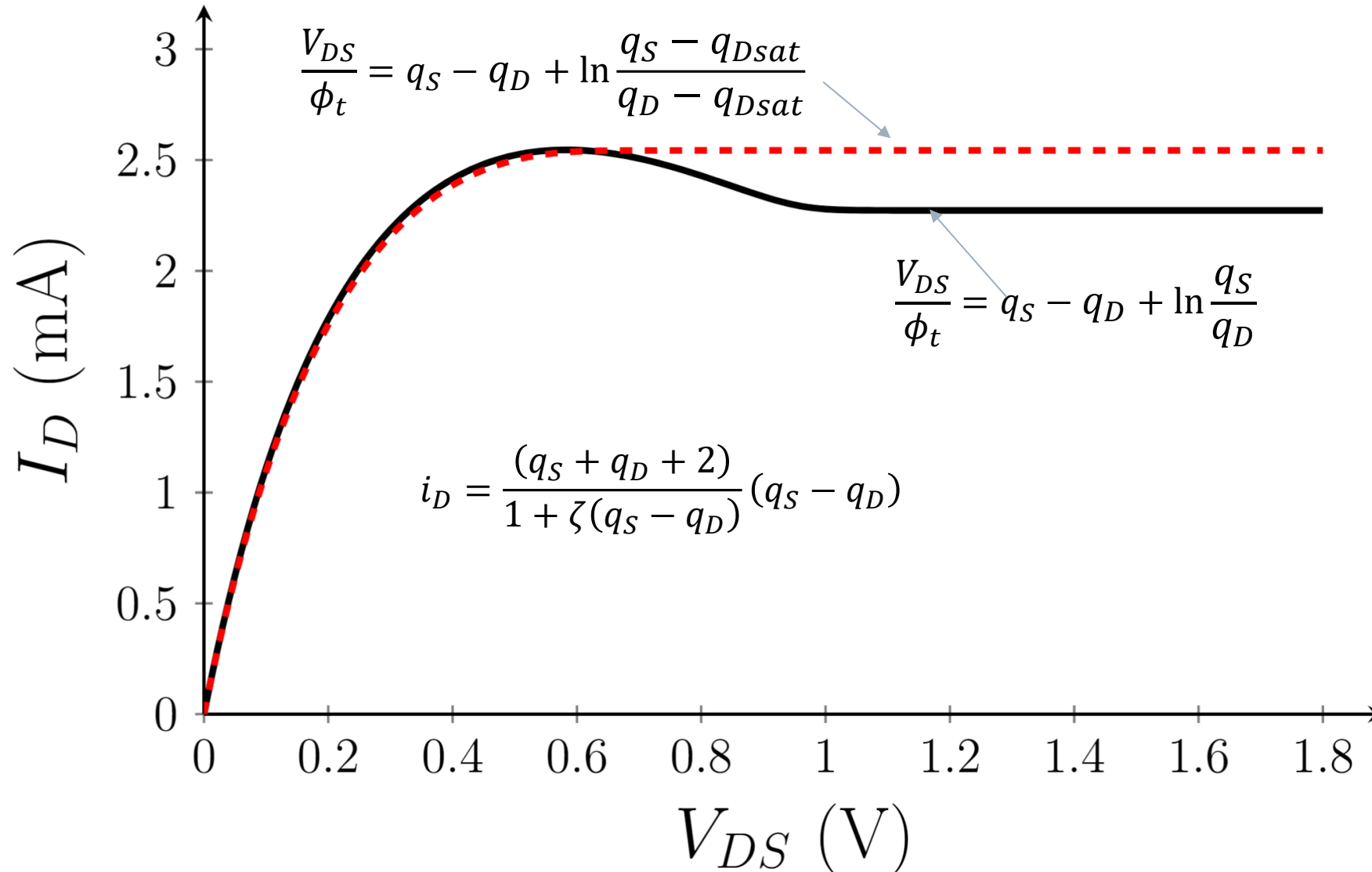
SI

$$q_I \gg 1 \rightarrow V_P - V_C \gg \phi_t$$

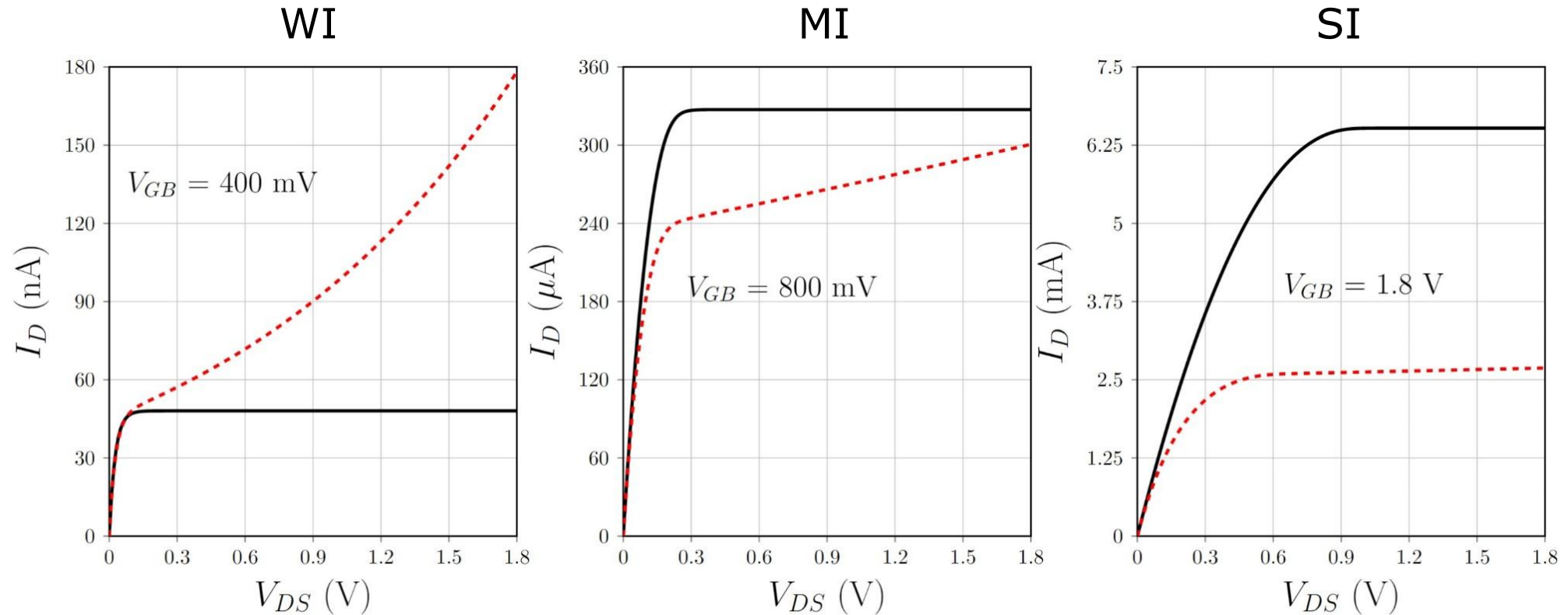
$$q_I \cong \frac{V_P - V_C + \phi_t}{\phi_t}$$

Error <10% for $q_I > 20$

UCCM including the effect of velocity saturation



Output curves including DIBL and v_{sat}



DIBL model: $V_T = V_{T0} - \sigma(V_S + V_D)$

— 3PM --- 5PM

Transistor	$W/L \text{ } [\mu\text{m}]$	$V_{T0} \text{ [mV]}$	$I_S \text{ } [\mu\text{A}]$	n	σ	ζ
NMOS	5/0.18	528	5.52	1.37	0.025	0.056

Total channel charge and capacitances

We can define a virtual charge that differs from the real charge by a constant term along the channel

$$Q_V = Q_I - nC_{ox} + \frac{I_D}{Wv_{sat}}$$

Thus $dQ_V = dQ_I$

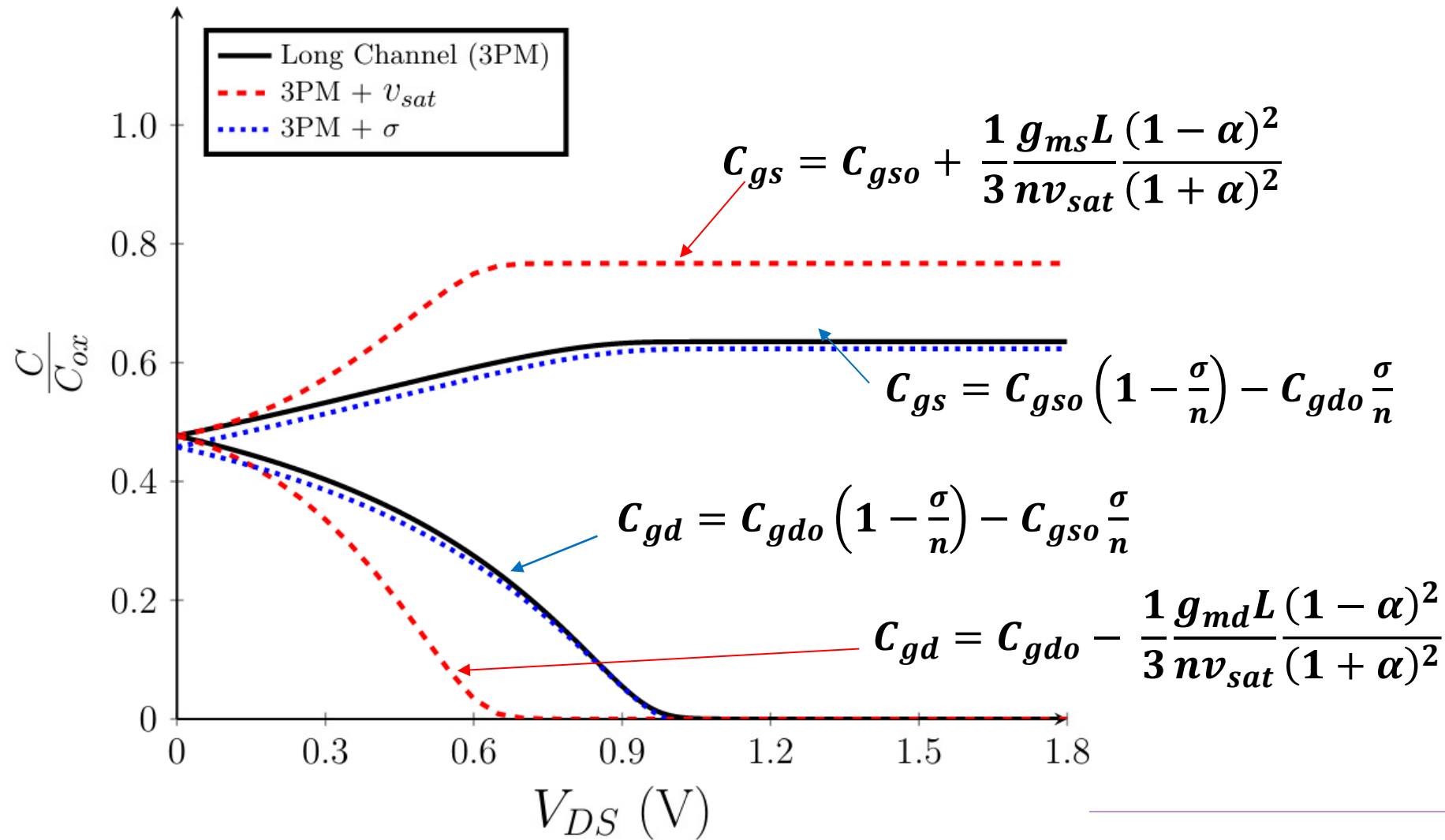
Consequently, the **virtual charge** varies along the channel as the **real charge in a long-channel transistor in strong inversion**

$$dy = -\frac{\mu_s W}{nC_{ox} I_D} Q_V dQ_V$$

$$Q_{ch} = W \int_0^L Q_I dy = WL \left(\frac{2}{3} \frac{Q_{VS}^2 + Q_{VS} Q_{VD} + Q_{VD}^2}{Q_{VS} + Q_{VD}} + nC_{ox} - \frac{I_D}{Wv_{sat}} \right)$$

$$C_{gs(d)} = -\frac{\partial Q_G}{\partial V_{S(D)}} = \frac{1}{n} \frac{\partial Q_{ch}}{\partial V_{S(D)}}$$

Intrinsic capacitances including velocity saturation and DIBL



$$\alpha = \frac{1 + q_D - \frac{\zeta}{2} i_d}{1 + q_S - \frac{\zeta}{2} i_d}$$

Thermal noise

Sah's formula for the thermal noise ¹

where Q_{ch} is the channel inversion charge $\frac{\overline{i_d^2}}{\Delta f} = -4kT\mu \frac{Q_{ch}}{L^2}$

In ACM2
$$Q_{ch} = WL \left(\frac{2}{3} \frac{Q_{VS}^2 + Q_{VS}Q_{VD} + Q_{VD}^2}{Q_{VS} + Q_{VD}} + nC_{ox} - \frac{I_D}{Wv_{sat}} \right)$$

$$Q_{VS(D)} = Q_{IS(D)} - nC_{ox} + \frac{I_D}{Wv_{sat}}$$

Design-oriented formula

For a short-channel transistor in saturation

$$\frac{\overline{i_d^2}}{\Delta f} \approx 4kT \frac{I_S}{\phi_t} \left(\sqrt{1 + i_{Dsat}} - 1 + \zeta i_{Dsat} \right)$$

¹C. T. Sah et al, *IEEE TED*, Apr. 1966

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Long-channel charge-based model

For short channel MOS

$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$

$$\frac{V_P - V_S}{\phi_t} = q_s - 1 + \ln q_s \qquad \frac{V_{DS}}{\phi_t} = q_s - q_d + \ln \frac{q_s - q_{dsat}}{q_d - q_{dsat}}$$

For long channel MOS $\zeta=0 \rightarrow q_{dsat}=0$

$$I_D = I_S(q_s + q_d + 2)(q_s - q_d) = I_S[(q_s^2 + 2q_s) - (q_d^2 + 2q_d)]$$

$$\frac{V_P - V_{S(D)}}{\phi_t} = q_{s(d)} - 1 + \ln q_{s(d)}$$

Obs: $\frac{dq_{s(d)}}{dV_G} = -\frac{1}{n} \frac{dq_{s(d)}}{dV_{S(D)}}; \quad \frac{dq_d}{dV_D} = -\frac{1}{\phi_t} \frac{q_d}{q_d + 1}; \quad \frac{dI_D}{dq_d} = -2I_S(q_d + 1)$

Small-signal transconductances

$$g_{md} = \frac{dI_D}{dV_D} = \frac{dI_D}{dq_d} \frac{dq_d}{dV_D} = 2I_S(q_d + 1) \frac{1}{\phi_t} \frac{q_d}{q_d + 1} = \frac{2I_S}{\phi_t} q_d$$

Symmetry $\downarrow$

$$g_{ms} = \frac{2I_S}{\phi_t} q_s$$

**Amazingly simple
(see Appendix)**

$$g_m = \frac{dI_D}{dV_G} = \frac{dI_D}{dq_s} \frac{dq_s}{dV_G} + \frac{dI_D}{dq_d} \frac{dq_d}{dV_G} \qquad \frac{dq_{s(d)}}{dV_G} = -\frac{1}{n} \frac{dq_{s(d)}}{dV_{S(D)}}$$

$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$g_m = \frac{g_{ms}}{n} \longrightarrow \text{in saturation}$$

Unified Current Control Model (UICM)-I

$$I_D = I_S[(q_s^2 + 2q_s) - (q_d^2 + 2q_d)] \quad (A)$$

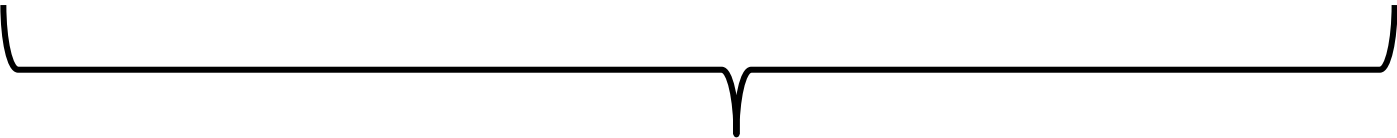
(A) can also be written as

$$I_D = I_F - I_R = I_S[i_f - i_r] \quad (B)$$

I_F, I_R : forward and reverse currents

$i_{f(r)} = q_{s(d)}^2 + 2q_{s(d)}$: forward (reverse) inversion coefficients

$$\downarrow$$
$$q_{s(d)} = \sqrt{1 + i_{f(r)}} - 1$$

$$\frac{V_P - V_{S(D)}}{\phi_t} = q_{s(d)} - 1 + \ln q_{s(d)} \quad q_{s(d)} = \sqrt{1 + i_{f(r)}} - 1$$


Normalized UICM

$$\frac{V_P - V_{S(D)}}{\phi_t} = \sqrt{1 + i_{f(r)}} - 2 + \ln \left(\sqrt{1 + i_{f(r)}} - 1 \right)$$



$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln \frac{q_s}{q_d} = \sqrt{1 + i_f} - \sqrt{1 + i_r} + \ln \left(\frac{\sqrt{1 + i_f} - 1}{\sqrt{1 + i_r} - 1} \right)$$

$$g_{ms(d)} = \frac{2I_S}{\phi_t} q_{s(d)} = \frac{2I_S}{\phi_t} \left(\sqrt{1 + i_{f(r)}} - 1 \right)$$

$$g_m = \frac{g_{ms} - g_{md}}{n}$$

$$\frac{g_m}{I_D} = \frac{2}{n\phi_t(q_s + q_d + 2)} = \frac{2}{n\phi_t(\sqrt{1 + i_f} + \sqrt{1 + i_r})}$$

For $V_{DS}/\phi_t \ll 1$ we have $i_f \cong i_r$

In saturation $i_f \gg i_r$

$$\frac{g_m}{I_D} \cong \frac{1}{n\phi_t(1 + q_s)} = \frac{1}{n\phi_t\sqrt{1 + i_f}}$$

$$\frac{g_m}{I_D} \cong \frac{2}{n\phi_t(q_s + 2)} = \frac{2}{n\phi_t(\sqrt{1 + i_f} + 1)}$$

- A. I. A. Cunha, M. C. Schneider and C. Galup-Montoro, "Derivation of the Unified Charge Control Model and Parameter Extraction Procedure", Solid-State Electronics, March 1999.
- O. C. Gouveia Filho, A. I. A. Cunha, M. C. Schneider and C. Galup-Montoro, "Advanced compact model for short-channel MOS transistors", IEEE Custom Integrated Circuits Conference, Orlando, FL, USA, May 2000.
- C. Galup-Montoro and M. C. Schneider, *MOSFET Modeling for Circuit Analysis and Design*, World Scientific, 2007.
- C. M. Adornes, D. G. Alves Neto, M. C. Schneider and C. Galup-Montoro, " Bridging the Gap between Design and Simulation of Low-Voltage CMOS Circuits ," Journal of Low Power Electronics and Applications, June 2022.
- D. G. Alves Neto *et al*, "Design-oriented single-piece 5-DC parameter MOSFET model," *IEEE Access*, 2024.

Appendix: the exact model of the long-channel MOSFET¹

$$I_D = -\frac{W}{L} \int_{V_S}^{V_D} \mu Q_I(V_C) dV_C$$

Consequently, the **exact** expressions for g_{ms} and g_{md} are

$$g_{md} = \frac{dI_D}{dV_D} = -\frac{W}{L} \mu Q_D = \frac{2I_S}{\phi_t} q_d$$

$$g_{ms} = -\frac{dI_D}{dV_S} = -\frac{W}{L} \mu Q_S = \frac{2I_S}{\phi_t} q_s$$

¹ H. C. Pao and C. T. Sah, 'Effects of diffusion current on characteristics of metal-oxide (insulator)-semiconductor transistors' Solid-State Electronics, Oct 1966.

Design and Simulation of analog/RF ICs with Open-Source CAD Tools and PDKs: Implementing the ACM2 model

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Outline



- ❑ The ACM2 model (a quick recap)
- ❑ Numerical solving of the UCCM equations
- ❑ acm 433 algorithm and Lambert function
- ❑ Analytical approximation of Lambert functions
- ❑ Towards a closed-form I-V MOSFET model

The ACM2 model (a quick recap)

The 5-parameter ACM2 model

- Only 5 parameters: n , I_{S0} , V_{T0} , σ and ζ

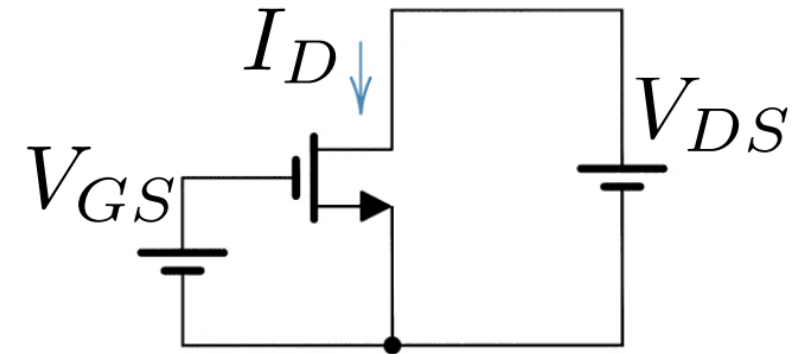
$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$

where

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$$

Alves Neto ACCESS 2024



with

$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

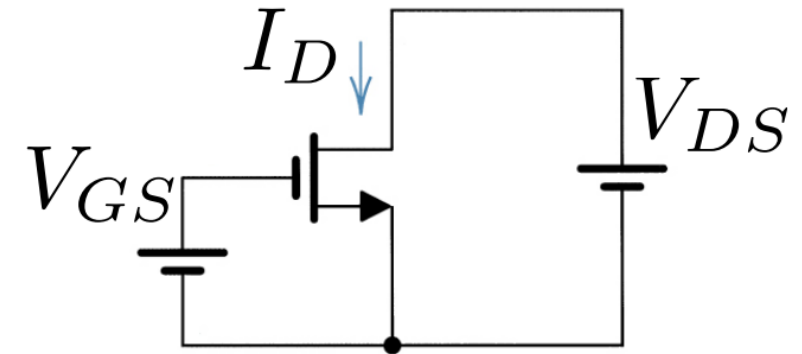
$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

The ACM2 model (a quick recap)

The 5-parameter ACM2 model

- Only 5 parameters: n , I_{S0} , V_{T0} , σ and ζ

$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$



where

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$$

Alves Neto ACCESS 2024

with

$$V_P = \frac{V_{GB} - V_{T0} + \frac{\sigma(V_{DB} + V_{SB})}{n}}{n}$$

$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

The ACM2 model (a quick recap)

Computing I_D in practice:

Given the model parameters and polarization voltages

$n, I_{S0}, V_{T0}, \sigma$ and ζ

V_G, V_D, V_S, V_B

Step 1: Compute V_P :
$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

Step 2: Compute q_s :
$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

Step 3: Compute q_{dsat} :
$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

Step 4: Compute q_d :
$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$$

Step 5: Compute I_D :
$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$



The ACM2 model (a quick recap)

Computing I_D in practice:

Given the model parameters and polarization voltages

$n, I_{S0}, V_{T0}, \sigma$ and ζ

V_G, V_D, V_S, V_B

Step 1: Compute V_P :
$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

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$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

Step 3: Compute q_{dsat} :
$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

Step 4: Compute q_d :
$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$$

Step 5: Compute I_D :
$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$



Transcendental
equations

Acm 443 algorithm and Lambert W Functions

How to solve $\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$ efficiently?

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s) \quad \Rightarrow \quad e^{\left(\frac{V_P - V_{SB}}{\phi_t} - 1\right)} = q_s e^{q_s} \quad \Rightarrow \quad x = we^w$$

Algorithm 443: Solution of the transcendental equation $we^w = x$

$$w_n = \begin{cases} \frac{x + \frac{4}{3}x^2}{1 + \frac{7}{3}x + \frac{5}{6}x^2}, & x < 0.7385 \\ \ln(x) - \frac{24(\ln(x)\ln(x) + 2\ln(x) - 3)}{7\ln(x)\ln(x) + 58\ln(x) + 127}, & x \geq 0.7385 \end{cases}$$

$$z_n = \ln(x) - w_n - \ln(w_n)$$

$$e_n = \frac{z_n}{1 + w_n} \frac{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - z_n}{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - 2z_n}$$

$$w = w_n(1 + e_n)$$

Fritsch et al. Communications of the ACM, 1973

Acm 443 algorithm and Lambert W Functions

How to solve $\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$ efficiently?

$$\frac{V_{DS}}{\phi_t} = q_s - q_{dsat} - q_d + q_{dsat} + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right) \Rightarrow$$

$$\Rightarrow q_s - q_{dsat} \frac{e^{(q_s - q_{dsat})}}{e^{\frac{V_{DS}}{\phi_t}}} = (q_d - q_{dsat}) e^{(q_d - q_{dsat})} \Rightarrow x = we^w$$

Algorithm 443

$$w_n = \begin{cases} \frac{x + \frac{4}{3}x^2}{1 + \frac{7}{3}x + \frac{5}{6}x^2}, & x < 0.7385 \\ \ln(x) - \frac{24(\ln(x)\ln(x) + 2\ln(x) - 3)}{7\ln(x)\ln(x) + 58\ln(x) + 127}, & x \geq 0.7385 \end{cases}$$

$$z_n = \ln(x) - w_n - \ln(w_n)$$

$$e_n = \frac{z_n}{1 + w_n} \frac{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - z_n}{2(1 + w_n) \left(1 + w_n + \frac{2}{3}z_n\right) - 2z_n}$$

$$w = w_n(1 + e_n)$$

$$q_D = w + q_{Dsat}$$

Fritsch et al. Communications of the ACM, 1973

Acm 443 algorithm and Lambert W Functions

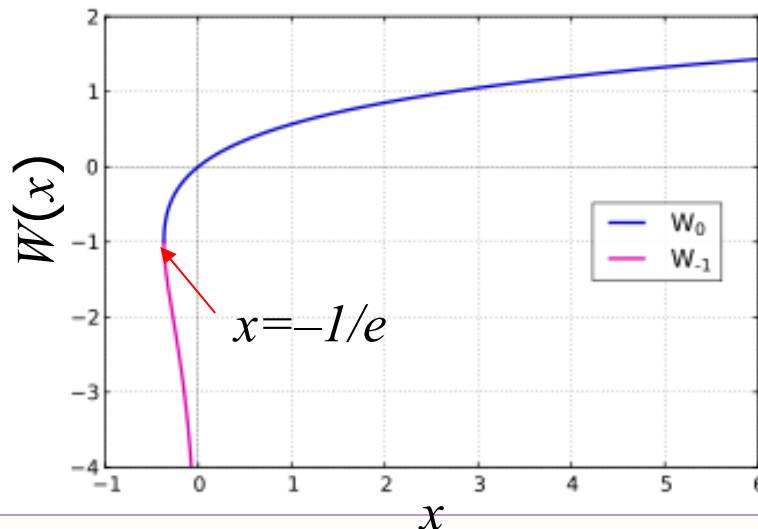
Can we consider the problem analytically?

- The Lambert W function $z = w e^w$ for w any complex number was studied by Johann Lambert in 1758, then properly defined by L. Euler in 1783

Acm 443 algorithm and Lambert W Functions

Can we consider the problem analytically?

- The Lambert W function $z = w e^w$ for w any complex number was studied by Johann Lambert in 1758, then properly defined by L. Euler in 178
- For real numbers x and $W(x)$, equation $x = W(x) e^{W(x)}$ has solutions W_0 and W_{-1} :



q_S , q_D and q_{Dsat} , and I_D can be expressed as a function of V_G , V_D , V_S using W_0

Acm 443 algorithm and Lambert W Functions

Expressing I_D using Lamber W function

$$I_D = I_{S0} \frac{q_S + q_D + 2}{1 + \zeta |q_S - q_D|} (q_S - q_D)$$

$$q_S = W_0 \left(e^{\left(\frac{V_P - V_{SB}}{U_T} - 1 \right)} \right)$$

$$q_D = q_{Dsat} + W_0 \left((W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat}) e^{(W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat}) - \frac{V_{DS}}{U_T}} \right)$$

$$q_{Dsat} = W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta} \right)^2 + \frac{2W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right)}{\zeta}}$$

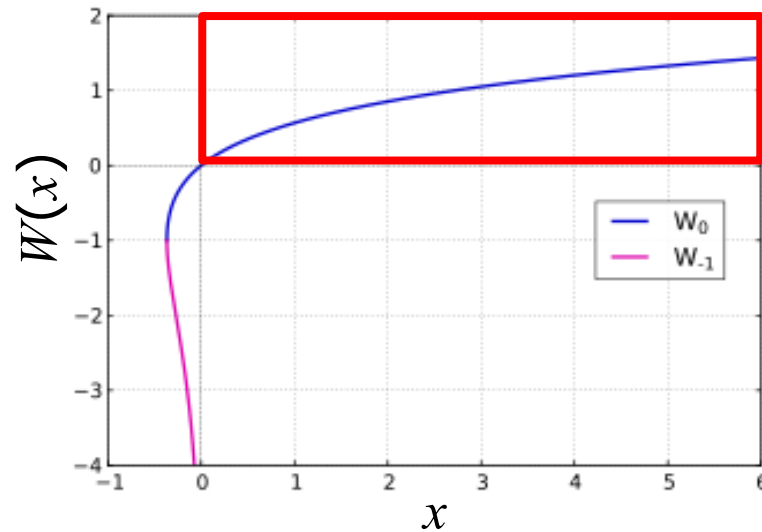
W_0 is available in main programming languages: Matlab (lambertw), Octave (lambertw), Python (lambertw in scipy lib), R (lambertW0), ...

...but can we approximate W_0 in terms of elementary functions?

Acm 443 algorithm and Lambert W Functions

Approximating the Lambert W function

For real numbers x and $W(x)$, $x = W(x) e^{W(x)}$



Approximation of $W_n(x)$ for $x > 0$

$$W_{0,0}(x) = \ln [1 + \alpha(x)x]$$

$$W_{0,n}(x) = \frac{W_{n-1}(x)}{1 + W_{n-1}(x)} \left[1 + \ln \left(\frac{x}{W_{n-1}(x)} \right) \right]$$

For $n=1,2,3,\dots$

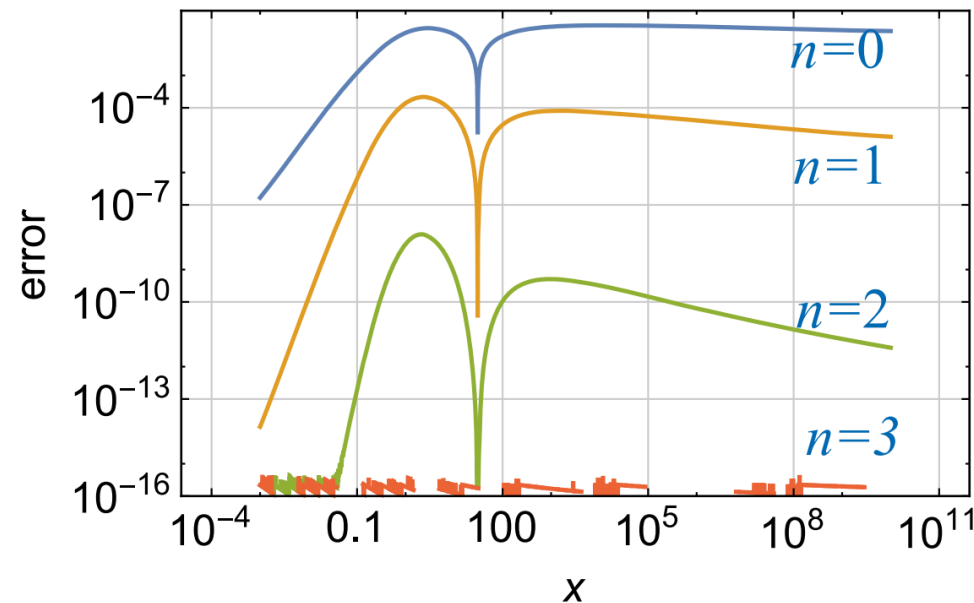
$$\text{where } \alpha(x) = \frac{1}{1 + \frac{\ln(1+x)}{2}}$$

R. Iacono et al. Advances in Computational Mathematics, 2017

Acm 443 algorithm and Lambert W Functions

Approximating the Lambert W function

For real numbers x and $W(x)$, $x = W(x) e^{W(x)}$



Approximation of $W_0(x)$ for $x > 0$

$$W_{0,0}(x) = \ln [1 + \alpha(x)x]$$

$$W_{0,n}(x) = \frac{W_{n-1}(x)}{1 + W_{n-1}(x)} \left[1 + \ln \left(\frac{x}{W_{n-1}(x)} \right) \right]$$

For $n=1,2,3,\dots$

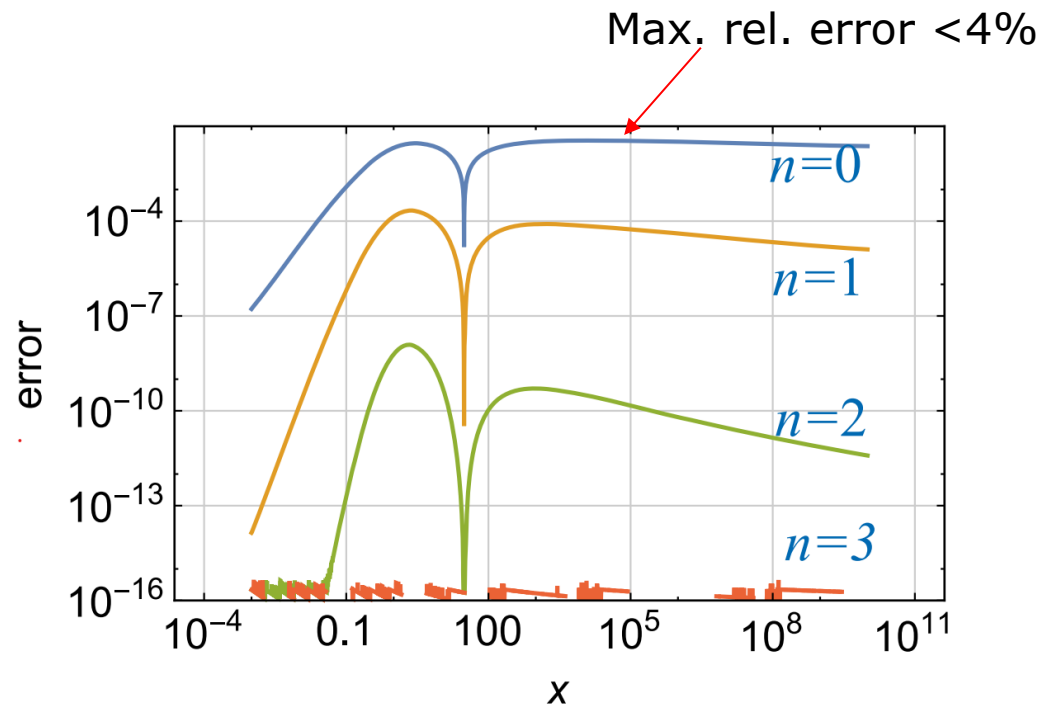
$$\text{where } \alpha(x) = \frac{1}{1 + \frac{\ln(1+x)}{2}}$$

R. Iacono et al. Advances in Computational Mathematics, 2017

Acm 443 algorithm and Lambert W Functions

Approximating the Lambert W function

For real numbers x and $W(x)$, $x = W(x) e^{W(x)}$



Approximation of $W_0(x)$ for $x > 0$

$$W_{0,0}(x) = \ln [1 + \alpha(x)x]$$

$$\text{where } \alpha(x) = \frac{1}{1 + \frac{\ln(1+x)}{2}}$$

R. Iacono et al. Advances in Computational Mathematics, 2017

Acm 443 algorithm and Lambert W Functions

Approximating the Lambert W function

Approximating q_S

$$q_S = W_0 \left(e^{\left(\frac{V_P - V_{SB}}{U_T} - 1 \right)} \right) \Rightarrow q_S \approx W_{0,0}(x) = \ln \left[1 + \frac{1}{1 + \frac{\ln[1 + e^{\left(\frac{V_P - V_{SB}}{U_T} - 1 \right)}]}{2}} e^{\left(\frac{V_P - V_{SB}}{U_T} - 1 \right)} \right]$$

Approximating q_D is similar from:

$$q_D = q_{Dsat} + W_0 \left(\left(W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat} \right) e^{\left(W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) - q_{Dsat} \right) - \frac{V_{DS}}{U_T}} \right)$$

$$q_{Dsat} = W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right) + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta} \right)^2 + \frac{2W_0 \left(e^{\frac{V_P - V_{SB}}{U_T} - 1} \right)}{\zeta}}$$

Towards an analytical I-V MOSFET model

Replacing $q_S(V_G, V_S, V_D)$ and $q_D(V_G, V_S, V_D)$ in I_D will give a closed-form I-V model... but the expression is complex

Proposal: modified empirical model targeting closed-form I-V

$$i_D = \frac{I_D}{I_{S0}} = \frac{(q_S - q_D)(q_S + q_D + 2)}{[1 + \zeta(q_S - q_D)][1 + \theta(q_S + q_D)]}$$

where

$$\frac{V_P - V_{S(D)B}}{U_T} = q_{S(D)} - q_{sat} - 1 + \ln(q_{S(D)} - q_{sat})$$

J. Poupon et al., IEEE Access 2024

Towards an analytical I-V MOSFET model

Replacing $q_S(V_G, V_S, V_D)$ and $q_D(V_G, V_S, V_D)$ in I_D will give a closed-form I-V model... but the expression is complex

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where

$$q_{S(D)} = \ln \left[1 + \frac{e^{v_P - v_{S(D)B}} + 1}{1 + \frac{1}{2} \ln \left(1 + e^{v_P - v_{S(D)B}} + 1 \right)} \right] \left[1 + \frac{\zeta}{2} \ln \left[1 + \frac{e^{v_P - v_{S(D)B}} + 1}{1 + \frac{1}{2} \ln \left(1 + e^{v_P - v_{S(D)B}} + 1 \right)} \right] + \zeta \right]$$

J. Poupon et al., IEEE Access 2024

Take aways

- ❑ The ACM2 model contains transcendental equations
- ❑ Numerical solving and/or approximations are needed
- ❑ acm 433 algorithm and Lambert W function are two useful mathematical tools in this context
- ❑ Happy coding!!

A graphic consisting of four overlapping speech bubbles in red, yellow, grey, and teal. The text "Thank you!" is written in white across the center of the bubbles.

Thank you!

Design and Simulation of analog/RF ICs with Open-Source CAD Tools and PDKs: Parameter Extraction and Circuit Simulation

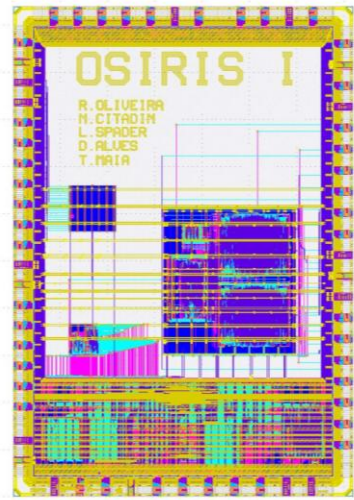
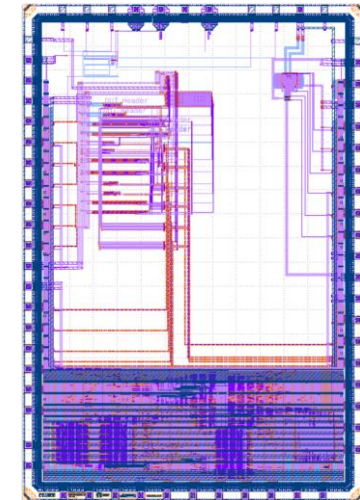
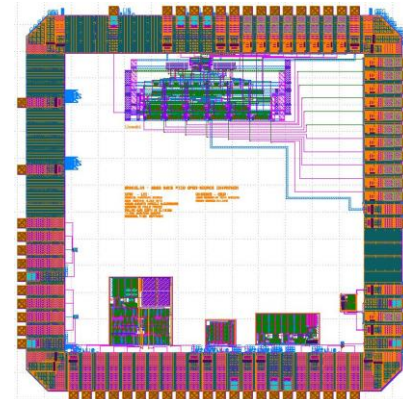
Deni Germano Alves Neto
PhD Student

Universidade Federal de Santa Catarina, Brasil
Université Grenoble Alpes, France

About me



- Universidade Federal de Santa Catarina – UFSC - Brasil
- Undergrad and Masters in IC design - 2022 $V_{DD} < 100\text{ mV}$
 - Dissertation: Ultra-Low-Voltage Standard Cell Library
 - ACM for low voltage circuits
- Contact with open-source IC design : Analog/Digital IC design
 - Chipathon-SSCS 2021/2023
 - UNIC-CASS 2023/2024
- Joint PhD in 2023 : UFSC & UGA
 - MOSFET Modeling – ACM2
 - RFIC design



Outline

- **Compact models**
- ACM2 model
 - Parameter Extraction
 - DC characteristics
 - Circuit examples

Why the need for a design-oriented MOSFET model ?

- Medium of information exchange between foundry and designer.
- Provides detailed information about device operation & characteristics.
- Simple enough to be incorporated in circuit simulators.
- Accurate enough to predict behavior of circuits.

Oversimplified models

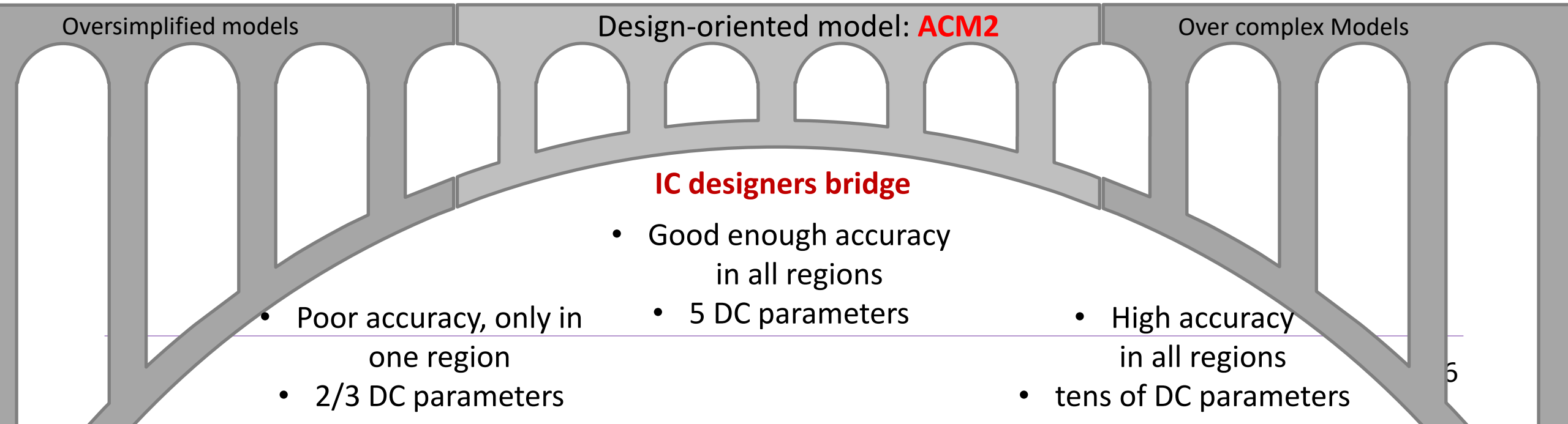
- Poor accuracy, only in one region
- 2/3 DC parameters

Over complex Models

- High accuracy in all regions
- tens of DC parameters

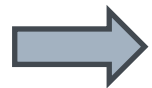
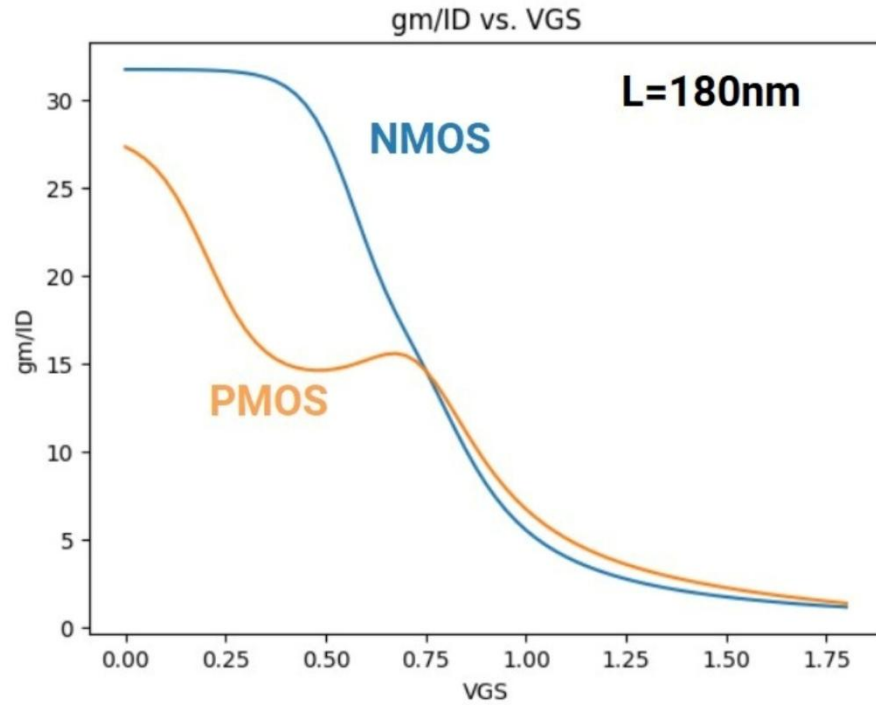
Why the need for a design-oriented MOSFET model ?

- Medium of information exchange between foundry and designer.
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- Simple enough to be incorporated in circuit simulators.
- Accurate enough to predict behavior of circuits.
- **Bridges the gap between design and simulation!**

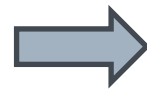


Problems in current open PDK models

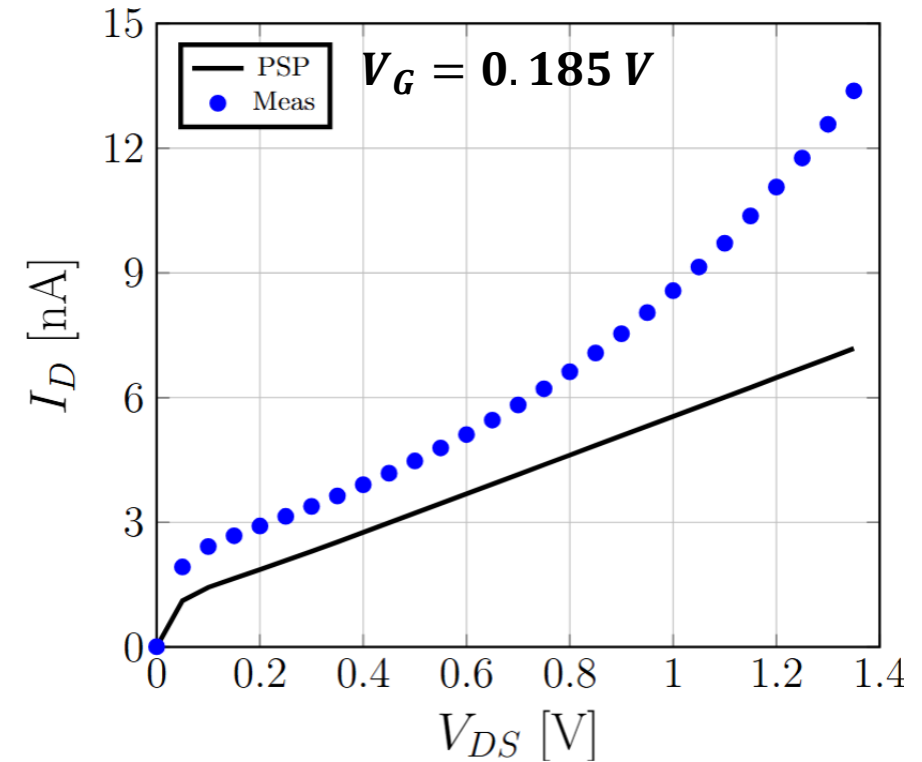
BSIM4 issues in sky130



Subthreshold region



PSP103 issues in IHP open PDK



**Modeling problem in weak inversion :
Effect of DIBL on the output
conductance not included**

Open-Source IC design

Open-source PDK



Open-source EDA tools



Few open-source IP & libraries



Available compact MOSFET models



We propose:

ACM2 : A Simple 5-DC-parameter MOSFET model



Outline

- Compact models
- **ACM2 model**
 - Parameter Extraction
 - DC characteristics
 - Circuit examples

ACM2: A simple 5-DC-parameter MOSFET model

Complete continuous all-region charge-based MOSFET model

$$V_P = \frac{V_{GB} - V_{T0} + \sigma(V_{DB} + V_{SB})}{n}$$

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln(q_s)$$

$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

$$\frac{V_{DS}}{\phi_t} = q_s - q_d + \ln\left(\frac{q_s - q_{dsat}}{q_d - q_{dsat}}\right)$$

$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$

- Used to calculate q_s
- Bridge between **WI** and **SI** regions

- Used to calculate q_d
- Bridge between **Triode** and **Saturation** regions

Specific
current
 $I_S (W, L)$

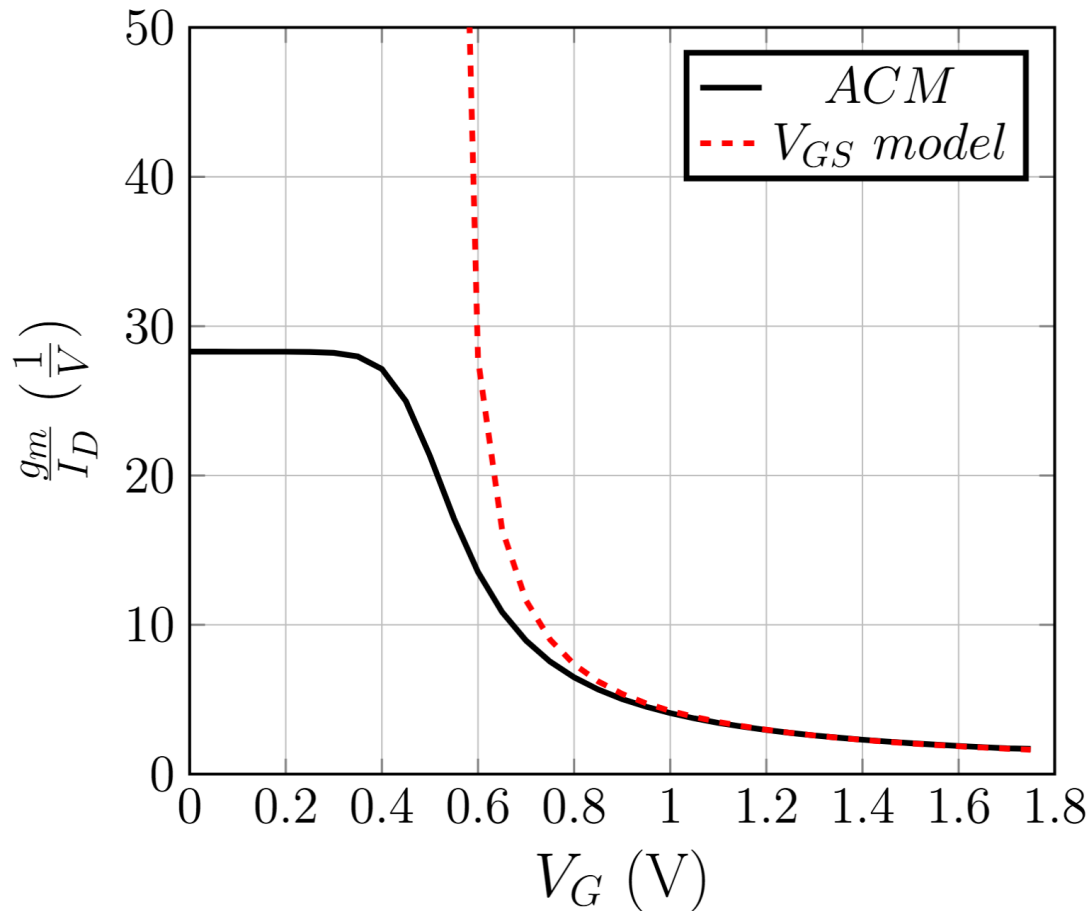
Threshold
voltage
 $V_{T0} (W, L)$

Slope
factor
 $n (W, L)$

DIBL
factor
 $\sigma (W, L)$

V_{sat}
effect
 $\zeta (W, L)$

Oversimplified model vs ACM model @ Saturation



Quadratic model:

$$I_D = \frac{\beta}{2} (V_{GS} - V_T)^2$$

$$\frac{g_m}{I_D} = \frac{1}{V_{GS} - V_T}$$

Unified **C**harge **C**ontrol **M**odel

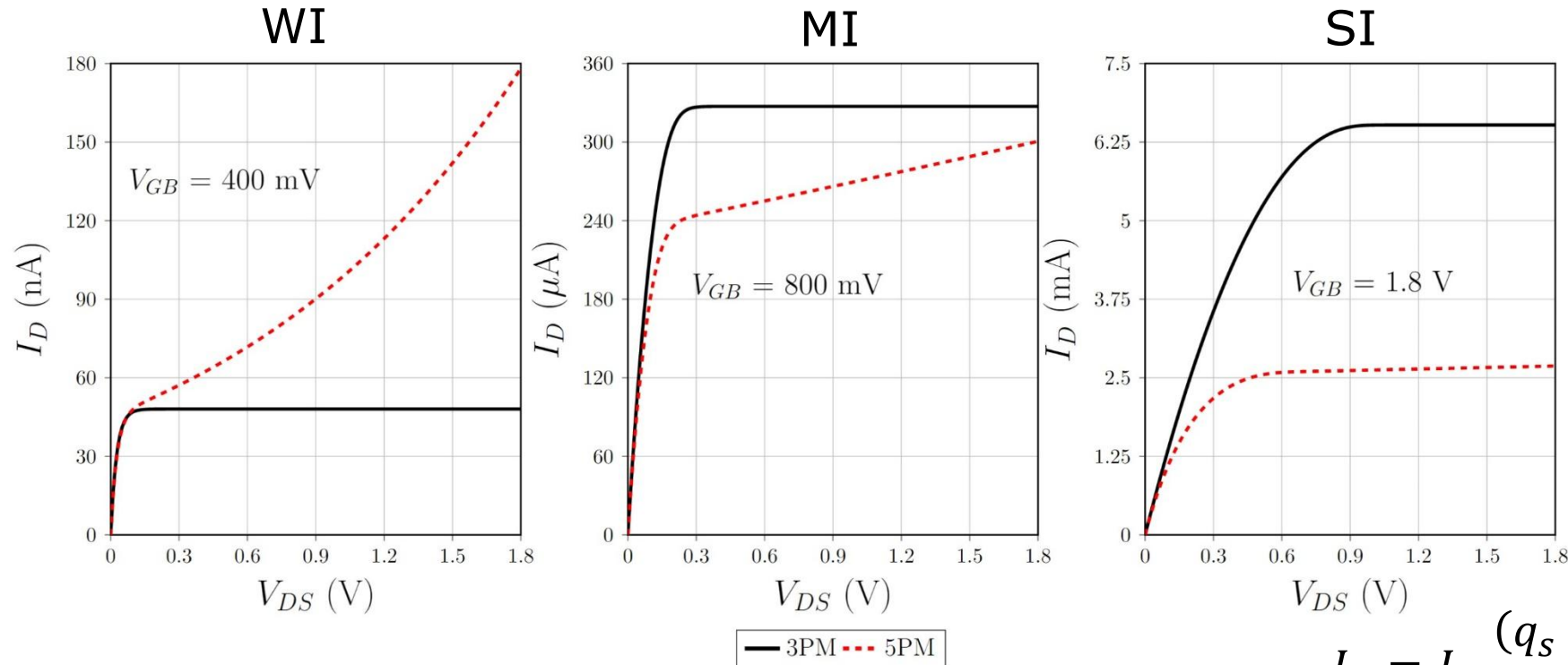
$$V_P = \frac{V_{GB} - \textcolor{blue}{V}_{T0}}{\textcolor{blue}{n}}$$

$$\frac{V_P - V_{SB}}{\phi_t} = q_s - 1 + \ln q_s$$

$$I_D = \textcolor{blue}{I}_s (q_s^2 + 2q_s)$$

$$\frac{g_m}{I_D} = \frac{2}{n\phi_t(1 + q_s)}$$

Output characteristics including DIBL and v_{sat}



DIBL model: $V_T = V_{T0} - \sigma(V_{SB} + V_{DB})$

$$I_D = I_S \frac{(q_s + q_d + 2)}{1 + \zeta(q_s - q_d)} (q_s - q_d)$$

Transistor	$W/L \text{ } [\mu\text{m}]$	$V_{T0} \text{ } [\text{mV}]$	$I_S \text{ } [\mu\text{A}]$	n	σ	ζ
NMOS	5/0.18	528	5.52	1.37	0.025	0.056

$$\zeta = \frac{(\mu_s \phi_t / L)}{v_{sat}}$$

Outline

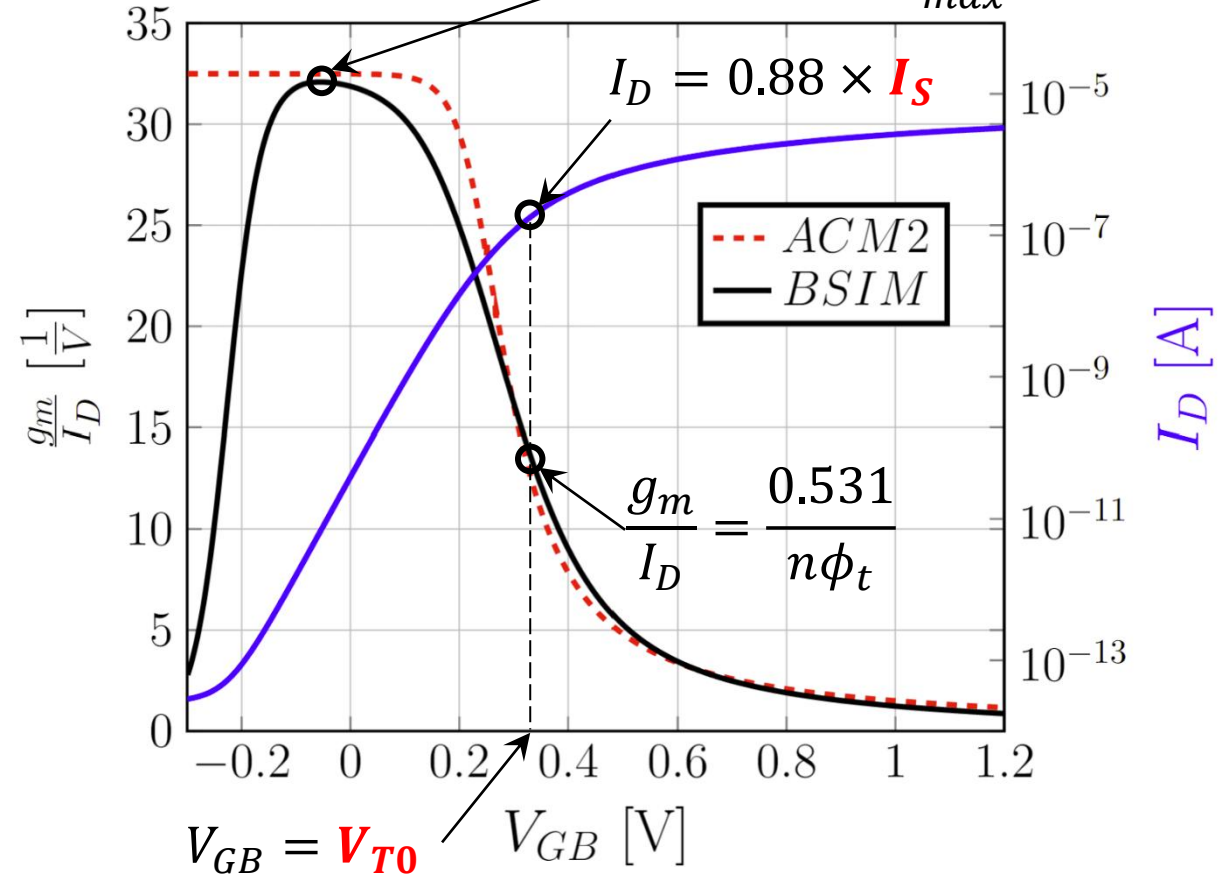
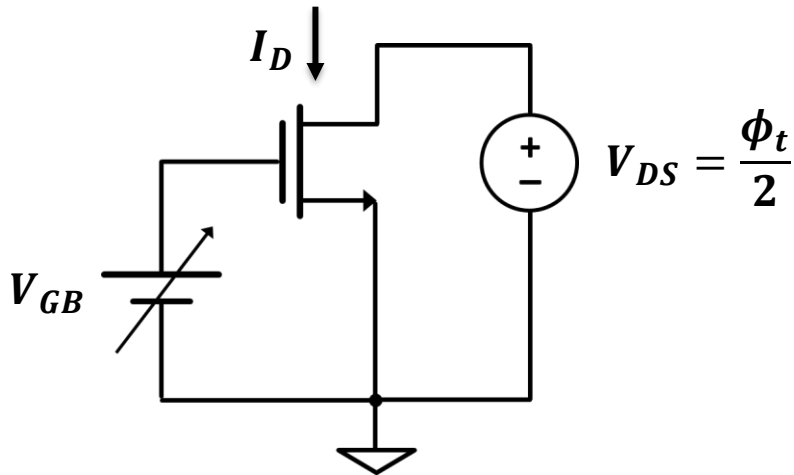
- Compact models
- ACM2 model
 - **Parameter Extraction**
 - DC characteristics
 - Circuit examples

I_S , V_{T0} and n extraction - g_m/I_D method

For $q_S = 1$, $V_{GB} = V_{T0}$ and g_m/I_D is at $\frac{1}{2}$ of its maximum value $\left(\frac{g_m}{I_D}\right)_{max} \cong \frac{1}{n\phi_t}$

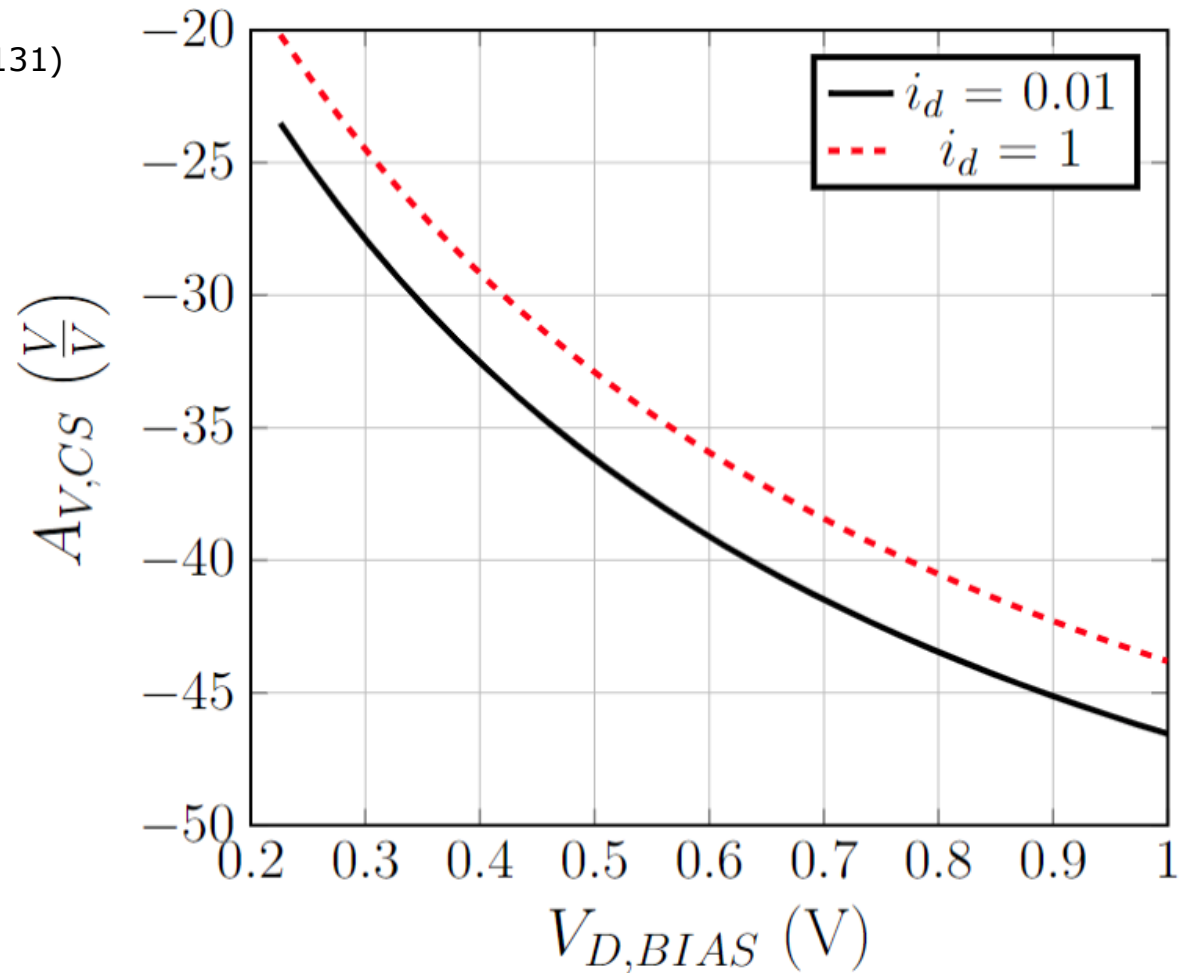
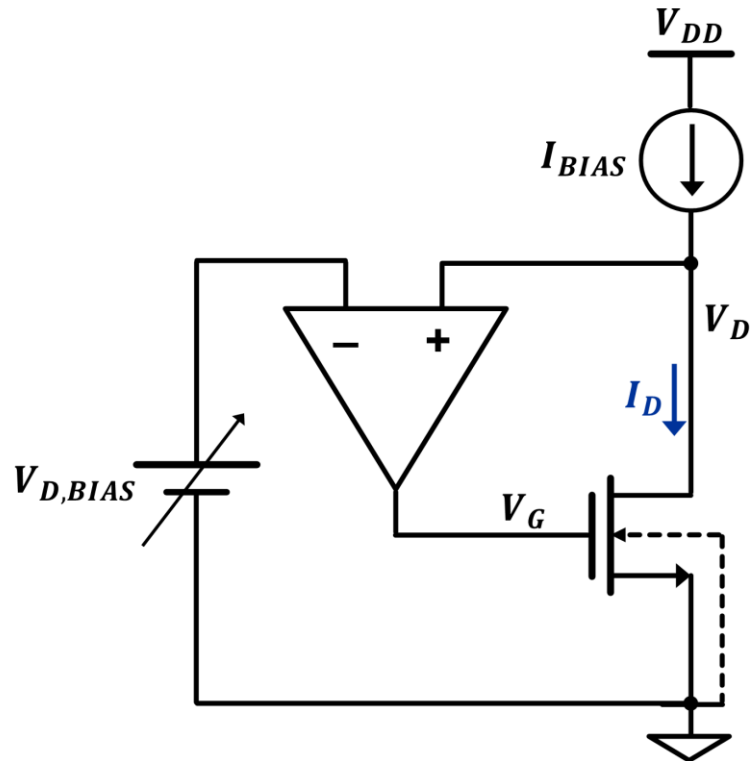
$$\frac{V_{GB} - V_{T0}}{n} = q_S - 1 + \ln q_S$$

$$\left.\frac{g_m}{I_D}\right|_{V_{DS} \rightarrow 0} = \frac{1}{n\phi_t(1 + q_S)}$$



σ extraction –CS Intrinsic gain method¹

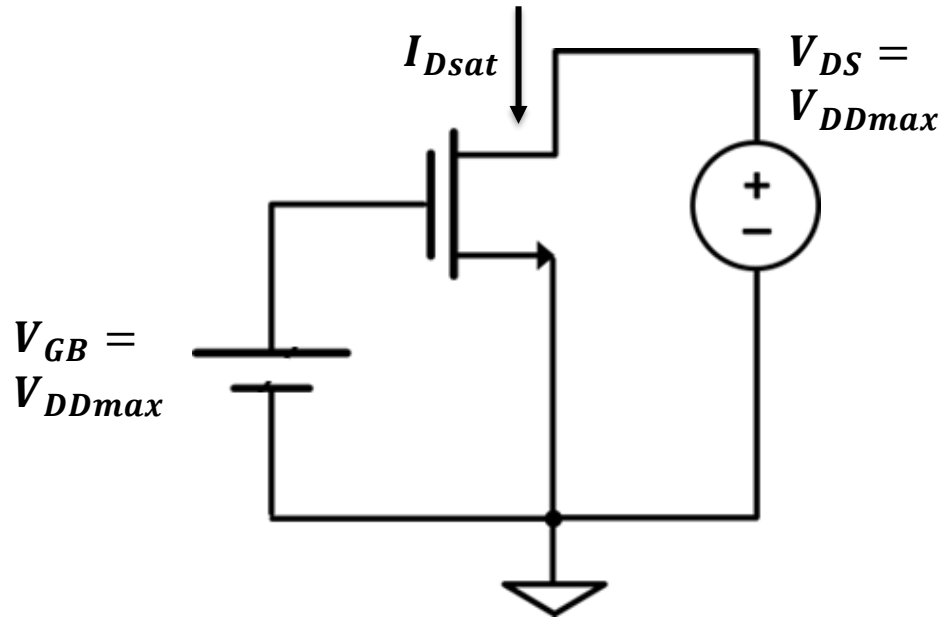
$$A_{V,CS} = -\frac{g_{msat}}{g_{dssat}} \cong -\frac{1}{\sigma} \quad (\text{Please see slide 131})$$



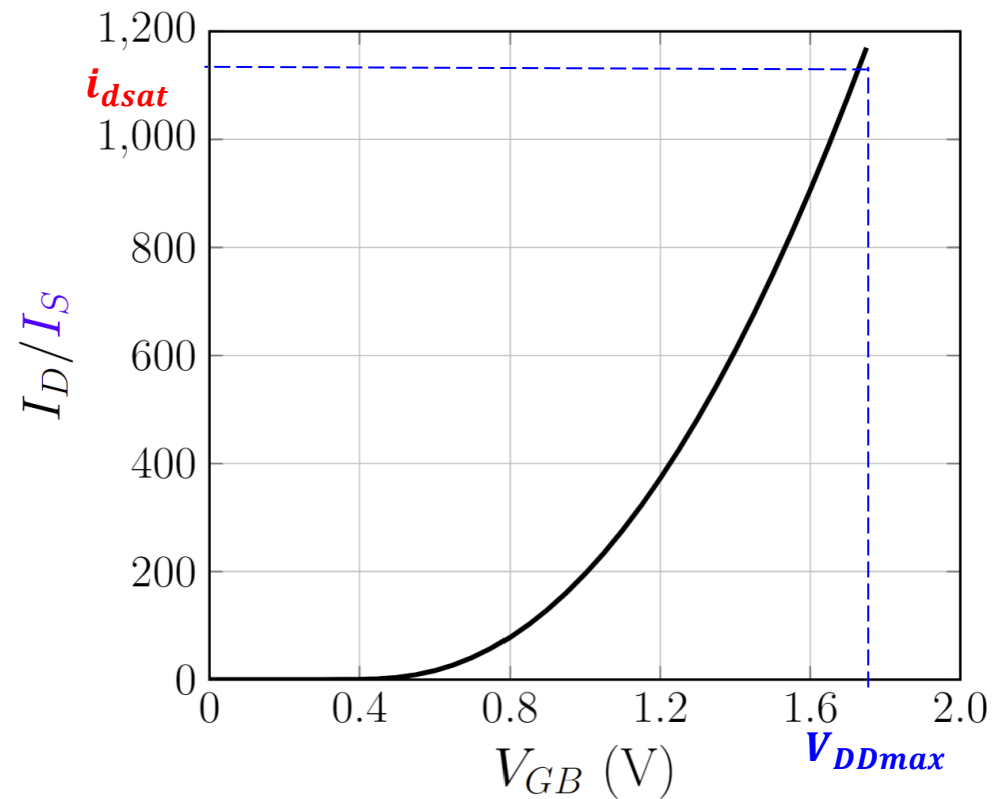
1 - C.M. Adornes et al, Journal of Low Power Electronics, 2022.

ζ extraction

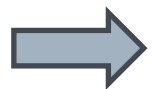
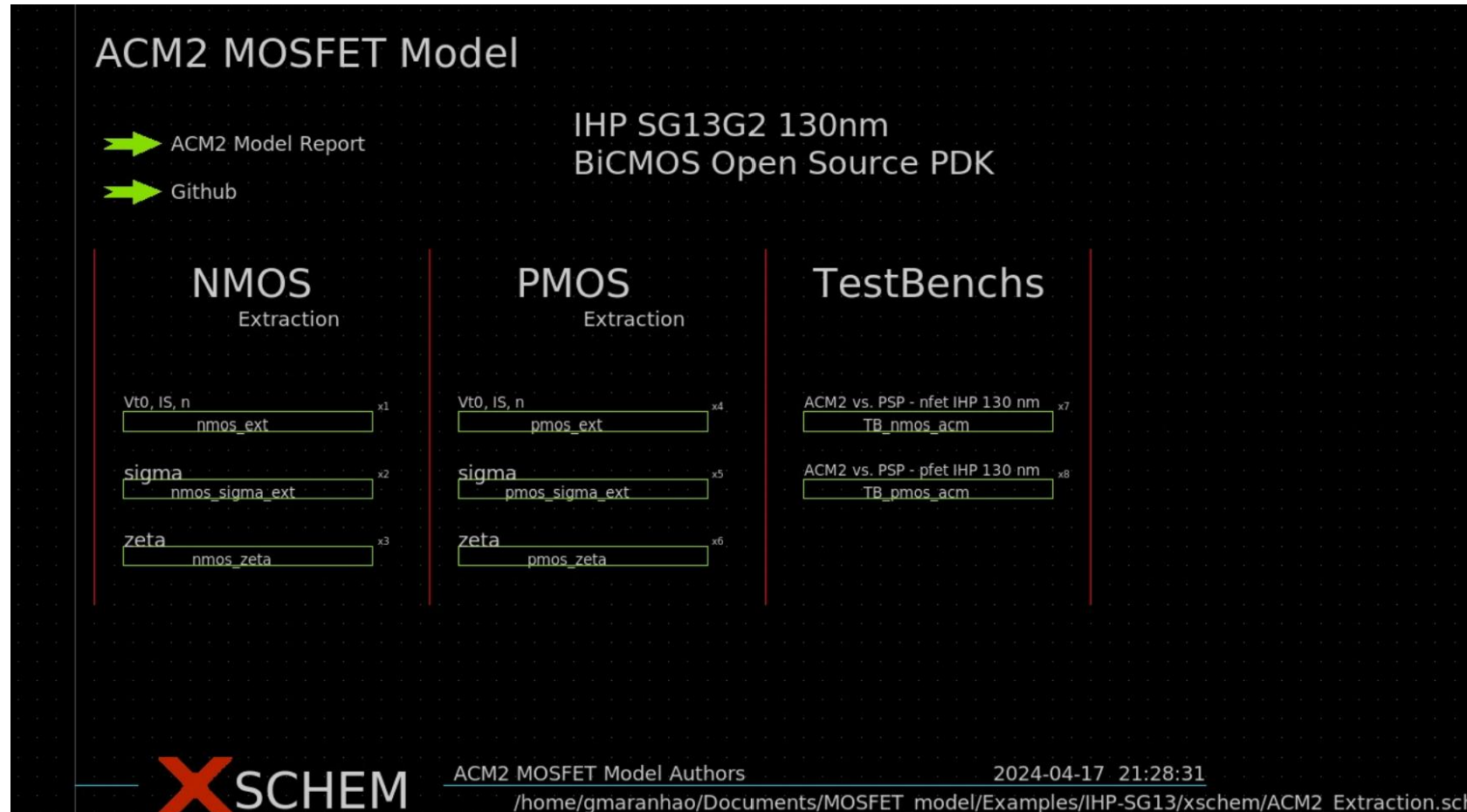
- q_s calculated from **UCCM** (V_{T0} , n , σ).
- Measure $i_{dsat} = I_{Dsat}/I_S$.



$$\zeta = \frac{2(q_s + 1 - \sqrt{1 + i_{dsat}})}{i_{dsat}}$$



Automatic parameter extraction tool @ XSCHEM

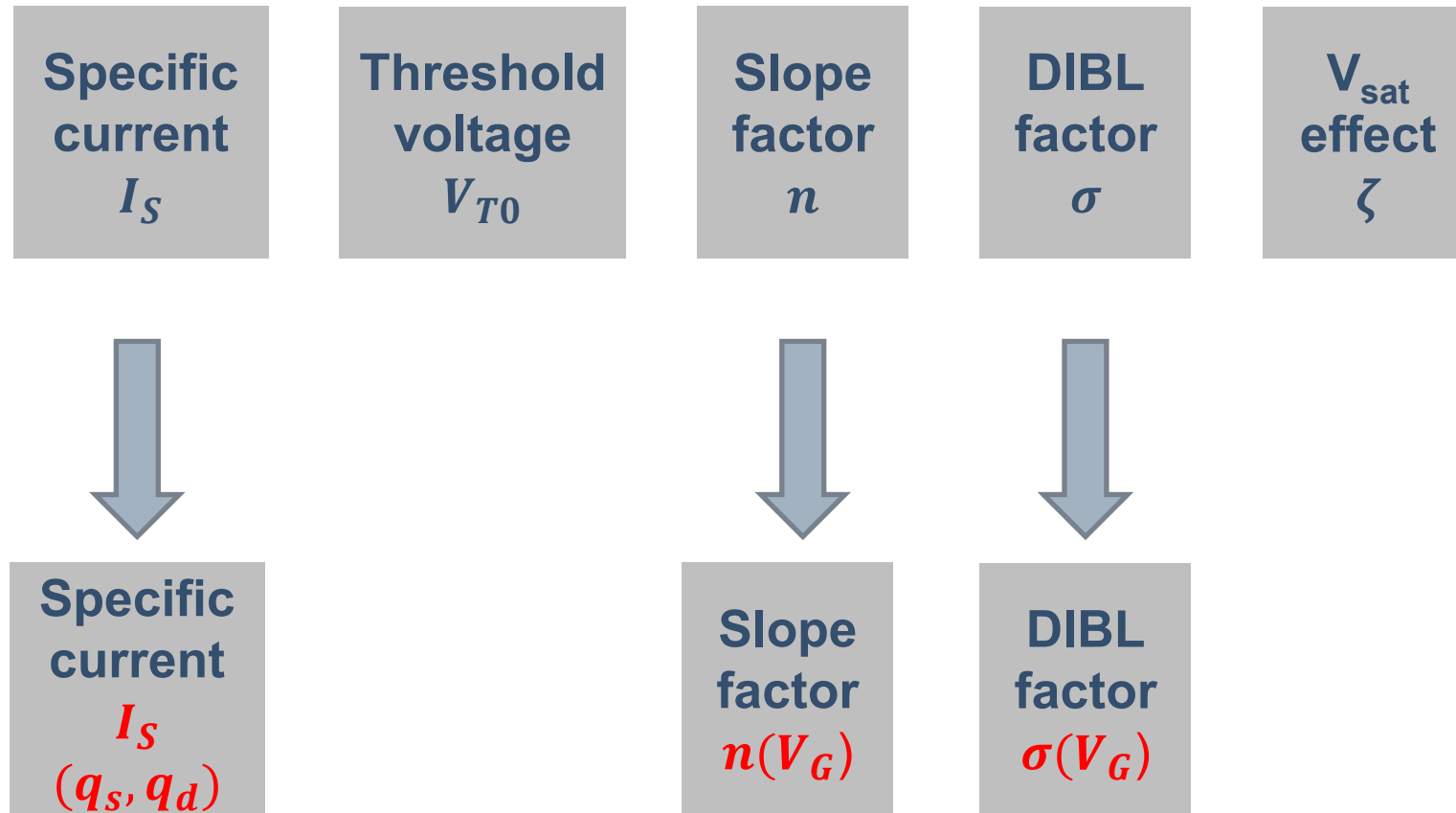


Available for IHP130, GF180 and SKY130

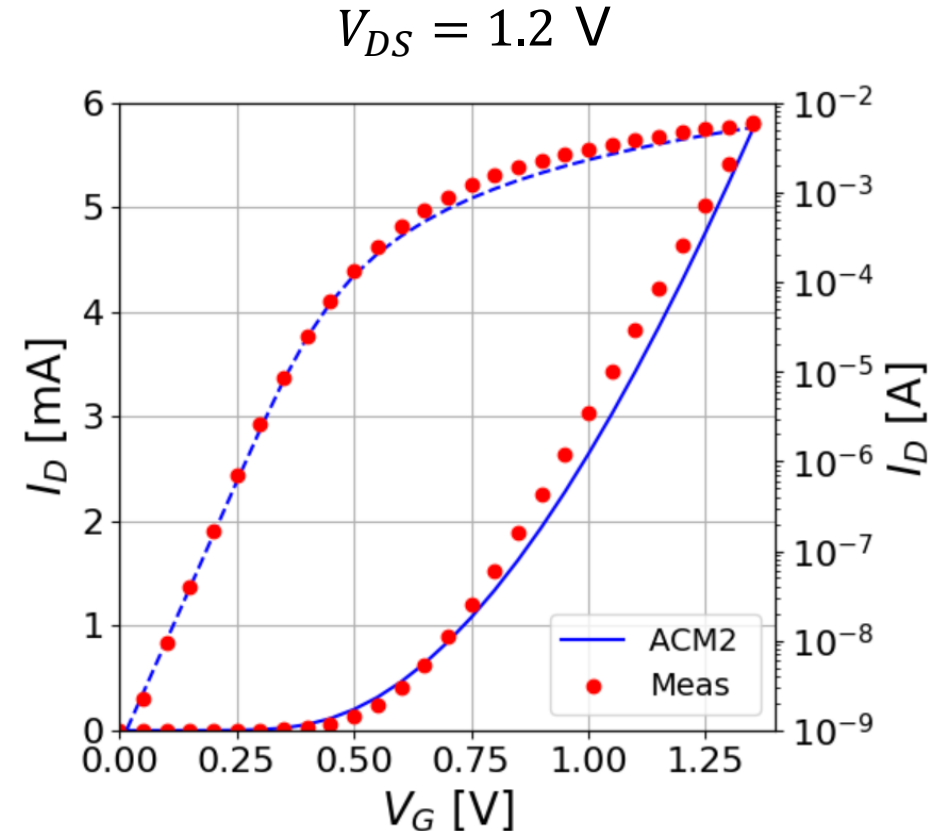
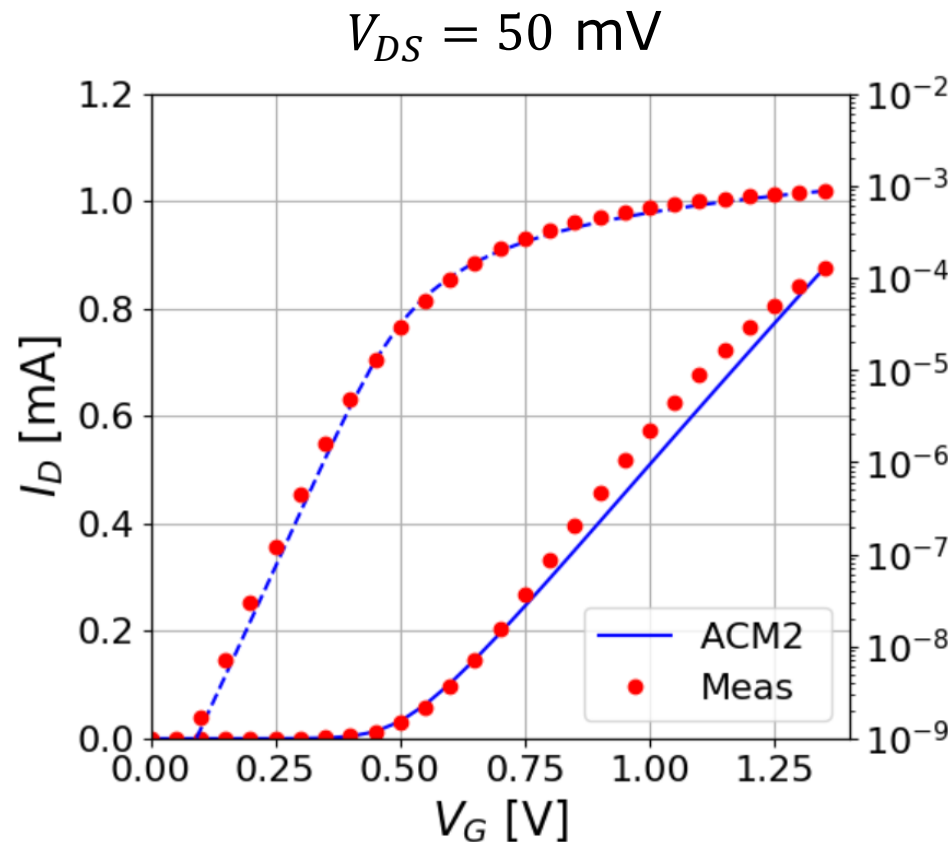
Outline

- Compact models
- ACM2 model
 - Parameter Extraction
 - **DC characteristics**
 - Circuit examples

ACM2: A simple 5-DC-parameter MOSFET model



ACM2¹ vs meas – 130 nm SiGe IHP²– I_D vs V_{GB}



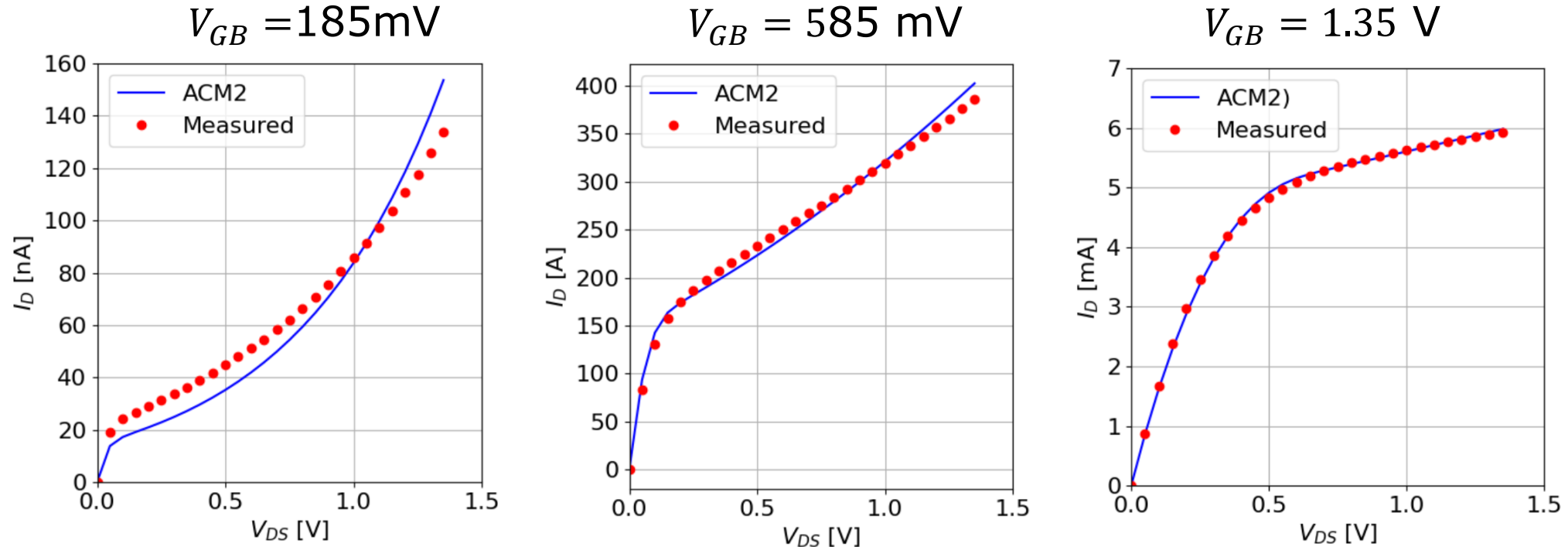
Characteristics of a LVT NMOS bulk transistor with $W/L = 10\mu\text{m}/120$ nm.

¹ ACM2 : implemented in verilog-A, compiled by OPENVAF, simulated in Ngspice

² Institut for High-Performance Microelectronics (IHP) open-source PDK

$V_S = V_B = 0$

ACM2¹ vs meas – 130 nm SiGe IHP² – I_D vs V_{DS}



OPENVAF compilation time ➡

	PSP103	BSIM4	EKV2.6	ACM2
time(s)	3.48	6.7	0.22	0.15

Characteristics of a LVT NMOS bulk transistor with $W/L = 10\mu\text{m}/120\text{ nm}$.

¹ ACM2 : implemented in verilog-A, compiled by OPENVAF, simulated in Ngspice

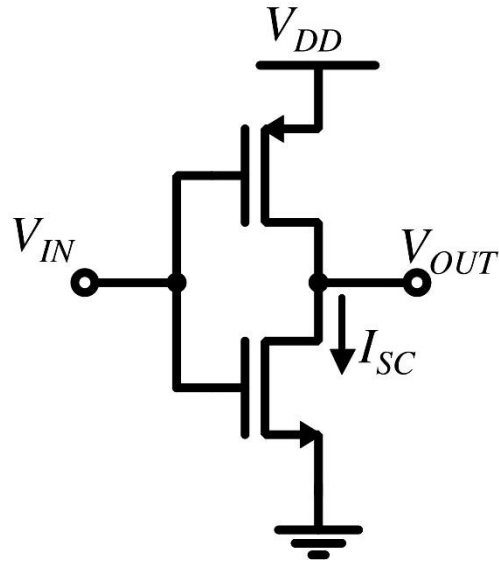
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$V_S = V_B = 0$

Outline

- Compact models
- ACM2 model
 - Parameter Extraction
 - DC characteristics
 - **Circuit examples**

CMOS Inverter in 130 nm bulk

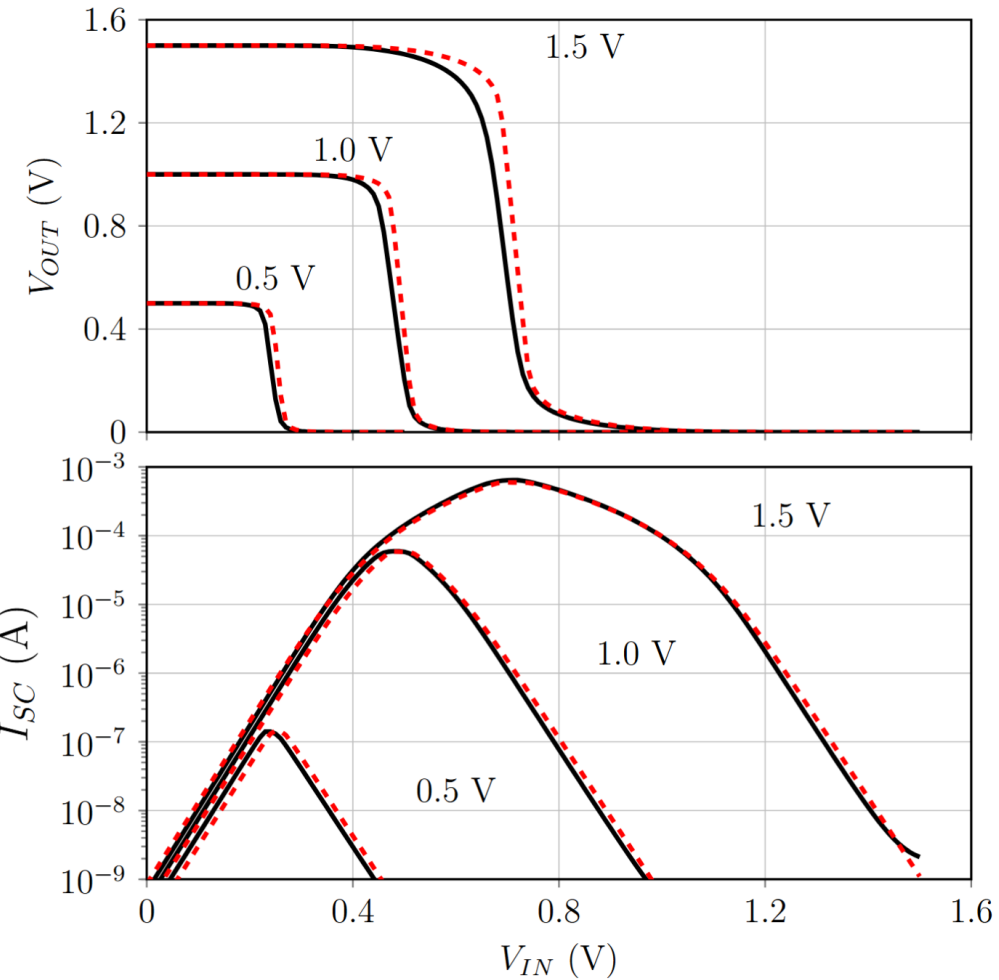


$$W_n = W_p = 10 \mu\text{m}$$
$$L_n = L_p = 120 \text{ nm}$$

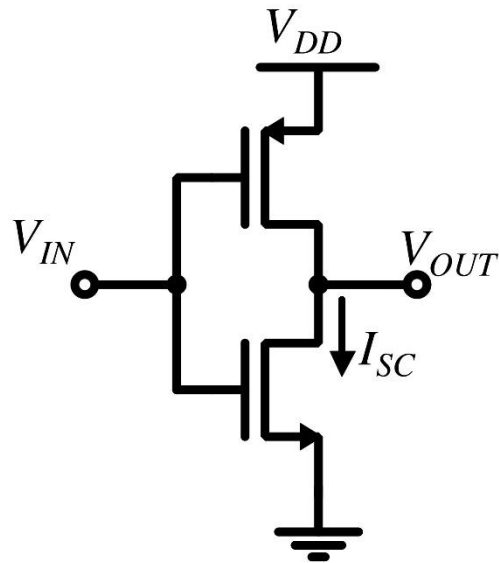
ACM

PSP103

VTC



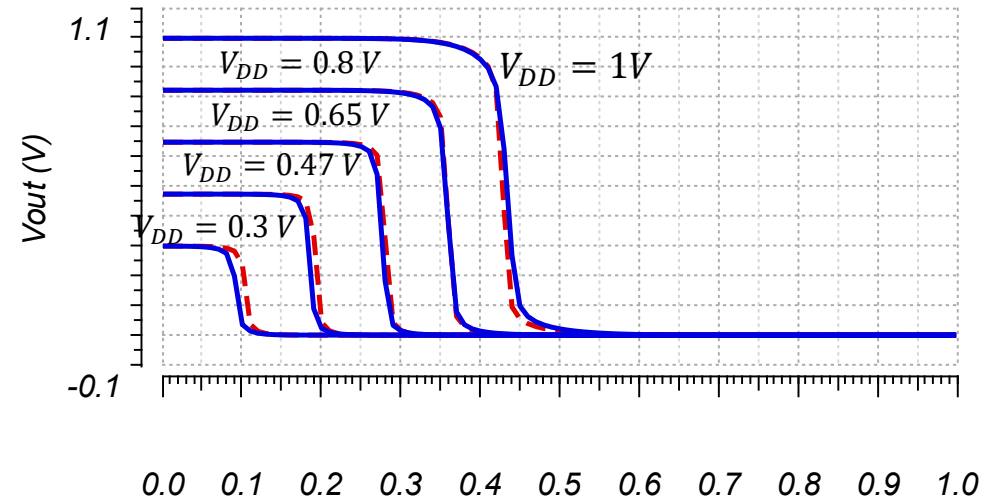
CMOS Inverter in 28 nm FD-SOI



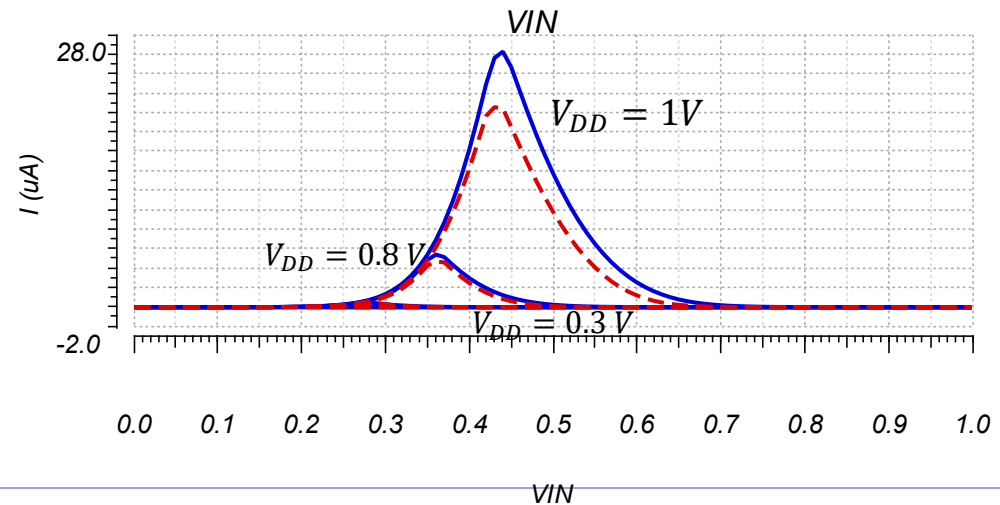
$$W_n = W_p = 1 \mu m$$
$$L_n = L_p = 60 nm$$

ACM	---
UTSOI2	---

VTC

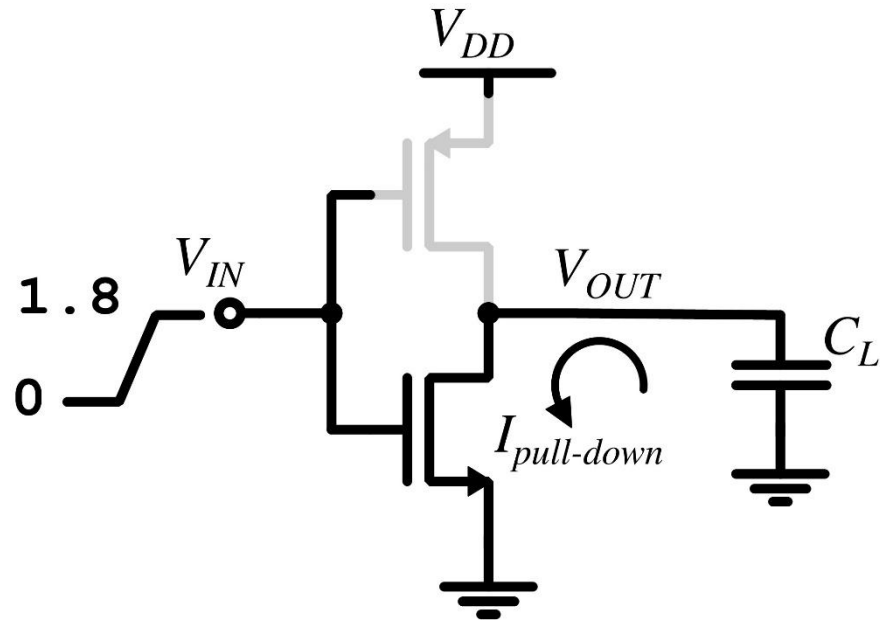


Short-circuit current

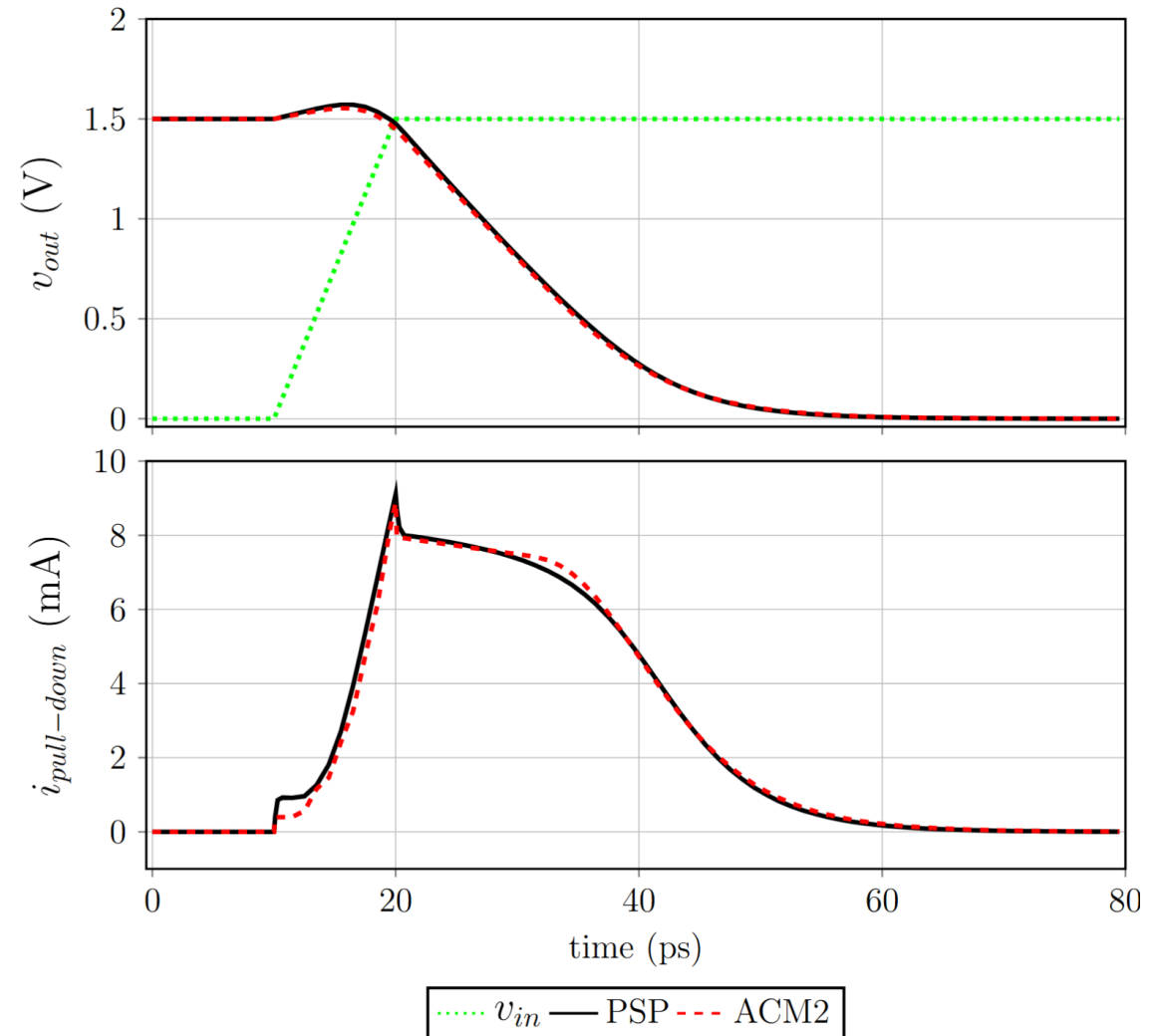


CMOS Inverter in 130 nm bulk

Output Voltage and pull-down current

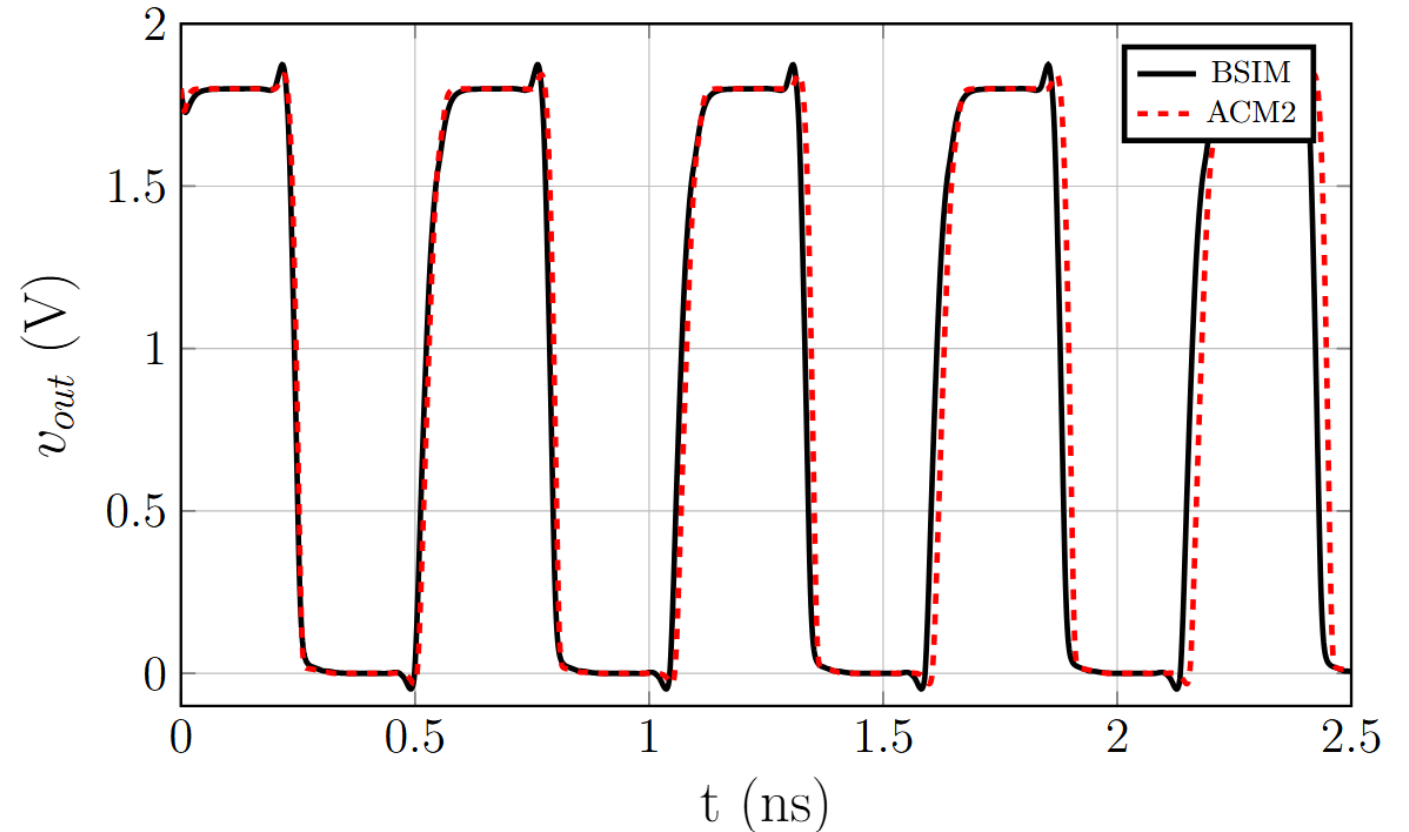
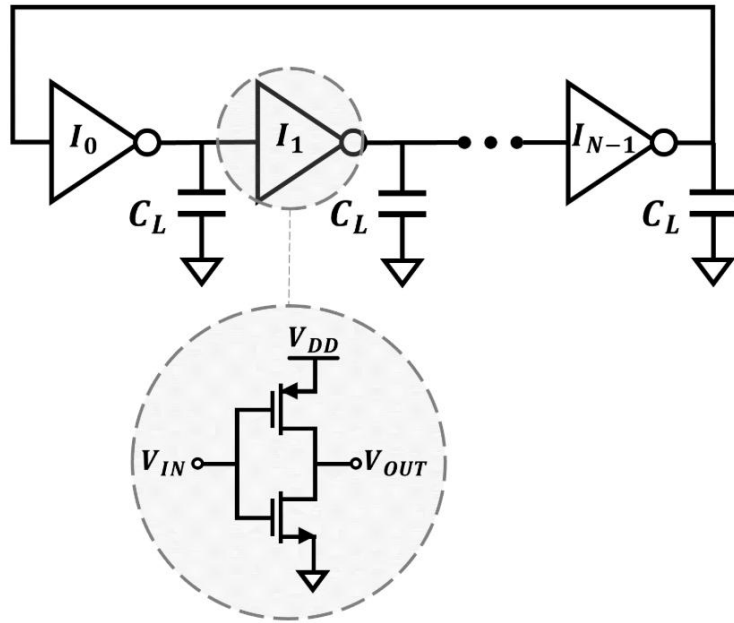


$$W_n = W_p = 10 \mu m$$
$$L_n = L_p = 120 nm$$

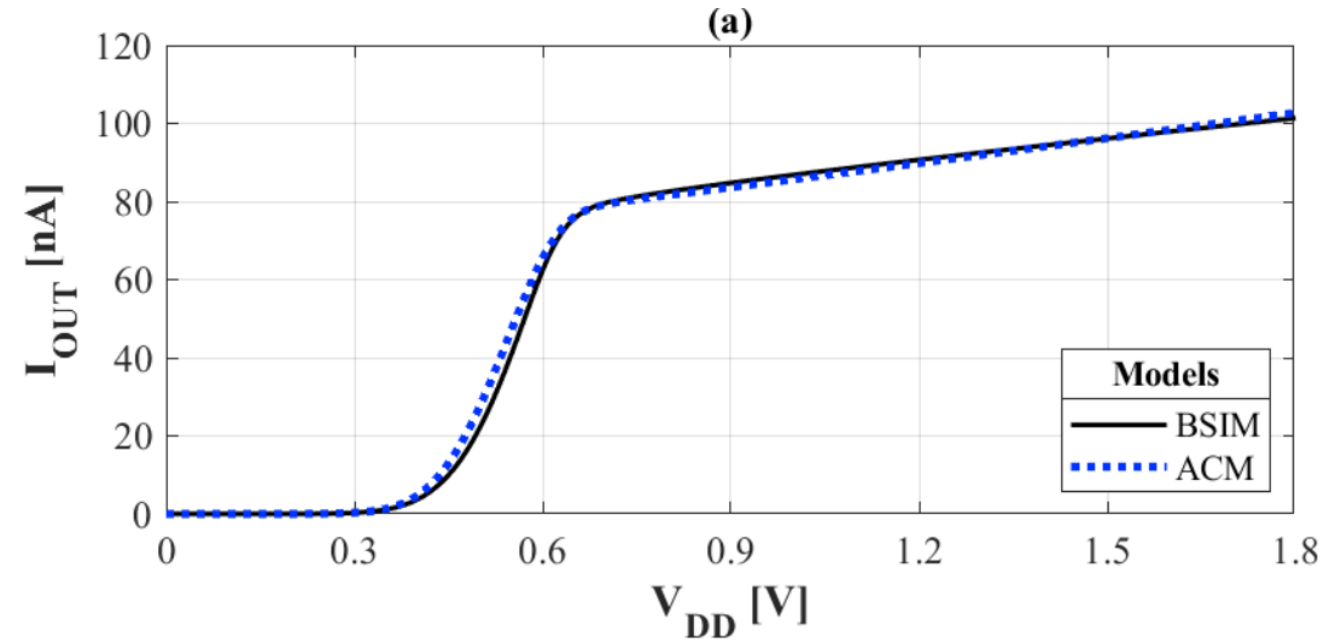
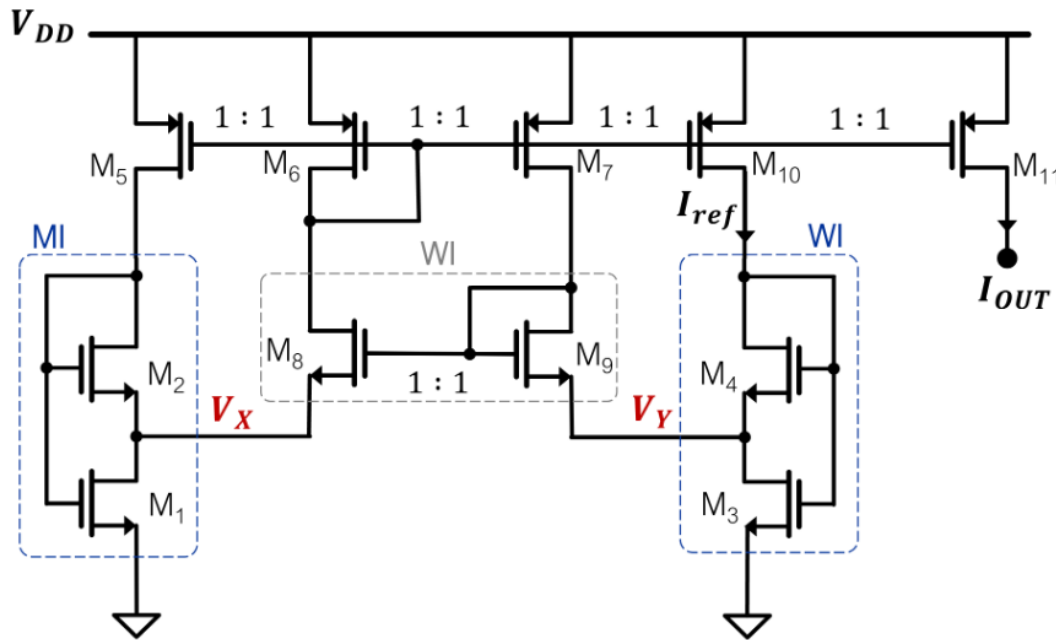


11-Stages Ring Oscillator 180 nm bulk

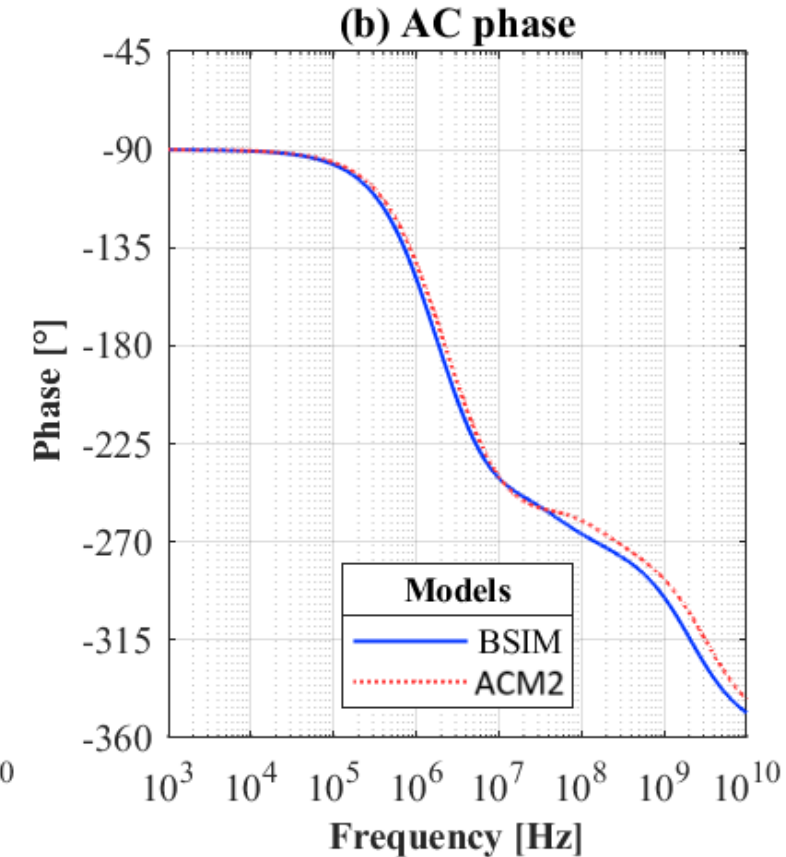
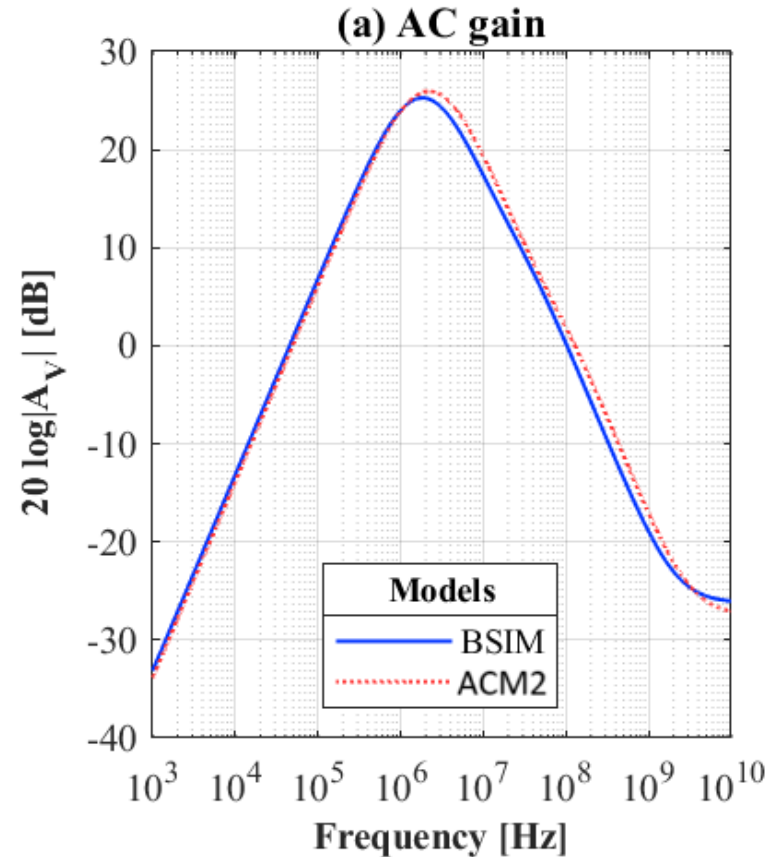
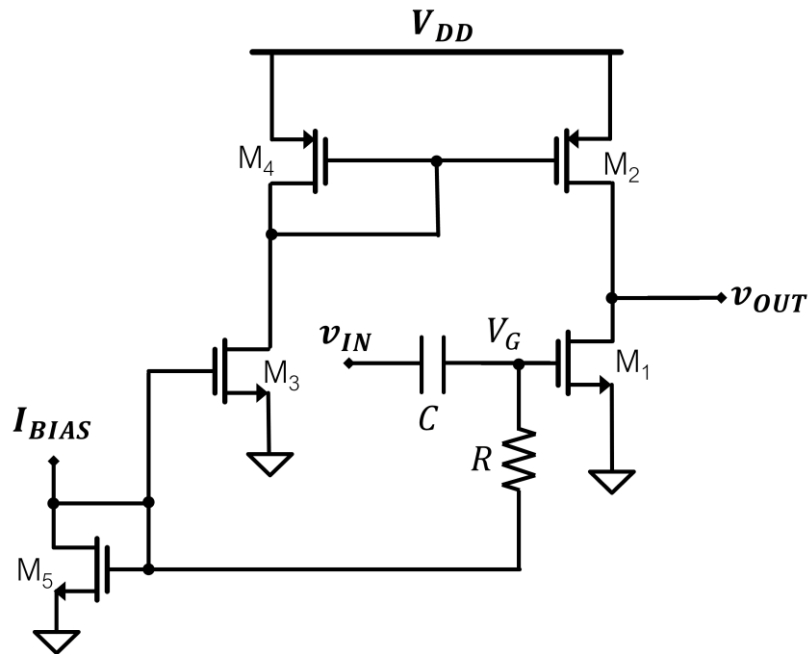
Output Voltage



Self-biased current source in 180 nm bulk



Common-source Amplifier in 180 nm bulk

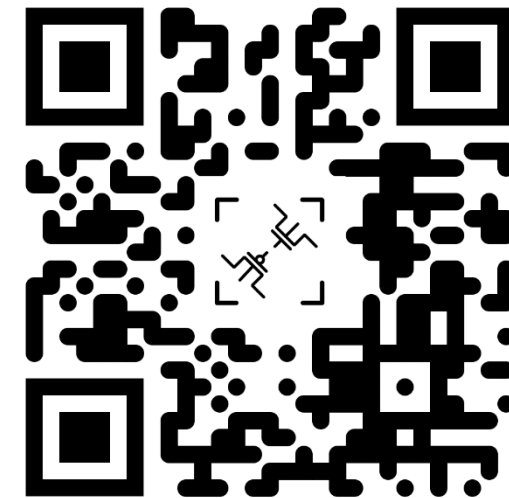
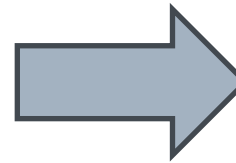


Summary – The ACM2 model

- A truly compact MOSFET model with single-piece functions
- Implemented in Verilog-A for simulation
 - Interchangeable between simulators (SPICE or SPECTRE)
- Verified in all three open-source PDKs (Sky130, GF180, IHP-SG13G2)
- Helpful to designers (only 5-DC-parameters)
- Simplified parameter extraction procedure
 - extracted from simulations or chip measurements

- Deni Germano Alves Neto et al., "Design-Oriented Single-Piece 5-DC-Parameter MOSFET Model," in IEEE Access, 2024.
- C.M. Adornes, D.G. Alves Neto, M.C Schneider, C. Galup-Montoro, "Bridging the Gap between Design and Simulation of Low-Voltage CMOS Circuits," Journal of Low Power Electronics, 2022.
- ACM2 Github: https://github.com/ACMmodel/MOSFET_model
- IHP Github : <https://github.com/IHP-GmbH/IHP-Open-SDK>

- **Available in Github:**
 - **DC model**
 - **Small-signal model**
 - **Dynamic model**
 - **Thermal & Flicker noise models (1/f)**



Design and Simulation of analog/RF ICs with Open-Source CAD Tools and PDKs: RF Circuit Design

Pr. Sylvain Bourdel
TIMA Laboratory
Grenoble INP, France

Agenda

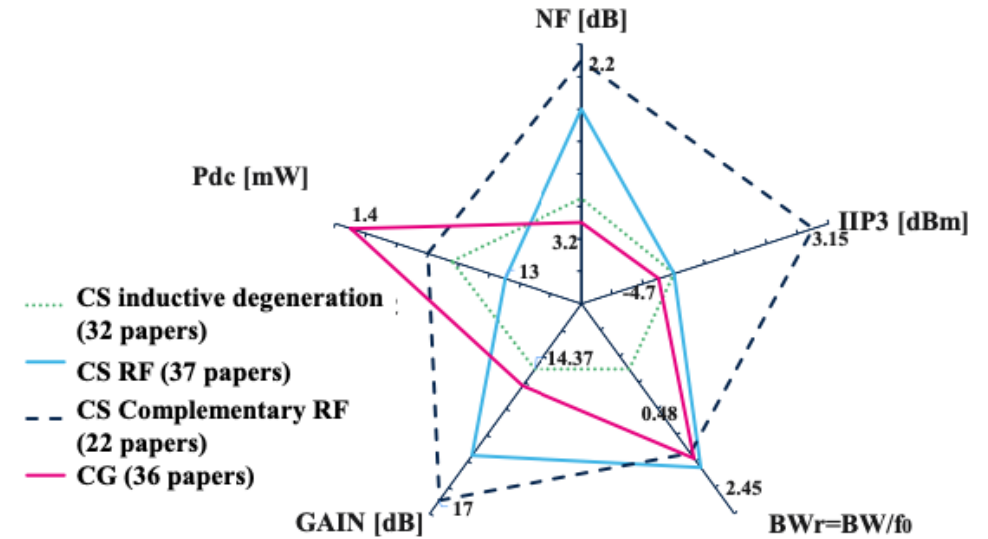
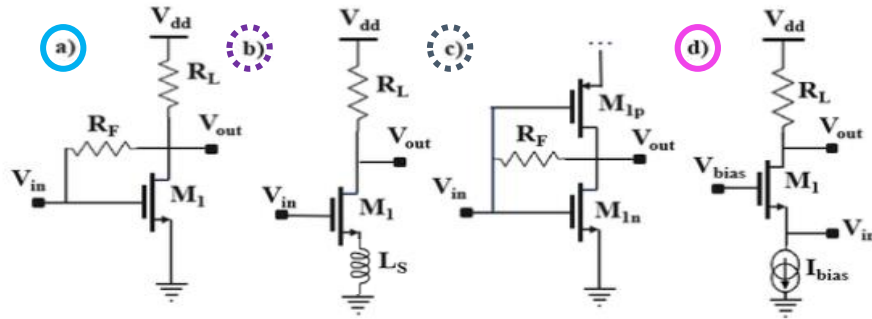
- Overview of Design Methods for RFIC
- LNA Design Considerations
- Resistive Feedback LNA
- What we need in ACM-2
- Inversion Level Based Method for R-Feedback LNA with ACM-2

Agenda

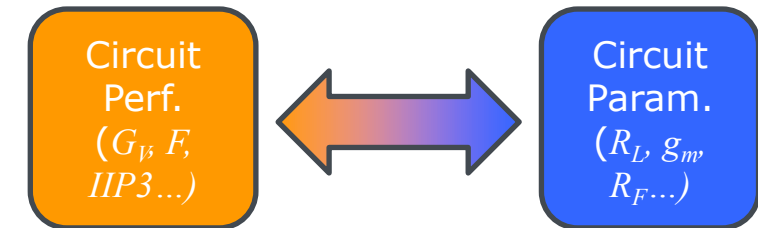
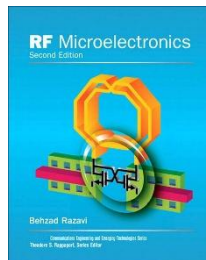
- **Overview of Design Methods for RFIC**
- LNA Design Considerations
- Resistive Feedback LNA
- What we need in ACM-2
- Inversion Level Based Method for R-Feedback LNA with ACM-2

Overview of Design Methods for RFIC General Principle

□ 1st Step: Architecture Choice



□ 2nd Step: Circuit Analysis

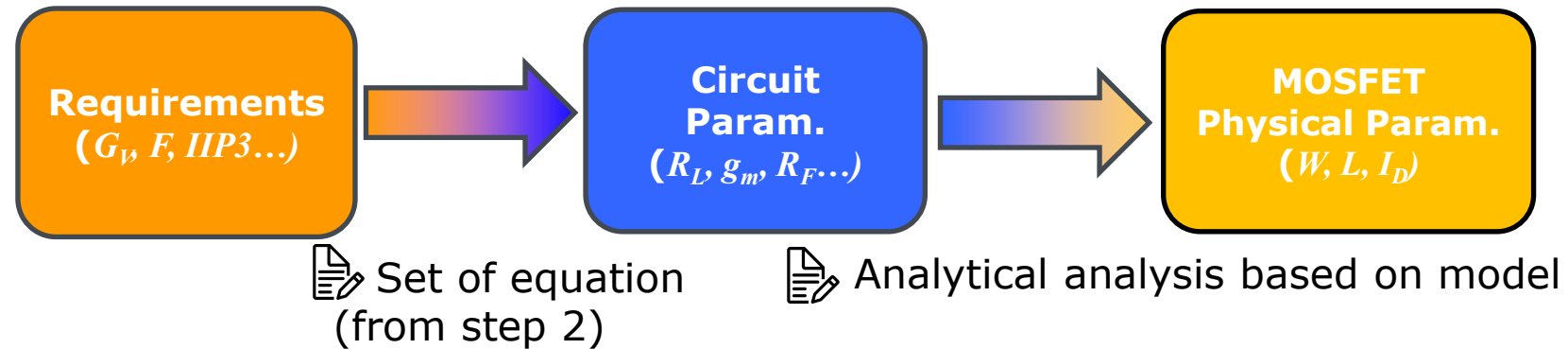


$$|G_T| = |G_v| Q_{IN} = \frac{(G_m R_F - 1) R_O}{(R_O + R_F)} \sqrt{1 + Q_P^2}$$

Overview of Design Methods for RFIC

General Principle

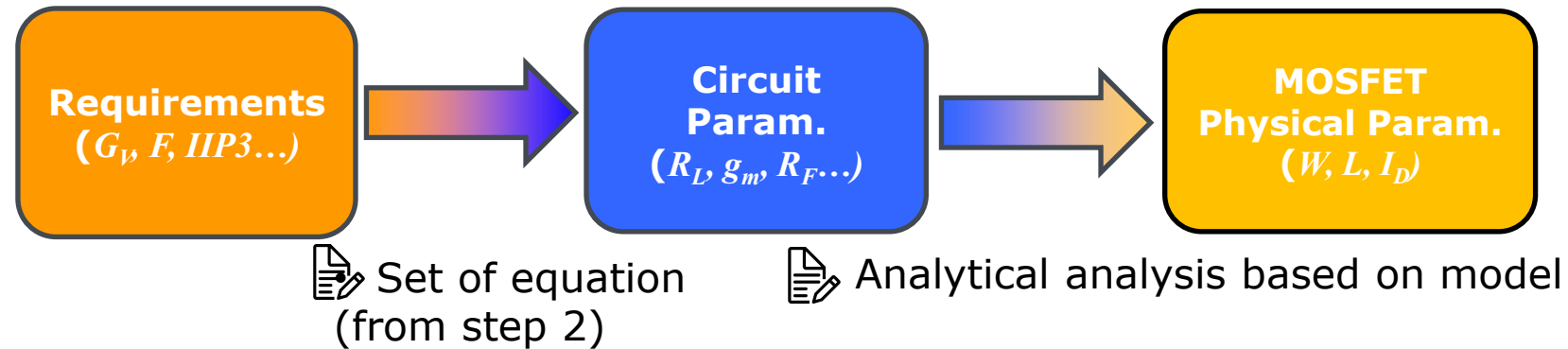
□ 3rd Step: Circuit Sizing



Overview of Design Methods for RFIC

General Principle

□ 3rd Step: Circuit Sizing



Region based model

$$sat. \quad I_d = 2.K_n \frac{W}{L} \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

$$lin. \quad I_d = K_n \frac{W}{L} (V_{gs} - V_t)^2 \left(1 + \frac{k_{en}}{L} V_{ds} \right)$$

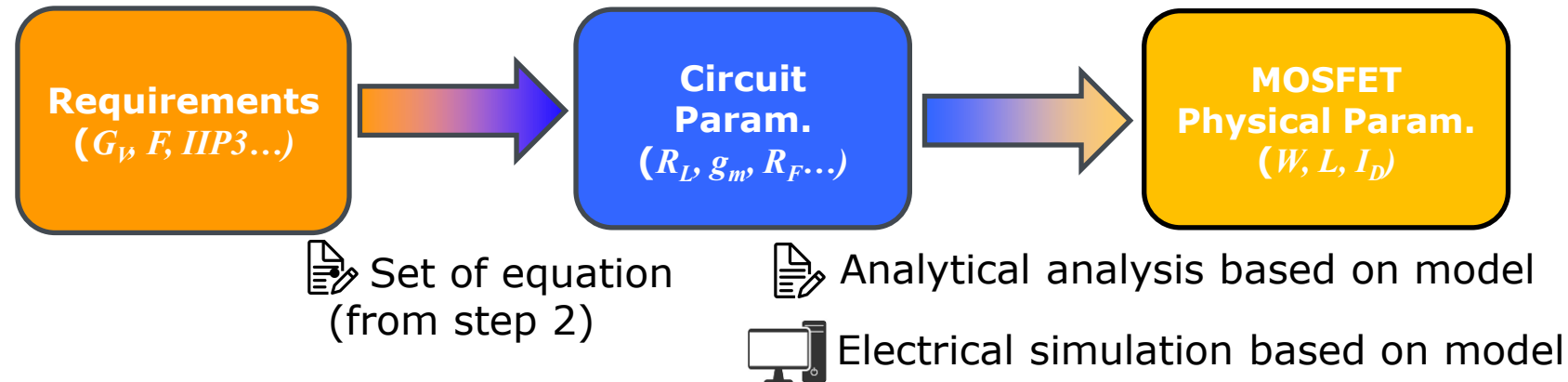
PROS: simple

CONS: Inaccurate with short channel MOS

Overview of Design Methods for RFIC

General Principle

□ 3rd Step: Circuit Sizing



BSIM3 / UTSOI compact model

PROS:

- Accurate
- Direct relationship between requirements and physical parameters

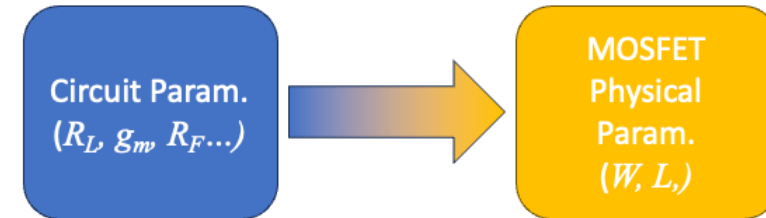
CONS:

- Increases the gap between physics and design
- Long optimization

Overview of Design Methods for RFIC

New Trends

□ Issue in the sizing step



? ■ How to maintain simplicity **and** accuracy with the scaling and especially Short Channel Effects (SCE) at the sizing step. ?

□ A possible solution

- g_m/I_d design approaches
 - LUT based
 - ACM or EKV based

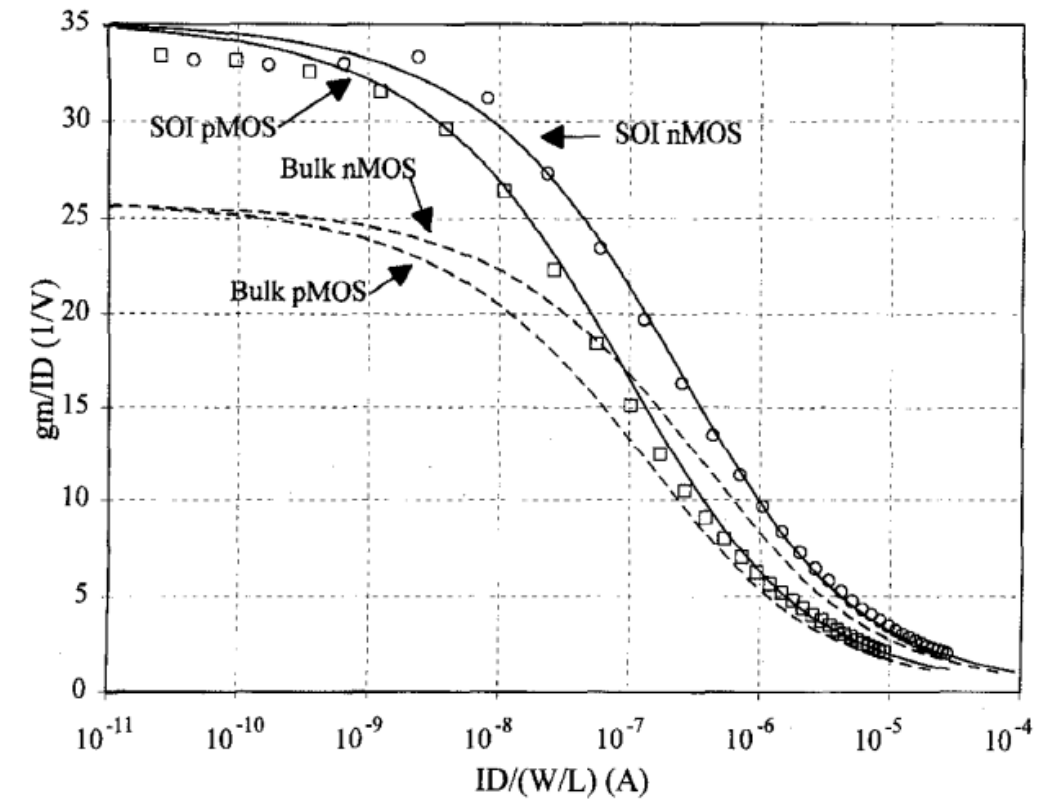
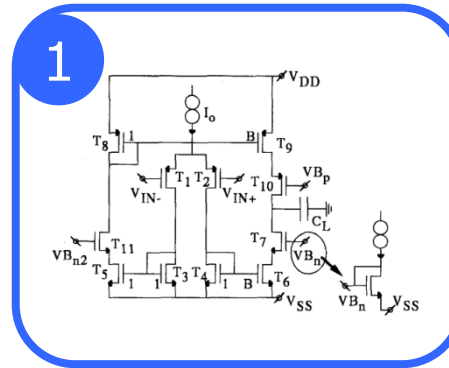
MUNICH
8-11 SEPTEMBER
ESERC
2025

Overview of Design Methods for RFIC

g_m/I_D approach based on LUT

General Principle

- For a given structure
- Methods are based on
 - Extraction of MOS parameters (LUT)
 - $I_D/(W/L)$ vs g_m/I_D
 - g_m/I_D vs V_{gs0}



Overview of Design Methods for RFIC

g_m/I_D approach based on LUT

General Principle

❑ For a given structure

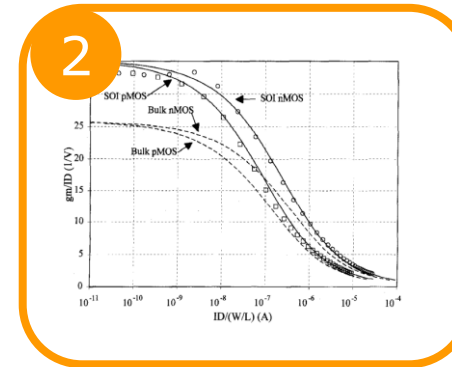
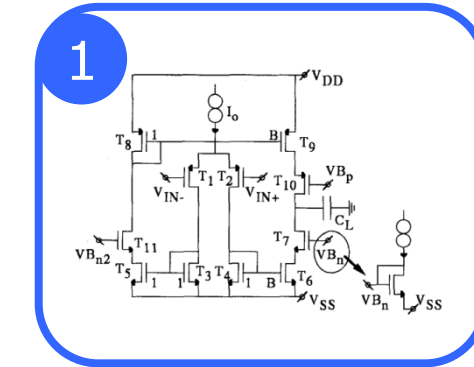
❑ Methods are based on

❑ Extraction of MOS parameters (LUT)

❑ $I_D/(W/L)$ vs g_m/I_D

❑ g_m/I_D vs V_{gs0}

❑ Set of equations



$$SR = \frac{B \cdot I_{D1}}{C_L}$$

$$f_T = \frac{B \cdot g_{m1}}{2\pi \cdot C_L} \quad (5)$$

$$A_0 = \left(\frac{g_m}{I_D} \right)_1 \cdot \left(\frac{g_m}{I_D} \right)_2 \cdot \frac{1}{\frac{1}{V_{A6} \cdot V_{A7}} + \frac{1}{V_{A9} \cdot V_{A10}}} \quad (4)$$

Overview of Design Methods for RFIC

g_m/I_D approach based on LUT

General Principle

□ For a given structure

□ Methods are based on

□ Extraction of MOS parameters (LUT)

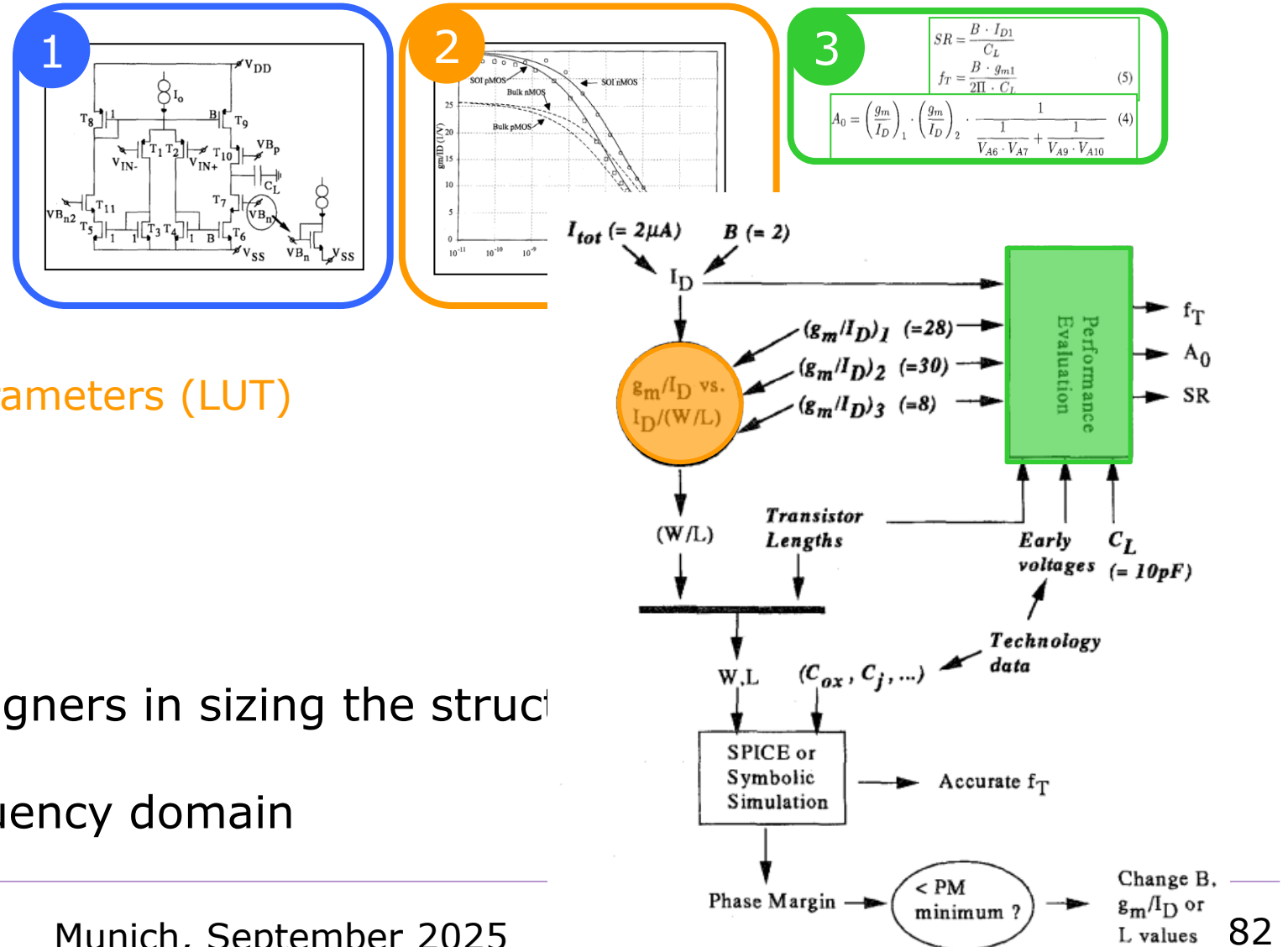
□ $I_D/(W/L)$ vs g_m/I_D

□ g_m/I_D vs V_{gs0}

□ Set of equations

□ Methods helps the designers in sizing the structure

□ Quite used in low frequency domain



Overview of Design Methods for RFIC

g_m/I_D approach based on LUT

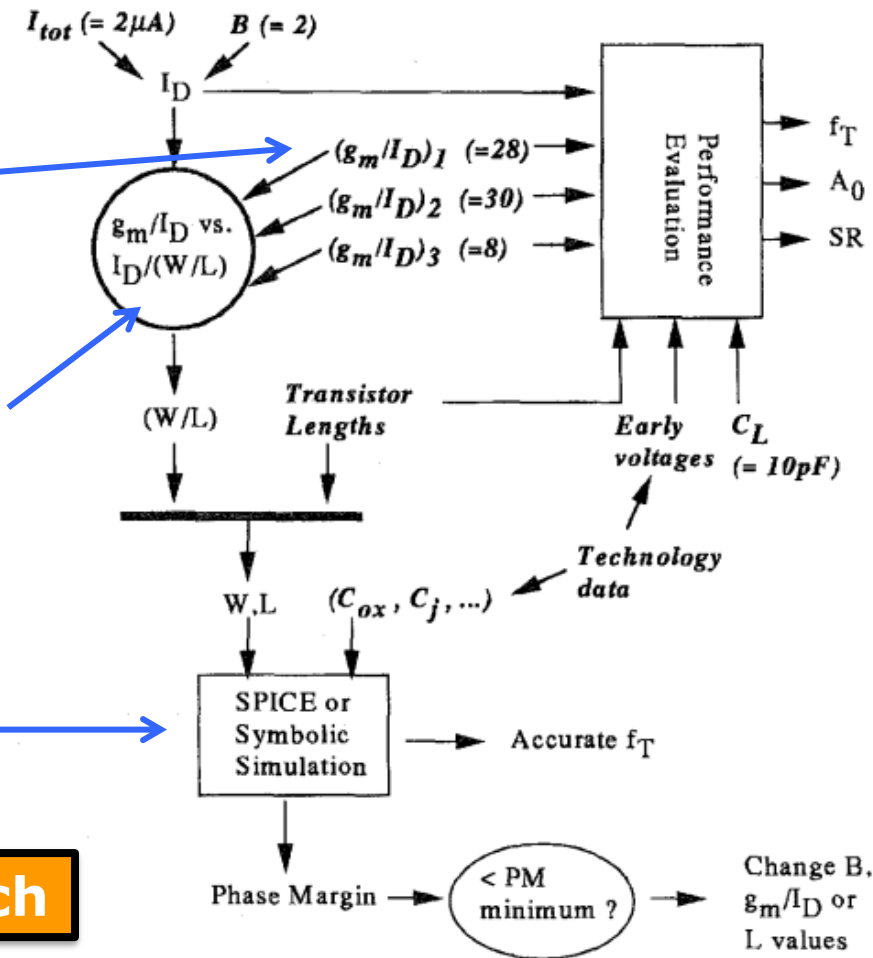
LIMITATIONS ?

❑ Time Consuming

❑ No explicit relationships between Physical parameters and circuit perf.

❑ Long optimization sequence

Need for a model based approach

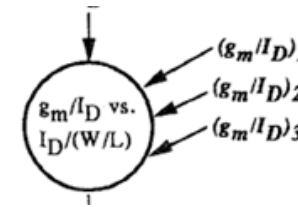


Overview of Design Methods for RFIC

g_m/I_D approach based All Region Models (ACM/EKV)

All Region 3PM model

- ❑ EKV – [Enz, Krummenacher, Vittoz]
- ❑ ACM – [Schneider, Galup]



$$\frac{I_D}{(W/L)} = i_f I_{S0}$$

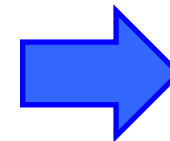
- ❑ With a 3PM model we have :

$$\frac{g_m}{I_D} = \frac{2}{n\phi_t \left(1 + \sqrt{1 + i_f}\right)}$$

$$g_m = \frac{2I_S}{\phi_t} \left(\sqrt{1 + i_f} - 1\right)$$

$$V_P - V_{S(D)} = \phi_t \left[\sqrt{1 + i_{f(r)}} - 2 + \ln \left(\sqrt{1 + i_{f(r)}} - 1 \right) \right]$$
$$V_P \cong \frac{V_G - V_{T0}}{n}$$

- ❑ PROS: The sizing is straightforward
- ❑ CONS: Inaccurate with SCE
 - ❑ Non-linearities can't be captured
 - ❑ gds effect can't be captured



Can we design RFICs ?

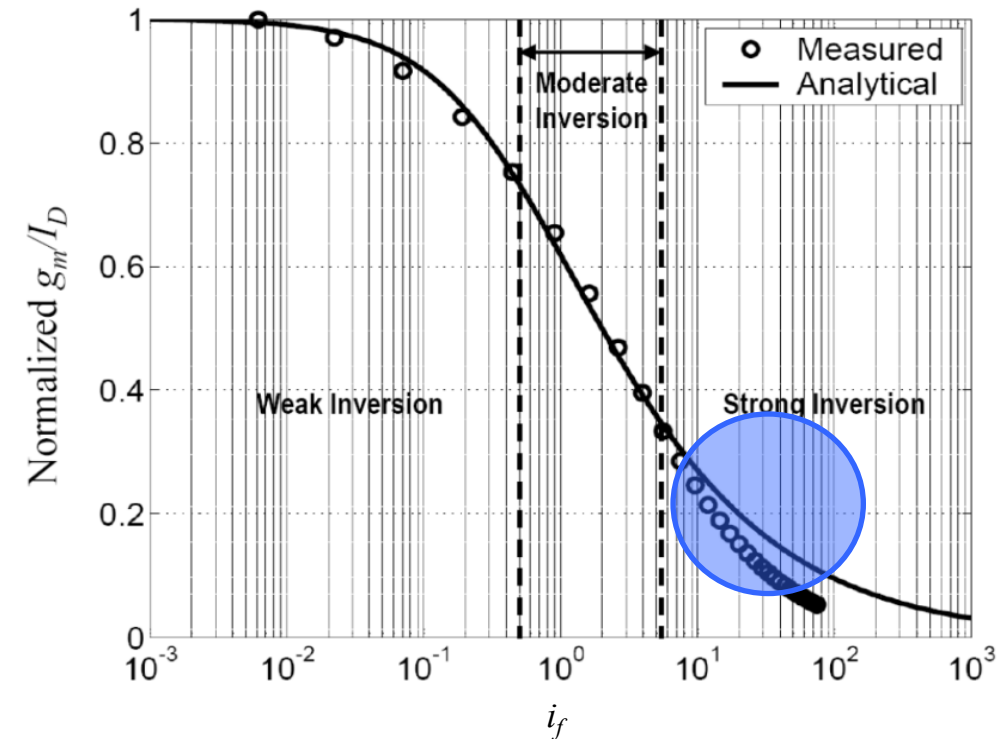
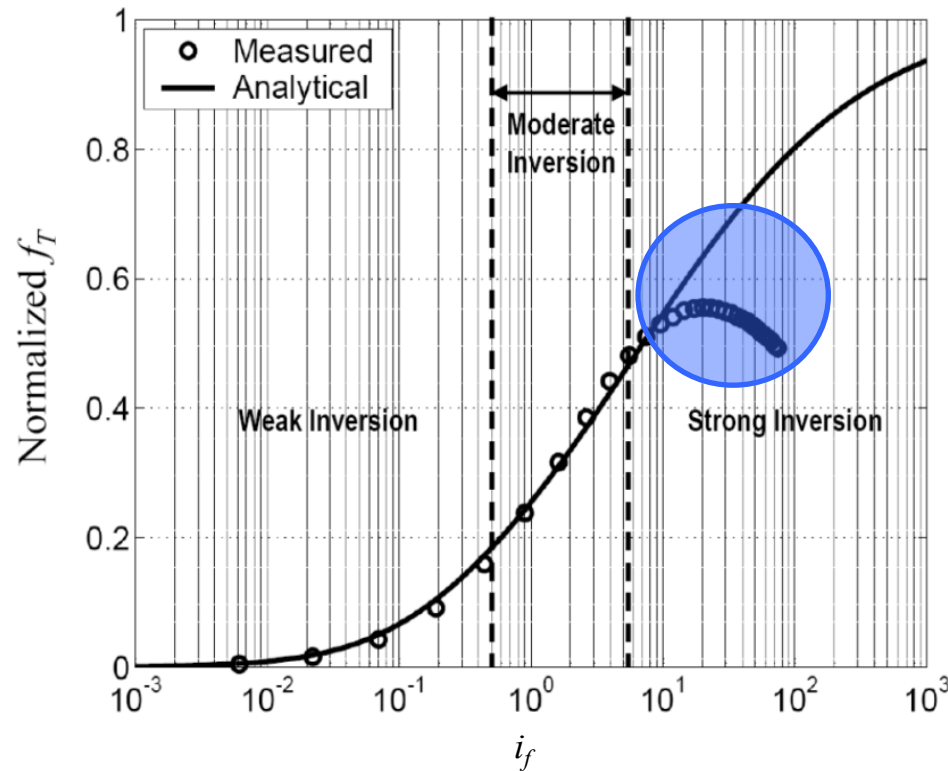
Overview of Design Methods for RFIC

g_m/I_D in RFIC design

Design Regions

□ f_t grows with i_f

● Year 2000 (RF-180nm)



Overview of Design Methods for RFIC

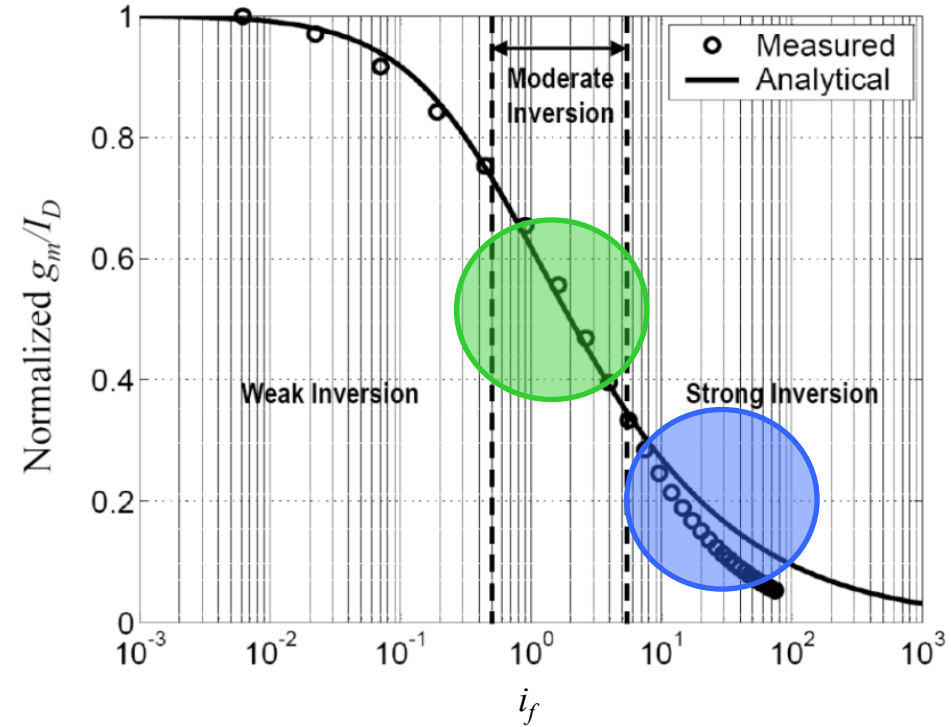
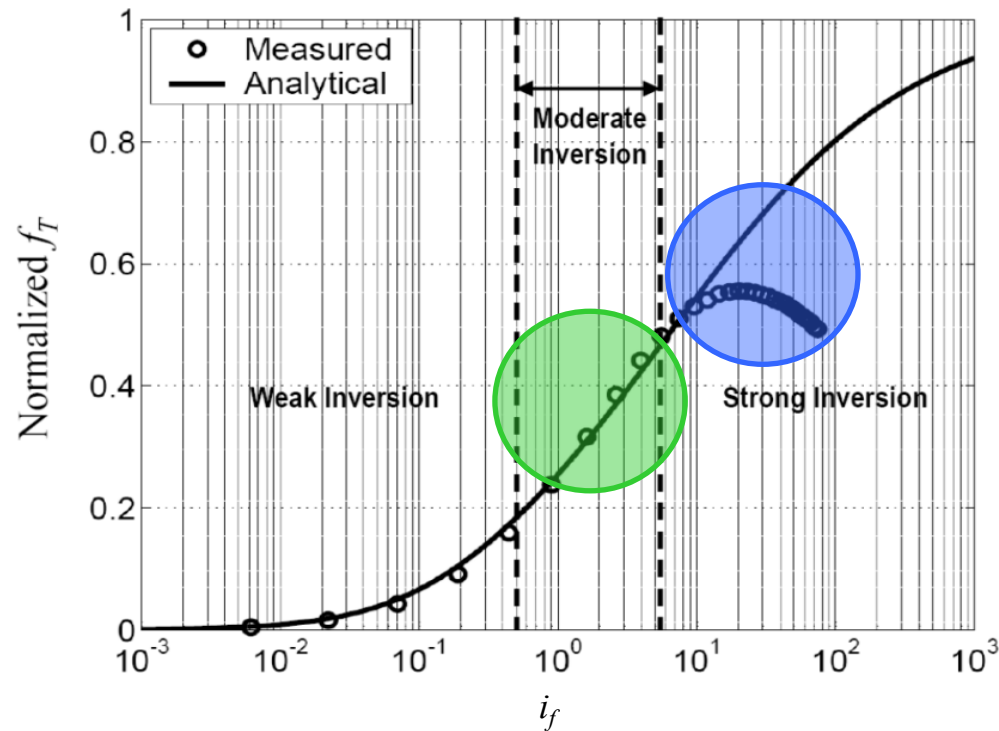
g_m/I_D in RFIC design

Design Regions

□ f_t grows with i_f **but** g_m/I_D reduces with i_f

● Year 2000 (RF-180nm)

● Year 2016 (RF-22nm) up to now



Overview of Design Methods for RFIC

g_m/I_D in RFIC design

❑ $g_m f_T/I_D$: A First Approach

A. Shameli et P. Heydari, « Ultra-Low Power RFIC Design Using Moderately Inverted MOSFETs: An Analytical/Experimental Study », *RFIC*, 2004.

❑ For RF design

❑ A FOM that maximize the gain bandwidth product

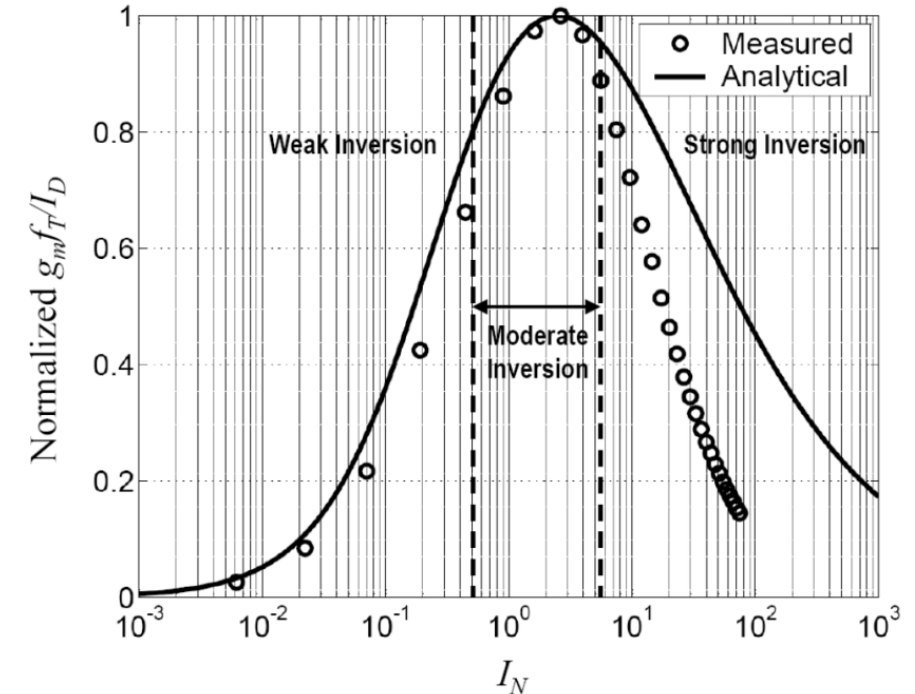
$$\frac{g_m f_T}{I_D}$$

❑ Gives the i_f that produces the best GBW product

❑ Not optimal :

❑ the gain or the bandwidth might be oversized.

❑ Do not depends on the topology



Overview of Design Methods for RFIC

g_m/I_D in RFIC design

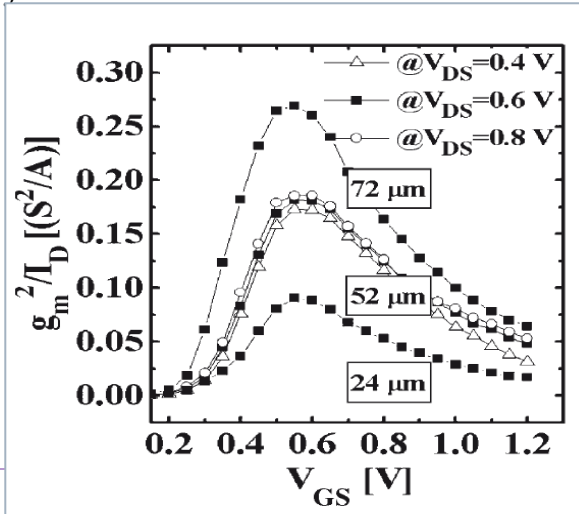
❑ **Function Based FoM**

❑ For a given Function

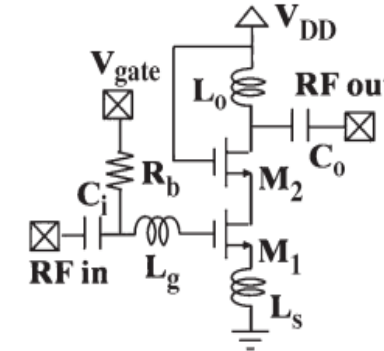
❑ Define a FoM for the function

$$\text{FoM}_{\text{LNA}} = \frac{G}{(F-1) \cdot P} \propto \frac{g_m^2}{\left(\frac{I_D}{g_m^2}\right) \cdot I_D} \propto \left(\frac{g_m^2}{I_D}\right)^2$$

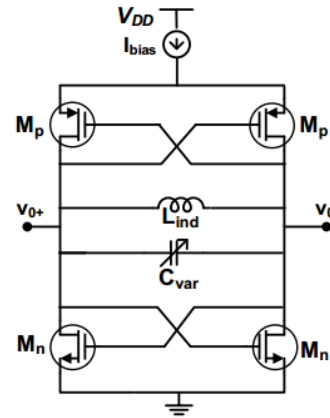
❑ Find the i_f that maximizes the FoM



LNA



LO



I. Song et B.-G. Park, « A Simple Figure of Merit of RF MOSFET for Low-Noise Amplifier Design », *Electron Device Lett. IEEE*, vol. 29(12), 2008.

$$PN(\Delta f) = 10 \log \left(\frac{k_B T \pi^2}{64 Q^2} \frac{g_m}{I_D} \frac{1}{I_D} \frac{f_0}{(\Delta f)^2} \lambda \right)$$

R. Fiorelli ; J. Núñez ; F. Silveira; "All-Rnversion region gm/I_D methodology for RF circuits in FinFET technologies" 2018 NEWCAS

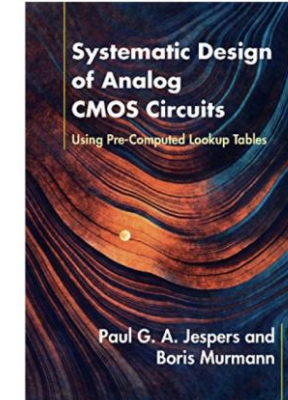
Overview of Design Methods for RFIC

g_m/I_D in RFIC design

Circuit Based Method

Circuit Sizing Based on LUT

- ❑ For a given Circuit
- ❑ Similar to the one introduced by Silveira, Flandre, Jespers in 1996
- ❑ Capacitance and noise effects can be captured in LUT for bandwidth and noise analysis
- ❑ Well known and well referenced
- ❑ But ...
 - ❑ Still complicated
 - ❑ The gap between physics and design remains
 - ❑ No analytical relationships to size « by hand »



Circuit Sizing Based on ACM

- ❑ Equations can be derived for a given circuit
- ❑ Quasi analytical approach
 - ❑ R-F LNA designed with 3PM-ACM [Bourdel - ICECS 2019]
 - ❑ Gds and gm3 (NL) are taken into account
 - R-F LNA designed with 7PM-ACM [Bouchoucha – ISCAS 2023]
 - ❑ CG LNA designed with 5PM-ACM2 [Alves Neto – ACCESS 2024]

A g_m/I_D Design Methodology for 28 nm FD-SOI CMOS Resistive Feedback LNAs

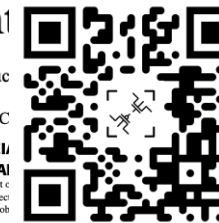


All Resources accessible for free
Resistive Feedback LNAs designed for advanced technologies
design-oriented

Mohamed Khalil Bouchoucha

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<https://github.com/ACMmodel/MOSTL-model>

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Agenda

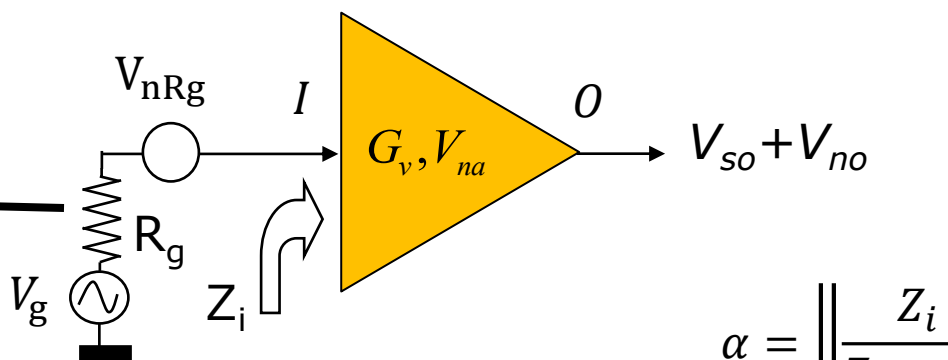
- Overview of Design Methods for RFIC
- **LNA Design Considerations**
- Resistive Feedback LNA
- What we need in ACM-2
- Inversion Level Based Method for R-Feedback LNA with ACM-2

LNA Design considerations

Noise Factor

- NF is the signal to noise ratio (SNR) degradation (Eq. 1)

$$F = \frac{(S/N)_i}{(S/N)_o} \quad (1)$$



$$\overline{v_{nRg}^2} = 4KTR_g$$

$$V_{RMS_nRg} = \sqrt{4KTR_g \Delta f}$$

$$\alpha = \left\| \frac{Z_i}{Z_i + R_g} \right\|$$

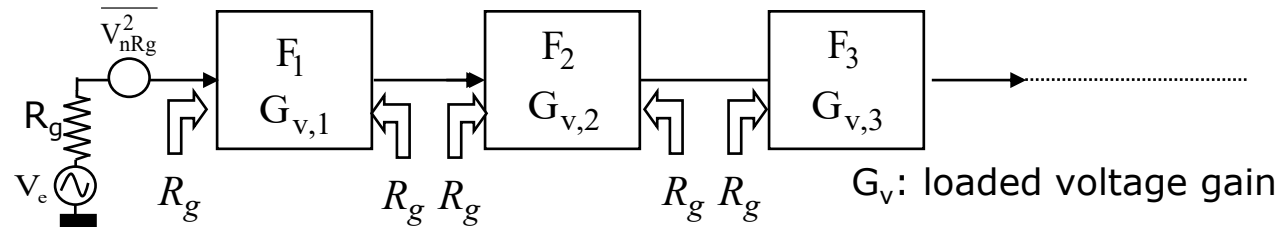
$$F = \frac{V_{si}^2/V_{ni}^2}{V_{so}^2/V_{no}^2} = \frac{V_g^2/V_{nRg}^2}{V_{so}^2/V_{no}^2} = \frac{V_{no}^2}{\alpha^2 \cdot G_v^2 \cdot V_{nRg}^2} = \frac{\overline{v_{no}^2}}{\alpha^2 \cdot G_v^2 \cdot \overline{v_{nRg}^2}}$$

$$F = \frac{\text{total output noise}}{\text{total output noise of the ideal (no additive noise) device}} \quad (2)$$

- NF also express the quantity of noise added by the stage regarding the noise delivered by the source (R_g) on Δf . (Eq. 2)

LNA Design considerations

Friss Formula



$$F = F_1 + \frac{(F_2 - 1)}{G_{v,1}^2} + \frac{(F_3 - 1)}{G_{v,1}^2 \cdot G_{v,2}^2} + \dots$$

This formula is valid only for real impedances and in case of power matching: ($Z_s = Z_e = R_g$)

In RFIC, impedances are complex and never equal.

This formula cannot be used as is.

However, the following statements remain valid in RFIC.

The gain of the first stages reduces the NF of the following ones.

=> The receiver chain must start by amplifiers

=> The NF of the first stage must be as low as possible

Note that lossy stages ($G_v < 1$) increase the NF, especially when they are in front of the receiver chain (***antenna filter, image rejection filter, antenna switch***).



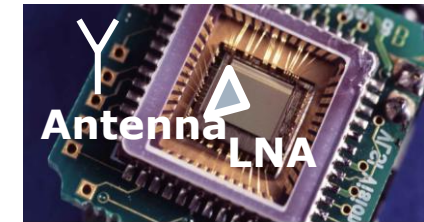
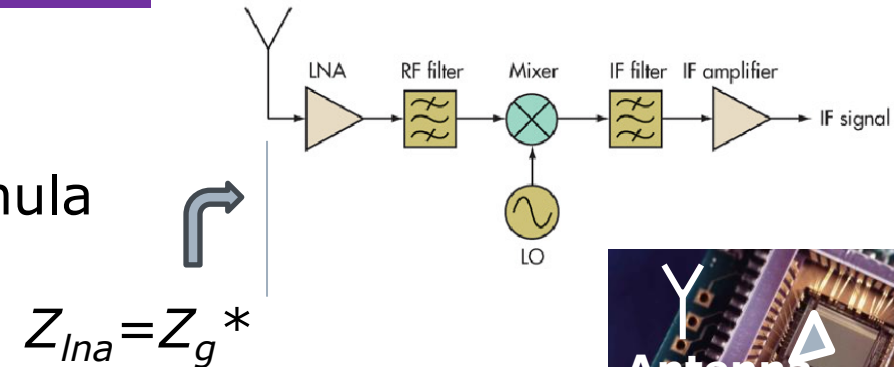
LNA

LNA Design considerations

Input patching

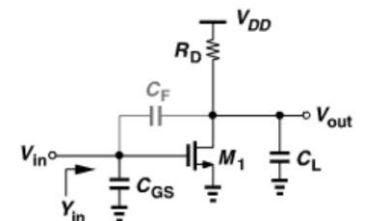
System point of vue

- ❑ LNA is the first device due to Friss formula
- ❑ Antenna is mainly 50 Ω .



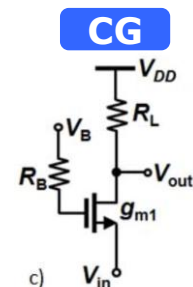
In CMOS

- ❑ Generally, the input impedance is capacitive $\Re(Y_{in})$ is very low
- ❑ For example :
CS amplifier

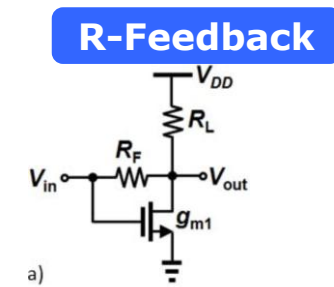


$$\Re(Y_{in}) = R_D \parallel 1/g_m$$

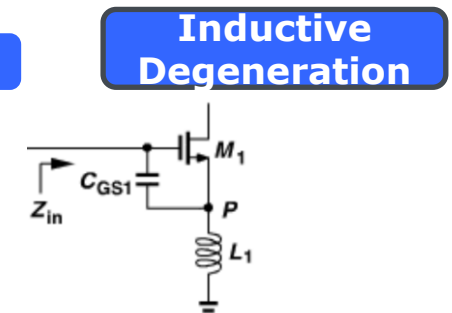
$$\Im(Y_{in}) = \frac{1}{C_{GS}C_F\omega}$$



$$\Re(Z_{in}) = \frac{1}{g_{m1}}$$



$$\Re(Z_{in}) = \frac{R_F}{1 + |A_V|}$$

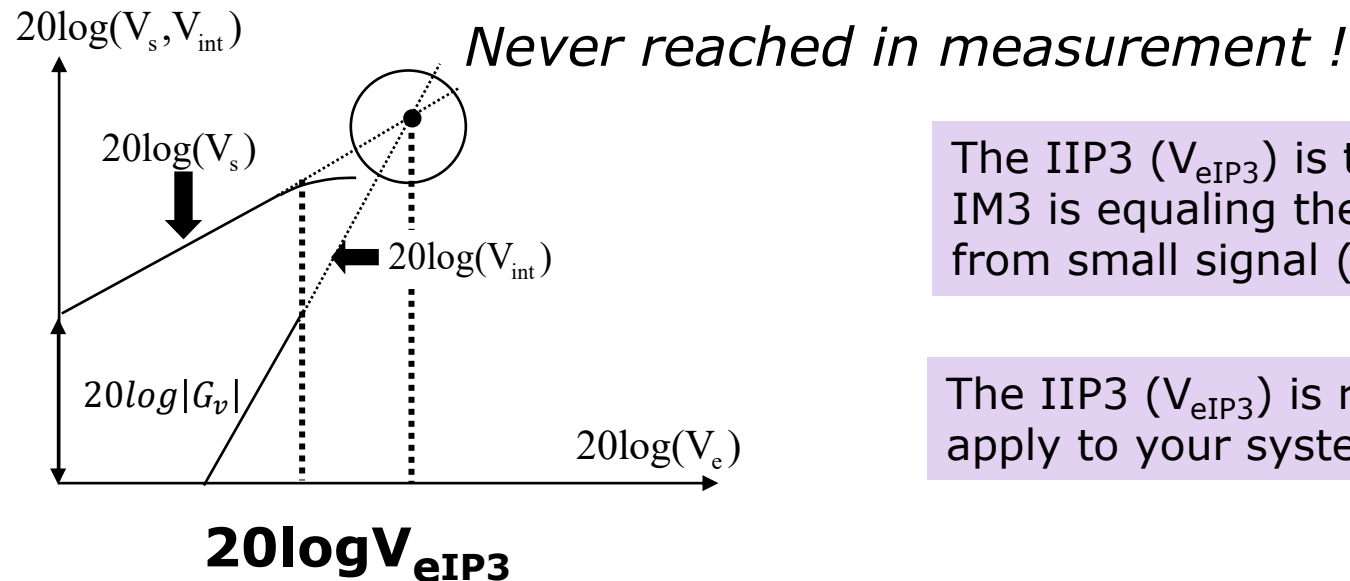
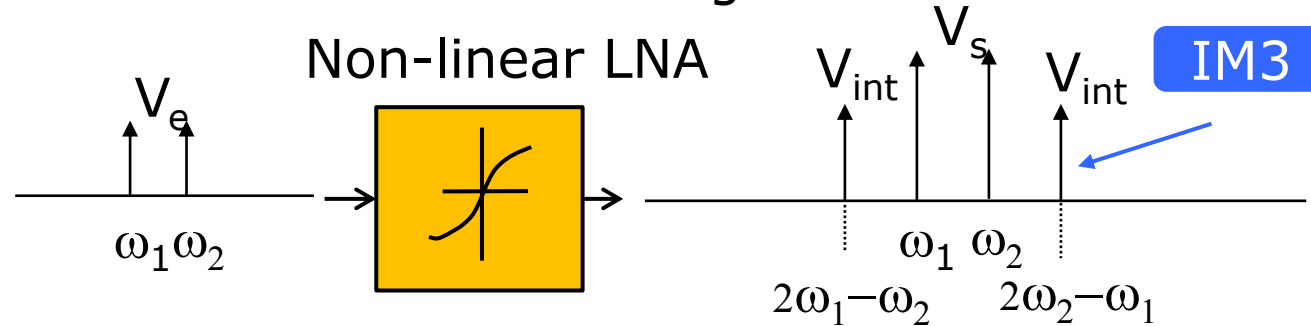


$$\Re(Z_{in}) = \frac{g_m L_1}{C_{GS}}$$

LNA Design considerations

Non-Linearities

ω_1 and ω_2 are 2 harmonics of the wanted signal



The IIP3 (V_{eIP3}) is the input signal level for which the output IM3 is equaling the fundamental harmonic (V_s) extrapolated from small signal (AC).

The IIP3 (V_{eIP3}) is related to the maximum power you can apply to your system

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Resistive Feedback LNA

Topology

R-Feedback general considerations

- ❑ Simple & Compact (No-inductors) & Wideband
- ❑ RF allows to synthesize a real part in Z_{in}
- ❑ L_G and C_{GS}' will help in cancelling the imaginary part
- ❑ while controlling Q_{in}

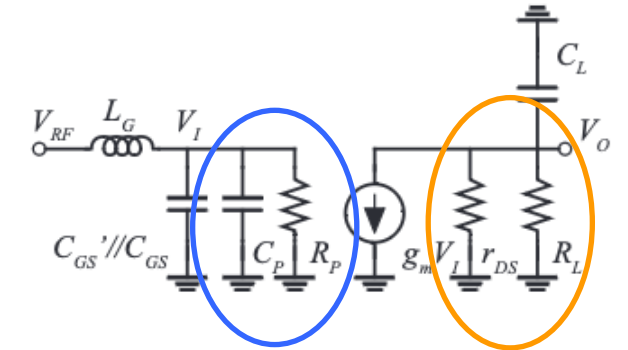
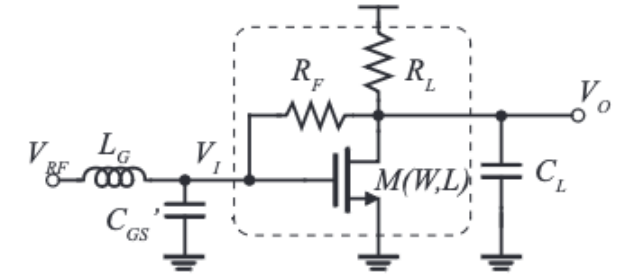
Input impedance

$$Z_{IN} = L_G s + \left(\frac{1}{(C_{GS} + C'_{GS})s} // Z_P \right)$$

$$\Re(Z_P) = R_P = \frac{R_O + R_F}{1 + G_m R_O},$$

$$C_P = \frac{R_O^2 C_L (G_m R_F - 1)}{(R_O + R_F)^2}$$

$$R_O = \frac{R_L r_{DS}}{R_L + r_{DS}}$$



$$\Re(Z_{IN}) = R_S = \frac{R_P}{(1 + Q_P^2)} = 50 \Omega$$

$$Q_P = R_P C_T \omega_0 \quad C_T = C_{GS} + C'_{GS} + C_P$$

$$\Im(Z_{IN}) = 0 = L_G s + \frac{Q_P^2}{C_T s (1 + Q_P^2)}$$

Note :

$$\Re(Z_P) = \frac{1 + \left(\frac{f}{f_c}\right)^2}{\frac{1 + G_m R_O}{R_O + R_F} + \frac{1}{R_F} \left(\frac{f}{f_c}\right)^2},$$

$$\Im(Z_P) = - \left(1 + \frac{R_F}{R_O}\right)^2 \frac{1 + \left(\frac{f}{f_c}\right)^2}{2\pi f C_L (G_m R_F - 1)}$$

Reduces for
 $\frac{f_c}{f} = N > 3$

Resistive Feedback LNA Topology

Voltage Gain

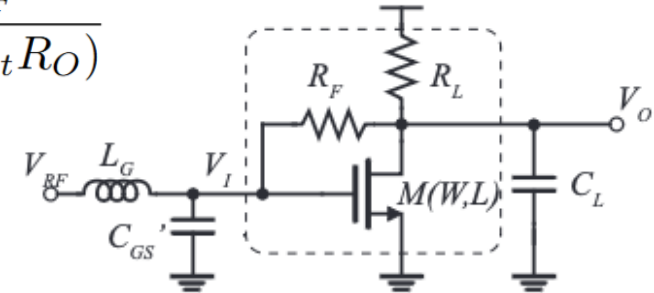
$$|G_o| = \left| \frac{V_o}{V_{RF}} \right| = \frac{|G_T|}{\sqrt{\left(1 + \left(\frac{f_0}{f_c}\right)^2\right)}}$$

$$|G_T| = |G_v| Q_{IN} = \frac{(g_{msat} R_F - 1) R_O}{(R_O + R_F)} \sqrt{1 + Q_P^2}$$

$$Q_{IN} = V_I / V_{RF}$$

$$Q_{IN} = \sqrt{1 + Q_P^2} = \sqrt{\frac{R_O + R_F}{R_S(1 + g_{msat} R_O)}}$$

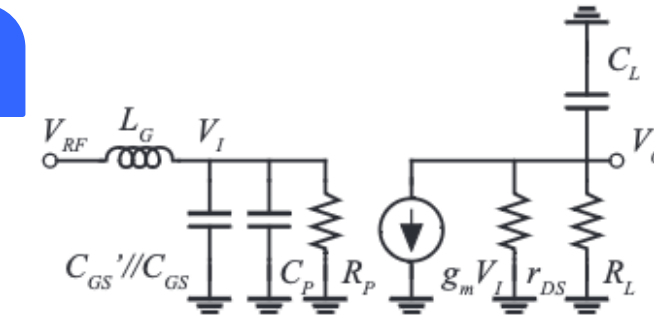
$$\frac{f_c}{f_0} = N > 3$$



BW

$$f_c = \frac{R_O + R_F}{2\pi R_O R_F C_L} = N f_0$$

R_S
 (the source resistance seen at the 50 Ω input)



Noise Figure

$$F = 1 + \frac{4 \left(\frac{R_F}{Q_{IN}^2} + R_S \right)^2}{Q_{IN}^2 g_{msat} R_S \left[\frac{R_F}{Q_{IN}^2} + R_S + \frac{g_{msat} R_S R_F R_L}{R_F + R_L} \right]^2} \left[\gamma + \frac{1}{g_{msat} R_L} + \frac{\left(1 + \frac{Q_{IN}^2 g_{msat} R_S R_F}{R_F + Q_{IN}^2 R_S} \right)^2}{g_m R_F} \right]$$

$$\left(F_{min} = 1 + \frac{(1 + g_{msat}) R_S^2 R_F}{R_S (1 - g_{msat} R_F)^2} + \frac{\gamma g_{msat} (R_S + R_F)^2}{R_S (1 - g_{msat} R_F)^2} + \frac{(R_S + R_F)^2}{R_S R_L (1 - g_{msat} R_F)^2} \right)$$

IIP₃

$$V_{IIP3} = \frac{2}{Q_{IN}} \sqrt{\frac{2g_{msat}}{g_{msat3}}}$$

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What we need in ACM-2

Small signal parameters

Small signal (AC) parameters

□ G_T , NF, IIP3 vs (g_m , g_{m3} , g_{ds})

$$g_m = \frac{\partial I_D}{\partial V_G} = \frac{I_S}{\phi_t} g_G = \frac{I_S}{n\phi_t} \frac{2(q_s - q_d) - i_d \zeta \left(\frac{q_s}{1+q_s} - \frac{q_d}{1+q_d} \right)}{1 + \zeta (q_s - q_d)} \quad g_{ds} = \frac{I_S}{\phi_t} g_D = \frac{I_S}{n\phi_t} \frac{2(q_s \sigma - q_d (\sigma - n)) - i_d \zeta \left(\sigma \frac{q_s}{1+q_s} - (\sigma - n) \frac{q_d}{1+q_d} \right)}{1 + \zeta (q_s - q_d)}$$

$$g_{m3} = \frac{I_S}{(n\phi_t)^3} \left\{ \frac{\frac{2q_s}{(1+q_s)^3} - \frac{2q_d}{(1+q_d)^3} - \zeta n^2 g_{G2} \left(\frac{q_s}{1+q_s} - \frac{q_d}{1+q_d} \right) - \zeta \left[2ng_G \left(\frac{q_s}{(1+q_s)^3} - \frac{q_d}{(1+q_d)^3} \right) + i_d \left(\frac{q_d(1-2q_d)}{(1+q_d)^5} - \frac{q_d(1-2q_d)}{(1+q_d)^5} \right) \right]}{1 + \zeta (q_s - q_d)} \right\}$$

□ Valid in all region (q_s and q_d shall be explored)

□ We consider only saturation

$$q_{dsat} = q_s + 1 + \frac{1}{\zeta} - \sqrt{\left(1 + \frac{1}{\zeta}\right)^2 + \frac{2q_s}{\zeta}}$$

□ It reduces the exploration to only q_s .

$$g_{msat} = \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

$$g_{dssat} = \sigma \frac{2I_S}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)}$$

$$g_{msat3} = \frac{16I_S}{(n\phi_t)^3} \frac{q_s}{(q_s + 1)^3} \frac{2 - 2\zeta q_s - 3\zeta q_s^2}{(\zeta q_s + 2)^4}$$

What we need in ACM-2

Small signal parameters

Large signal (DC) parameters

- To compute the final voltages

$$V_T = V_{T0} - \sigma(V_{SB} + V_{DB})$$

$$V_P = \frac{V_{GB} - V_T}{n} \quad \frac{V_P - V_{S(D)B}}{\phi_t} = q_{s(d)} - 1 + \ln q_{s(d)}$$

- Sometimes we'll use i_f in the code

$$i_d = i_f - i_r \quad \longleftrightarrow \quad q_{S(D)} = \sqrt{1 + i_{f(r)}} - 1 \quad \longleftrightarrow \quad i_d = \frac{(q_s + q_d + 2)}{1 + \zeta|q_s - q_d|} (q_s - q_d)$$
$$I_D = I_S \cdot i_d \quad I_S = \frac{\mu C'_{ox} n (U_T)^2 W}{2 L}$$

Agenda

- Overview of Design Methods for RFIC
- LNA Design Considerations
- Resistive Feedback LNA
- What we need in ACM-2
- **Inversion Level Based Method for R-Feedback LNA with ACM-2**

Inversion Level Based Method for R-Feedback LNA with ACM-2

General Approach :

Requirements

- ☐ GT, NF, IIP3, IDC  Possible Tradeoff
- ☐ fo, CL, BW  No Tradeoff

Inversion Level Based Method for R-Feedback LNA with ACM-2

Reducing the variables :

Starting from : $|G_T| = \frac{(g_{msat}R_F - 1)R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S(1 + g_m R_O)}}$
 $G_T(g_{msat}; R_0; R_F; R_S) = G_T(\underbrace{W; q_s; I_D; V_{DSsat}}_{\text{variables}}; \underbrace{V_{T0}; I_S; n; \sigma; \zeta; f_c; C_L}_{\text{parameters}})$

$$\rightarrow g_{msat} = \frac{2I_S^* W}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \rightarrow g_{msat}(q_s; W)$$

$$\rightarrow g_{dssat} = \sigma \frac{2I_S^* W}{n\phi_t} \frac{q_s}{1 + \zeta(q_s + 1)} \rightarrow g_{dssat}(q_s; W)$$

$$R_O = \frac{R_L r_{ds}}{R_L + r_{ds}} \rightarrow g_{dssat}(q_s; W) \rightarrow V_{DSsat} = \phi_t(\sqrt{1 + i_f} + 3)$$

$$R_L(I_D) = \frac{V_{DD} - V_{DSsat}}{I_D}$$

$$R_F = \frac{R_0}{2\pi R_0 C_L f_c - 1} \rightarrow R_F(q_s; W) \Big|_{f_c}$$

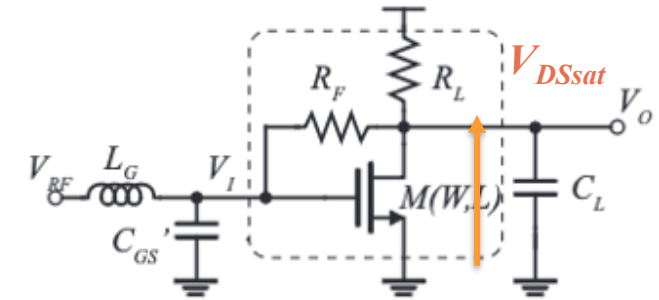
$$f_c = \frac{R_O + R_F}{2\pi R_O R_F C_L}$$

$$I_D = I_S^* W i_f$$

$$i_f = (q_s + 1)^2 - 1 \rightarrow R_0(q_s; W)$$

$$I_D(q_s; W)$$

$$G_T(g_{msat}; R_0; R_F; R_S) = G_T(W; q_s)$$



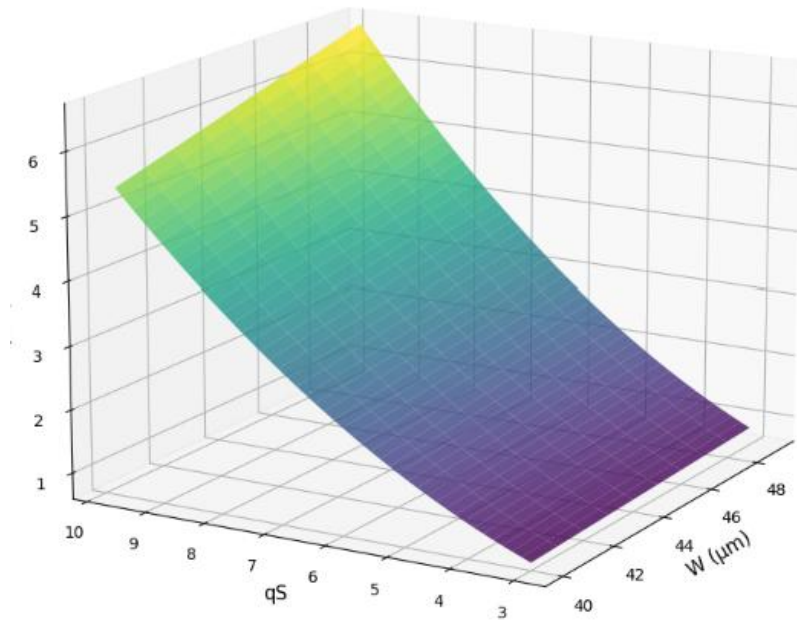
Inversion Level Based Method for R-Feedback LNA with ACM-2

Computing I_D and G_T

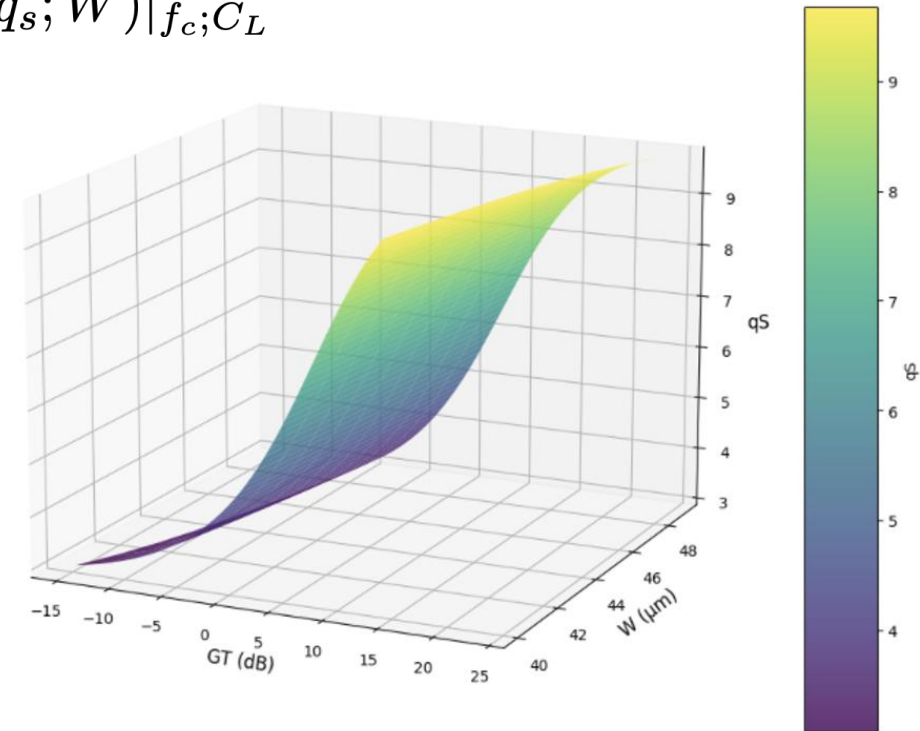
$$I_D(q_S; W) = I_S W ((q_S + 1)^2 - 1)$$

$$|G_T| = \frac{(g_{msat} R_F - 1) R_O}{(R_O + R_F)} \sqrt{\frac{R_O + R_F}{R_S (1 + g_{msat} R_O)}}$$

$I_D(q_S; W)$



↳ $G_T(q_S; W)|_{f_c; C_L}$



Inversion Level Based Method for R-Feedback LNA with ACM-2

Computing F and $IIP3$:

$$F = 1 + \frac{4 \left(\frac{R_F}{Q_{IN}^2} + R_S \right)^2}{Q_{IN}^2 g_{msat} R_S \left[\frac{R_F}{Q_{IN}^2} + R_S + \frac{g_{msat} R_S R_F R_L}{R_F + R_L} \right]^2} \left[\gamma + \frac{1}{g_{msat} R_L} + \frac{\left(1 + \frac{Q_{IN}^2 g_{msat} R_S R_F}{R_F + Q_{IN}^2 R_S} \right)^2}{g_m R_F} \right]$$

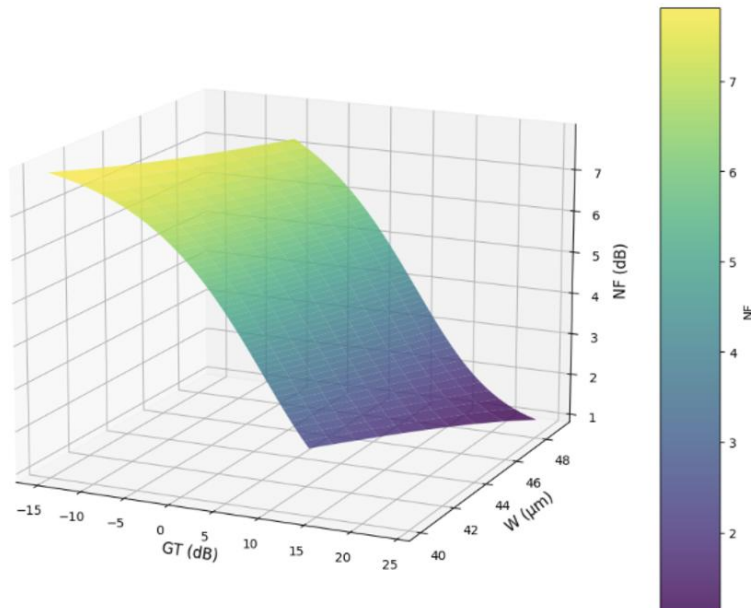
Remember :

$$Q_{IN} = V_I / V_{RF} = \sqrt{1 + Q_P^2} = \sqrt{\frac{R_O + R_F}{R_S(1 + g_{msat} R_O)}}$$

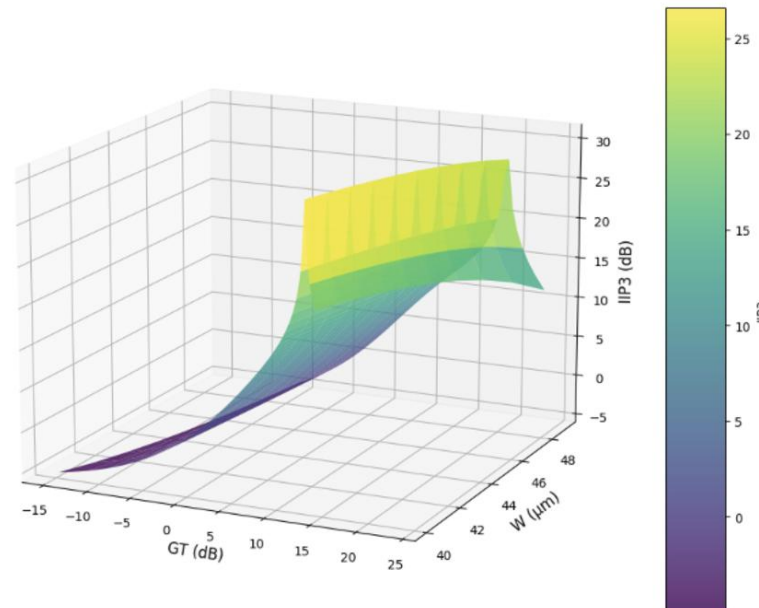
$$\hookrightarrow F(q_s; W) |_{f_c; C_L; R_S}$$

$$V_{IIP3} = \frac{2}{Q_{IN}} \sqrt{\frac{2g_{msat}}{g_{msat3}}} \hookrightarrow IIP3(q_s; W) |_{f_c; C_L; R_S}$$

NF vs W and Gain (GT)



IIP3 vs W and Gain (GT)



Inversion Level Based Method for R-Feedback LNA with ACM-2

Exploring the design space:

Setting GT (or NF, or IIP3) helps to build 2D plots.

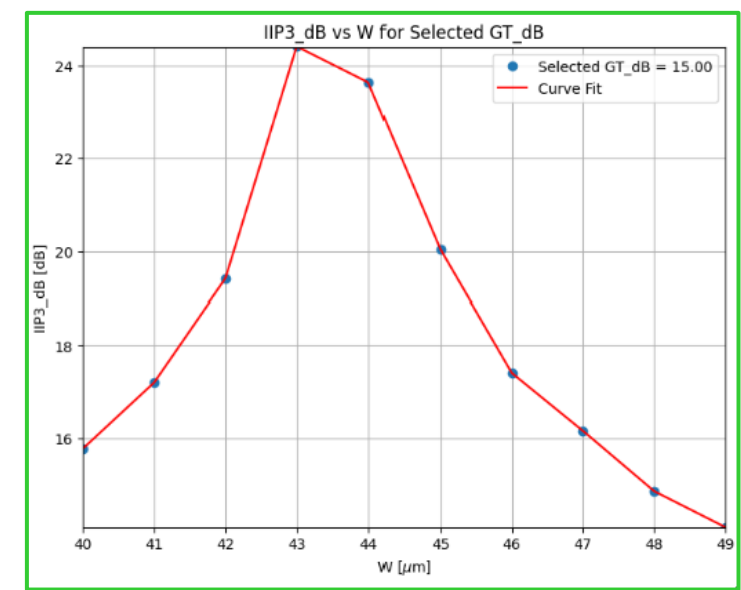
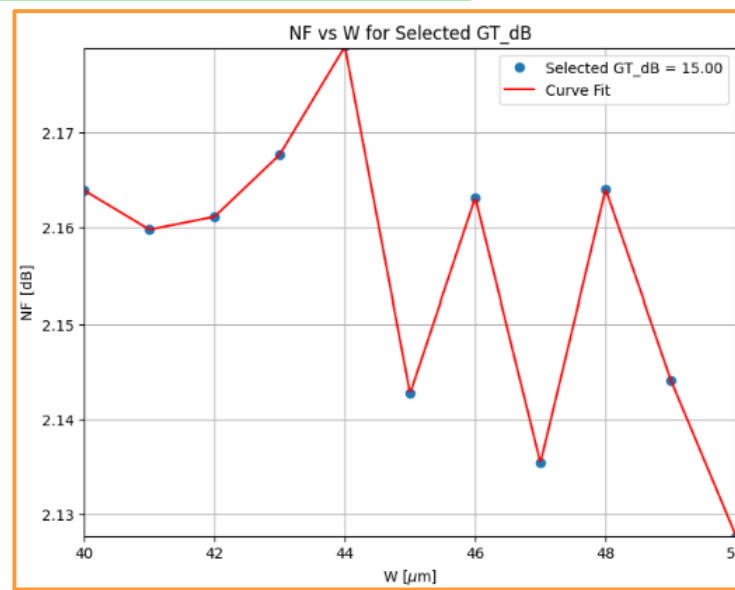
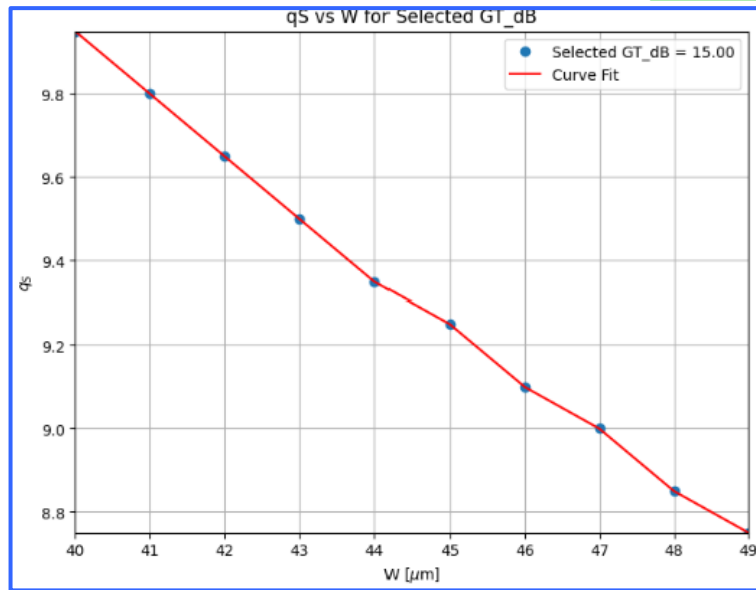
For a particular G_T

=> We get the relationship $q_S \Leftrightarrow W$

$$G_T(g_{msat}; R_0; R_F; R_S) = G_T(W; q_S)$$

=> We plot $NF(W)$

=> We plot $IIP3(W)$



$W(\mu m)$

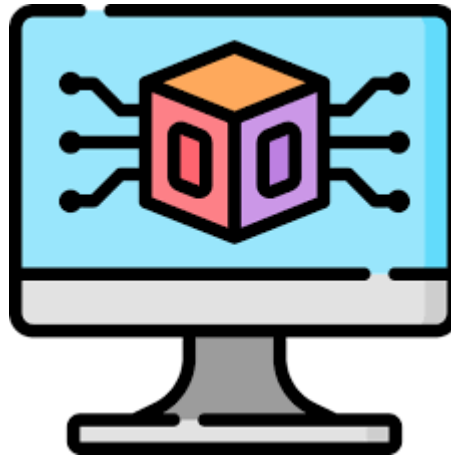
Setting the final value:

- We choose W , that gives q_S for a given G_T .
- $(W; q_S)$ gives $R_0, R_L, I_D, g_{msat}, V_{DSsat}$ and V_G

Input Matching

- With $R_0, R_F, g_m \Rightarrow R_P$ $\Re(Y_P) = \frac{1}{R_P} = \frac{1 + g_{msat}R_O}{R_O + R_F}$
- With R_S and R_P calculate Q_P $\Re(Z_{IN}) = R_S = \frac{R_P}{(1 + Q_P^2)} = 50 \Omega$
- With Q_P calculate C_T and C_{GS}' $Q_P = R_P C_T \omega_0 \quad C_T = C_{GS} + C_{GS}' + C_P \quad C_P = \frac{R_O^2 C_L (1 - g_{msat} R_F)}{(R_O + R_F)^2}$
- With C_T calculate L_G $\Im(Z_{IN}) = 0 = L_G s + \frac{Q_P^2}{C_T s (1 + Q_P^2)}$

Now, let's simulate with a real PDK



Design and Simulation of analog/RF ICs with Open-Source CAD Tools and PDKs:

Analog Circuit Design

Pr. Márcio Cherem Schneider
Department of electrical engineering
Universidade Federal de Santa Catarina, Brasil

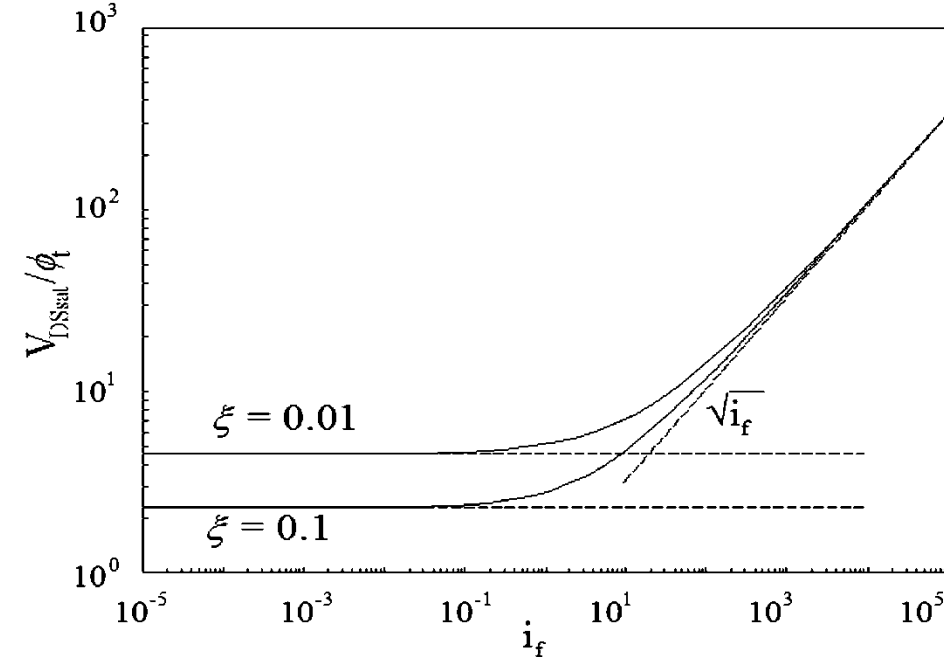
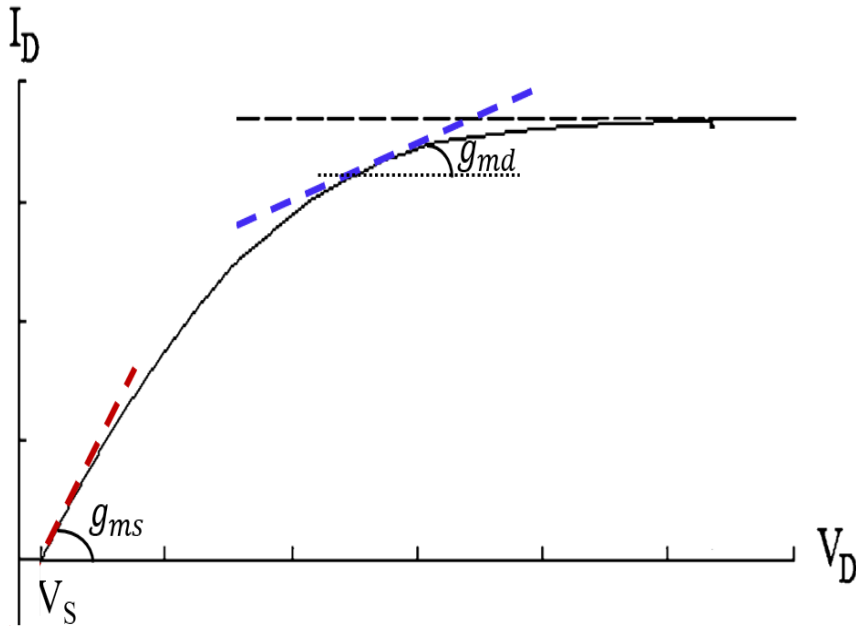
- ☐ **Current mirrors**
- ☐ **High-swing cascode current mirror**
- ☐ **Current gain schemes**
- ☐ **Current reference**
- ☐ **Common-source amplifier**

Saturation voltage

$V_{DSsat} = V_{DS}$ such that
 $g_{md}/g_{ms} = q_{ID}/q_{IS} = \xi$

UCCM

$$V_{DSsat} = \phi_t \left[\ln \left(\frac{1}{\xi} \right) + (1 - \xi) (\sqrt{1 + i_f} - 1) \right]$$



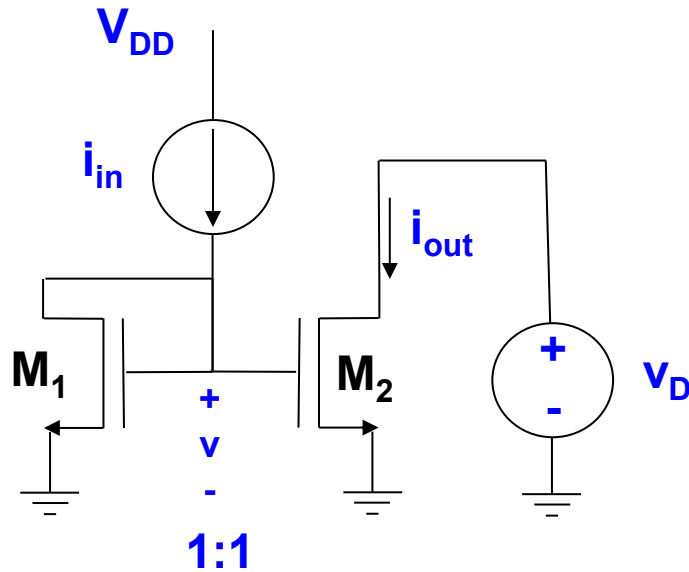
$(1 - \xi)$ is the saturation level

Saturation voltage versus inversion level

A good approximation for the saturation voltage:

$$V_{DSsat} \cong \phi_t \left(\sqrt{1 + i_f} + 3 \right)$$

Current mirrors

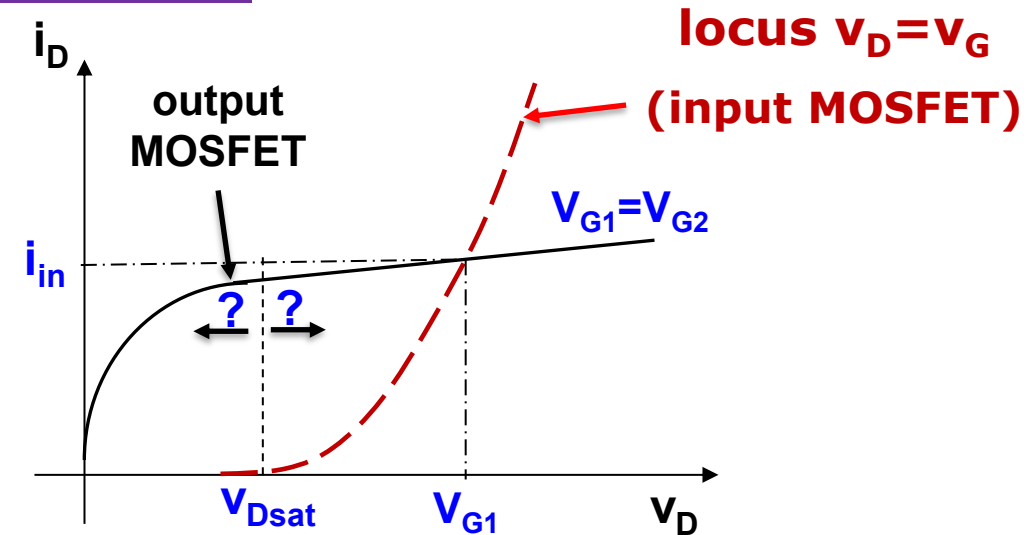


$$M_1 = M_2$$

M_1 : $i \rightarrow v$ converter

M_2 : $v \rightarrow i$ converter

An approximation for the saturation voltage:



Basic principle $V_{G1} = V_{G2}$; $V_{S1} = V_{S2}$; $V_{B1} = V_{B2}$;
 $v_{out} > V_{Dsat} \rightarrow i_{out} \cong i_{in}$

$$V_{DSSat} \cong \phi_t \left(\sqrt{1 + i_f} + 3 \right)$$

ACM2 applied to weak inversion

ACM2 parameters: V_{T0} , I_S , n , σ , ζ

$$V_P = \frac{V_{GB} - (V_{T0} - \sigma(V_{DB} + V_{SB}))}{n}$$

$$q_{S(D)} = Q_{S(D)} / (-nC_{ox}\phi_t)$$

$$I_D/I_S = i_f - i_r = (I_F - I_R)/I_S$$

$$I_S = \frac{W}{L} \mu_s n C_{ox} \frac{\phi_t^2}{2} = S I_{SH}$$

$$\mathbf{WI : } q_{S(D)} \ll 1$$

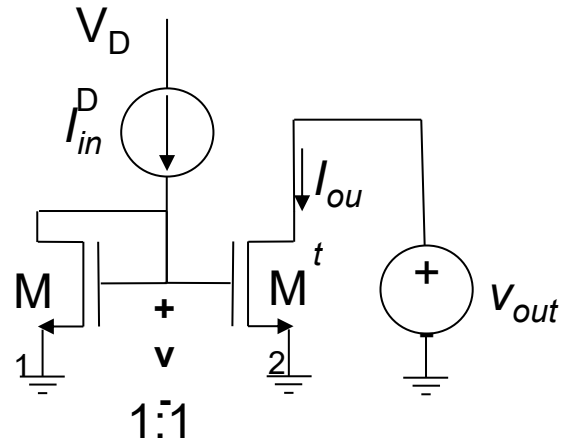
UCCM

$$q_{S(D)} = e^{(\frac{V_P - V_{S(D)B}}{\phi_t} + 1)}$$

$$i_D = I_D/I_S = 2(q_S - q_D)$$

Current mirrors

Error due to difference in drain voltages



V_A is non-physical

A simple case: error for weak inversion using ACM2

$$I_{in} = I_{F1} - I_{R1} = I_{F1} [1 - e^{-V_{in}/\varphi_t}]$$

$$I_{out} = I_{F2} - I_{R2} = I_{F2} [1 - e^{-V_{out}/\varphi_t}]$$

$$I_{F2} = I_{F1} e^{\sigma(V_{out}-V_{in})/n\varphi_t}$$

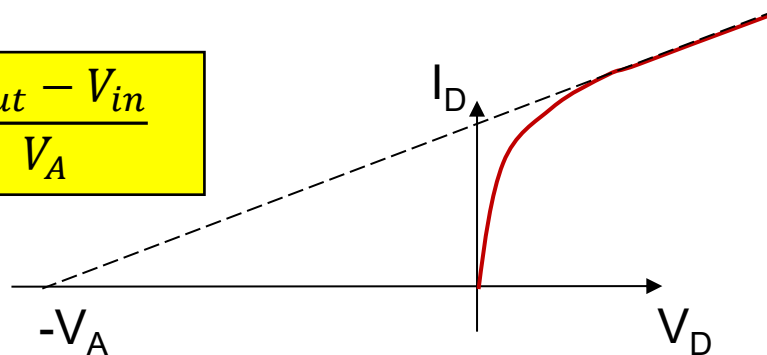
$$\frac{I_{out}}{I_{in}} = e^{\frac{\sigma(V_{out}-V_{in})}{n\varphi_t}} \frac{1 - e^{-V_{out}/\varphi_t}}{1 - e^{-V_{in}/\varphi_t}}$$

$$\frac{I_{out}}{I_{in}} \cong e^{\frac{\sigma(V_{out}-V_{in})}{n\varphi_t}} \quad \text{for } V_{out} \text{ \& } V_{in} \gg \varphi_t$$

Linear approximation

$$\frac{I_{out}}{I_{in}} = \frac{I_S f(V_G - V_T, V_S) (1 + V_{out}/V_A)}{I_S f(V_G - V_T, V_S) (1 + V_{in}/V_A)}$$

$$\frac{I_{out} - I_{in}}{I_{in}} \cong \frac{V_{out} - V_{in}}{V_A}$$



Current mirrors

Dependence of the output current (conductance) on the output voltage

Weak inversion $i_f < 1$

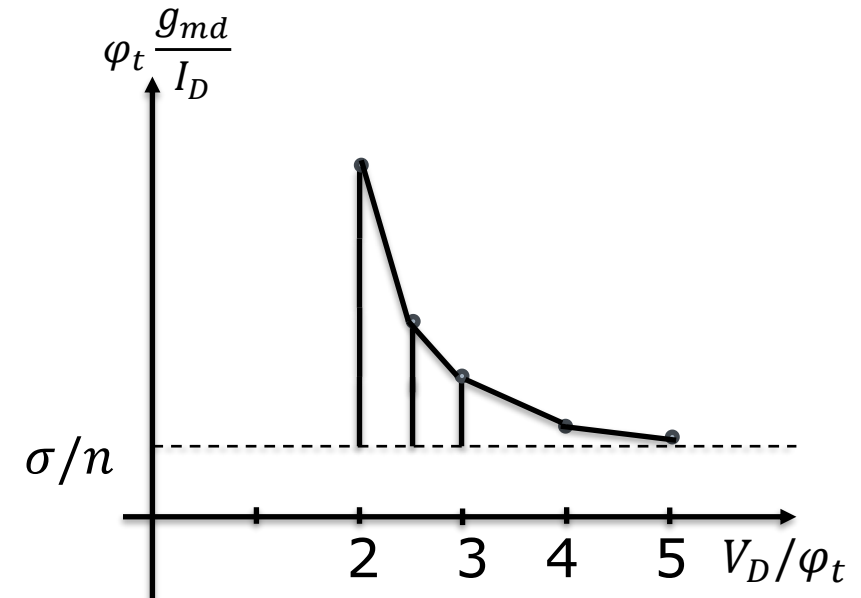
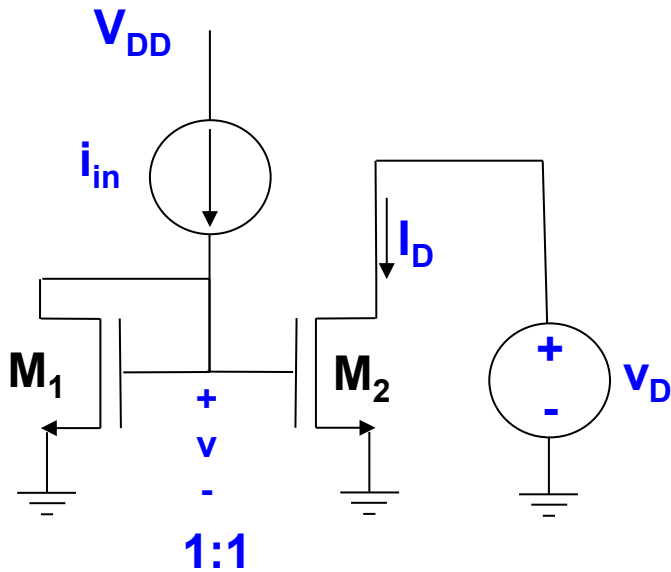
$$I_D = \underbrace{I_0 e^{(V_G - V_T)/\varphi_t}}_{I_F} [1 - e^{-V_D/\varphi_t}] = I_F \left[1 - \frac{I_R}{I_F} \right]$$

$$I_0 = 2I_S e^1$$

$$I_F$$

$$V_T \cong V_{T0} - \sigma[V_{SB} + V_{DB}]$$

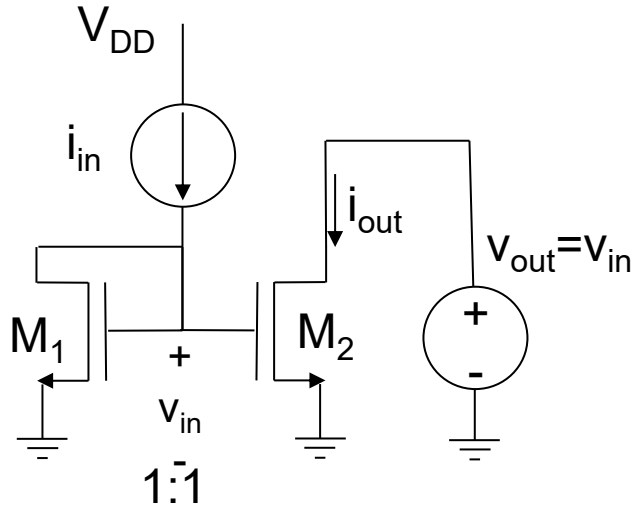
$$\varphi_t \frac{g_{md}}{I_D} = \frac{1}{I_D} \frac{dI_D}{dV_D} = \frac{d \ln I_D}{dV_D} = \frac{\sigma}{n} + \frac{1}{e^{V_D/\varphi_t} - 1} \quad (\text{A})$$



Current mirrors

Error due to mismatch

Main contributors to mismatch: I_S & V_T



Using Pelgrom's model

$$g_m/I_D$$

$$\Delta I_D \cong \frac{dI_D}{dI_S} \Delta I_S + \frac{dI_D}{dV_T} \Delta V_T$$

$$\frac{dI_D}{dV_T} = -\frac{dI_D}{dV_G} = -g_m$$

For uncorrelated mismatch parameters:

$$I_D = I_S f(V_G - V_T) \quad (B)$$

$$\frac{\Delta I_D}{I_D} \approx \frac{\Delta I_S}{I_S} - \frac{g_m}{I_D} \Delta V_T \quad (C)$$

$$\frac{\sigma^2(I_D)}{I_D^2} \approx \frac{\sigma^2(I_S)}{I_S^2} + \left(\frac{g_m}{I_D} \right)^2 \sigma^2(V_T) \quad (D)$$

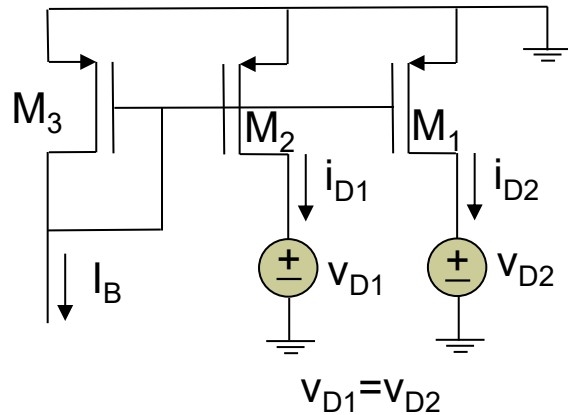
$$\frac{\sigma^2(I_D)}{I_D^2} \approx \frac{2}{WL} \left\{ A_{\beta}^2 + \left(\frac{A_{VT}}{n\phi_t} \right)^2 \left[\frac{2}{(\sqrt{1+i_f} + 1)} \right]^2 \right\}$$

Better matching in strong inversion (for the same WL)

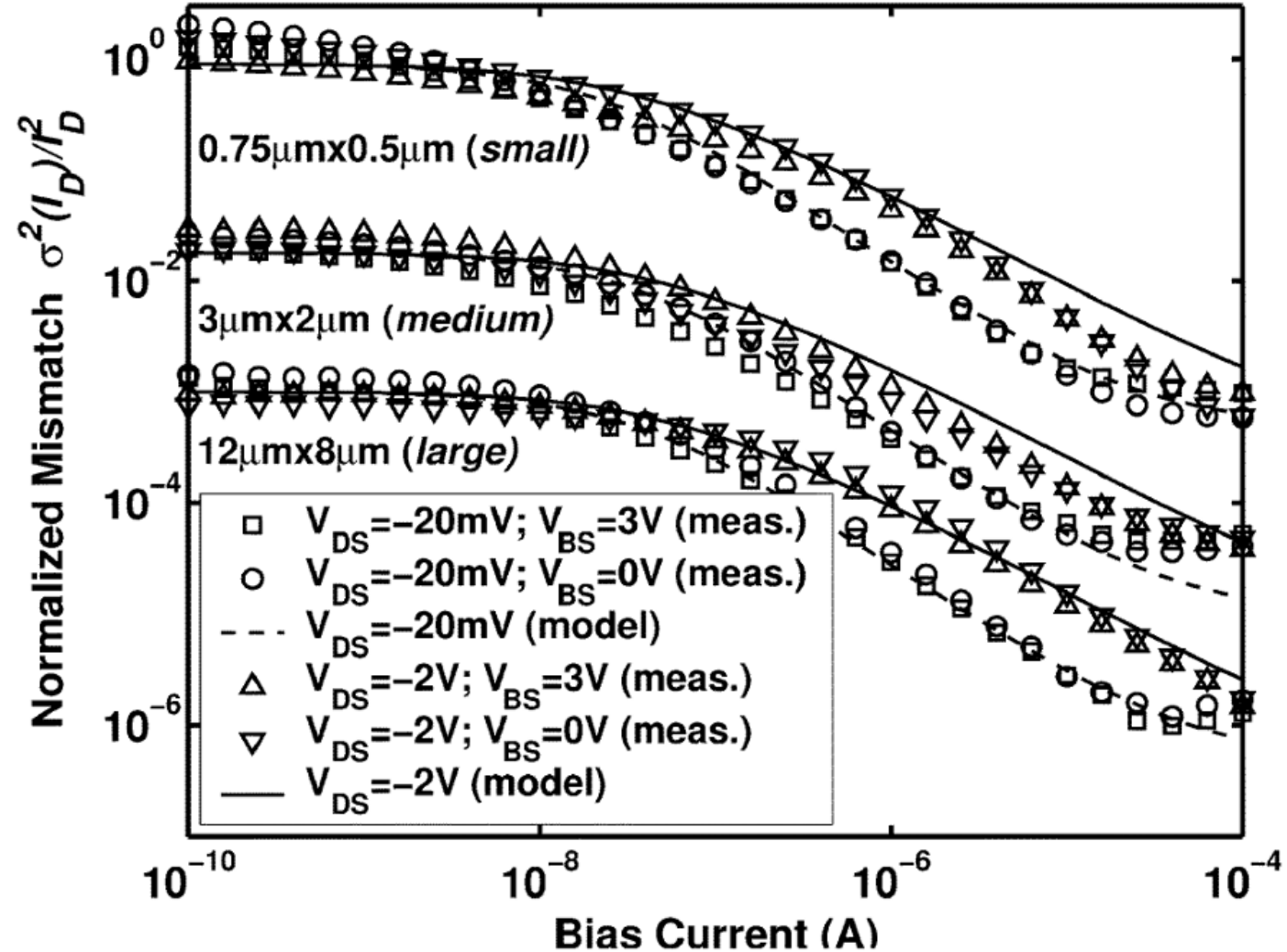
In general, this is the prevailing term

Current mirrors

Error due to mismatch

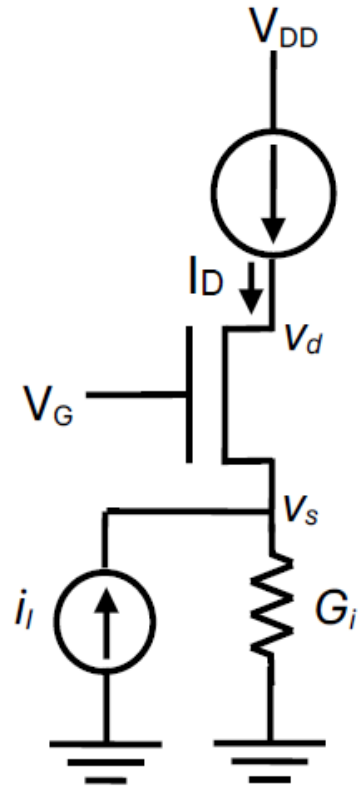


Statistics determined
for 18 pairs of transistors.



Current mirrors

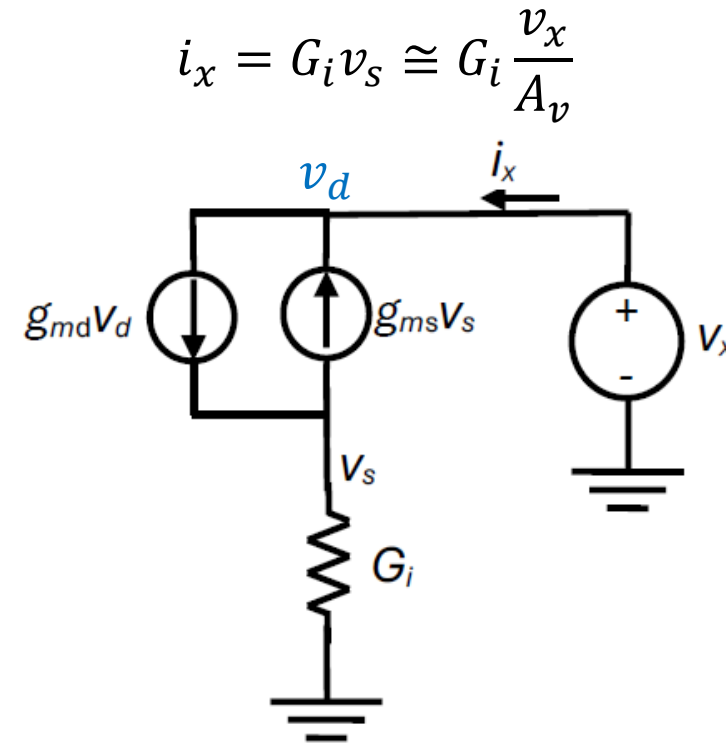
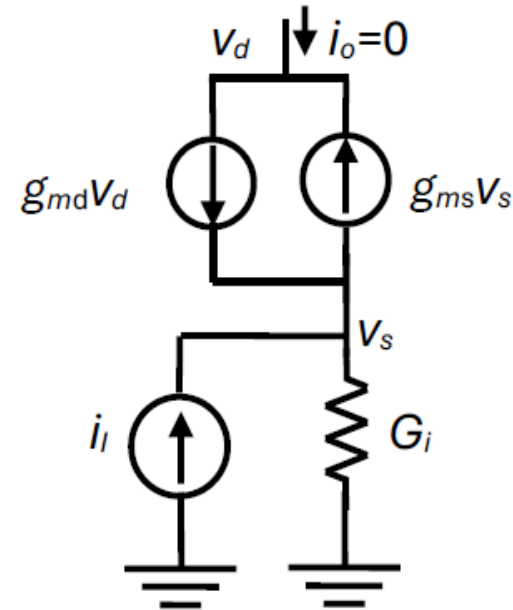
Common gate amplifier



Voltage gain

$$A_v = \frac{v_d}{v_s} = \frac{g_{ms}}{g_{md}}$$

$$g_{md}v_d = g_{ms}v_s$$

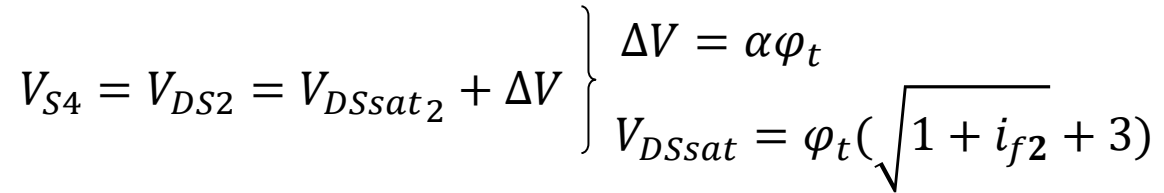


Output conductance

$$G_o = \frac{i_x}{v_x} \cong \frac{G_i}{A_v} = G_i \frac{g_{md}}{g_{ms}}$$

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Goal: Design V_{G5} to bias M2 (M1) at the edge of saturation



- Input constraint:

$$V_{DD} > V_{CS,min} + V_{GS1} = V_{CS,min} + V_{DS1,sat} + \Delta V + V_{DS3}$$

- Output constraint:

$$V_{out} > V_{DS2,sat} + \Delta V + V_{DS4,sat} \text{ for saturation of } M_4$$

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For simplicity, choose identical $M_1 - M_4$


$$V_X = V_{DS1} = V_{DSSat1} + \Delta V \quad V_{DSSat1} = \varphi_t(\sqrt{1 + i_{f1}} + 3)$$

$$V_{P5} = V_{P3} \rightarrow \varphi_t F(i_{f5}) = \varphi_t F(i_{f3}) + V_x$$

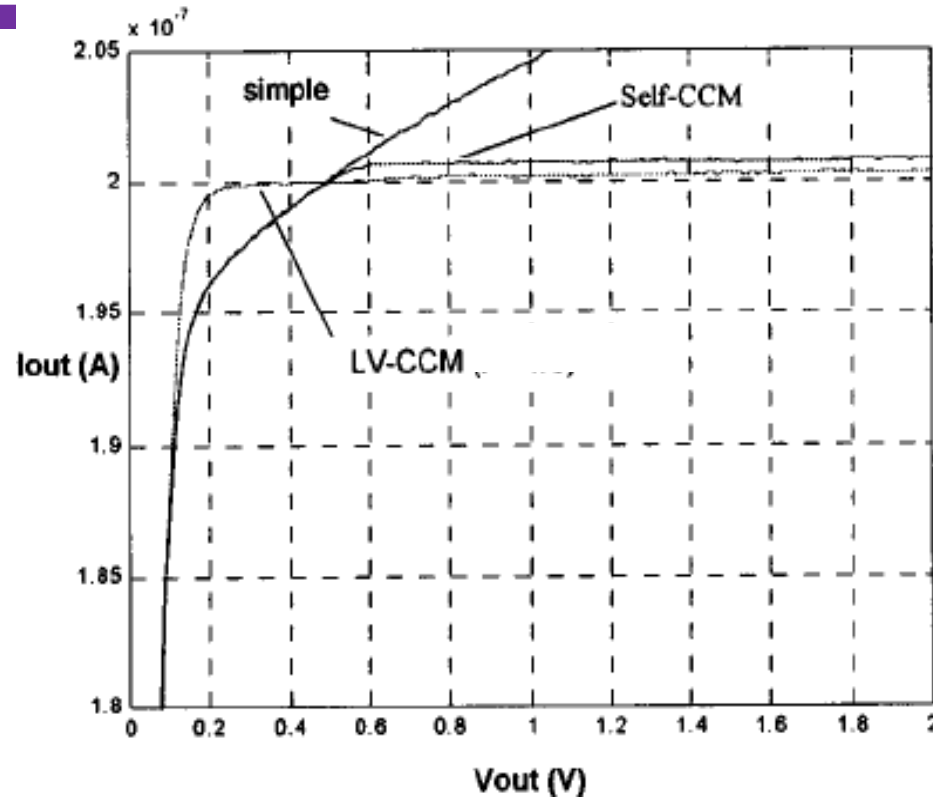


$$I_{B5} = I_{SH} \left(\frac{W}{L} \right)_5 i_{f5}$$

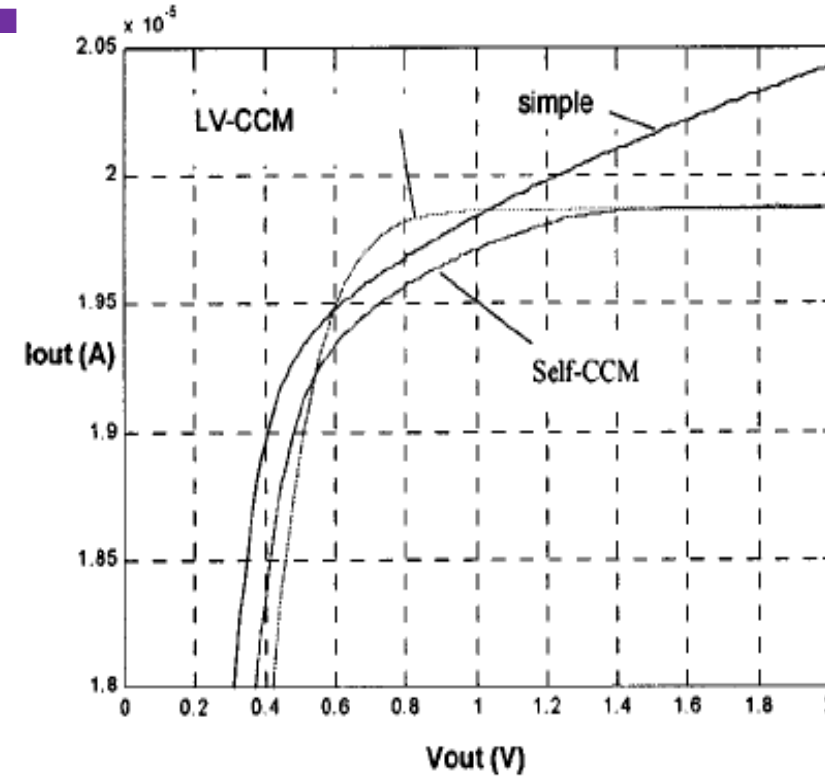
$$F(i_f) = \sqrt{1 + i_f} - 2 + \ln \left(\sqrt{1 + i_f} - 1 \right)$$

Cascode current mirrors

Low-voltage cascode current mirror



$(i_f = 1)$.



$(i_f = 100)$.

Fig. 7. Detail of experimental output characteristics of the current mirrors

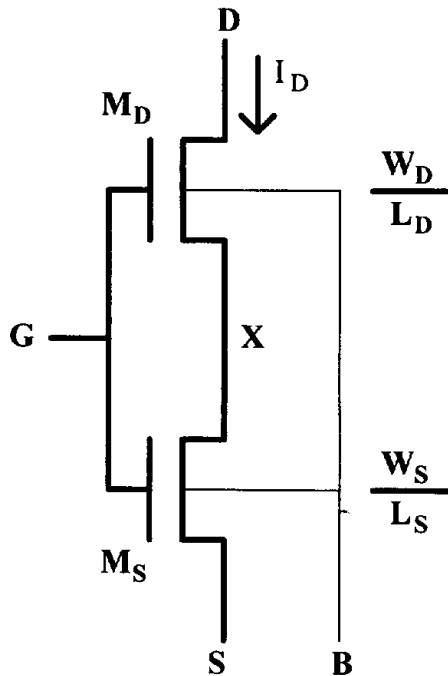
2 μ m CMOS technology - $V_{T0N} \approx 0.55$ V

- V. C. Vincence, C. Galup-Montoro and M. C. Schneider, *IEEE Trans. Circuits Syst. II*, 2000.
- P. Aguirre and F. Silveira, *Proc. of SBCCI* 2006.

Series association of transistors

DC equivalence between series-connected and single transistors. The gate and bulk terminals are common to both transistors.

In the triode region for a long channel transistor



$$I_D = \frac{W}{L} [g(V_G, V_S) - g(V_G, V_D)]$$

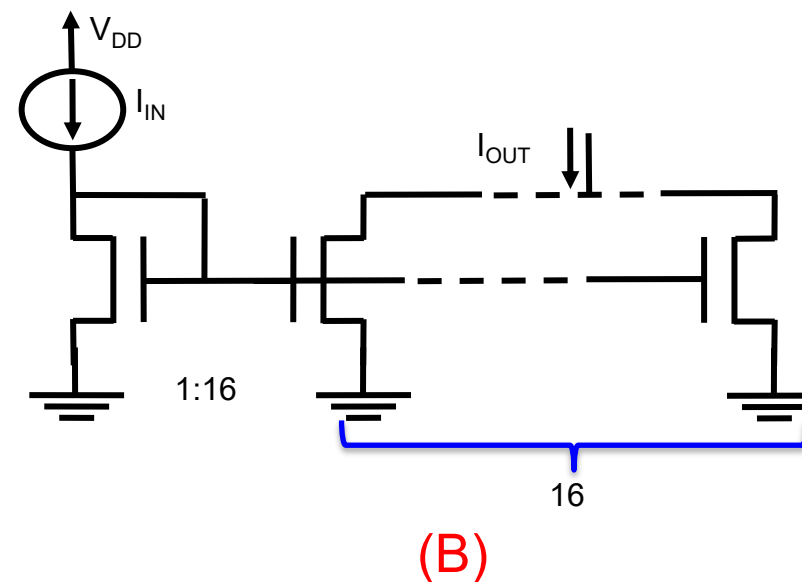
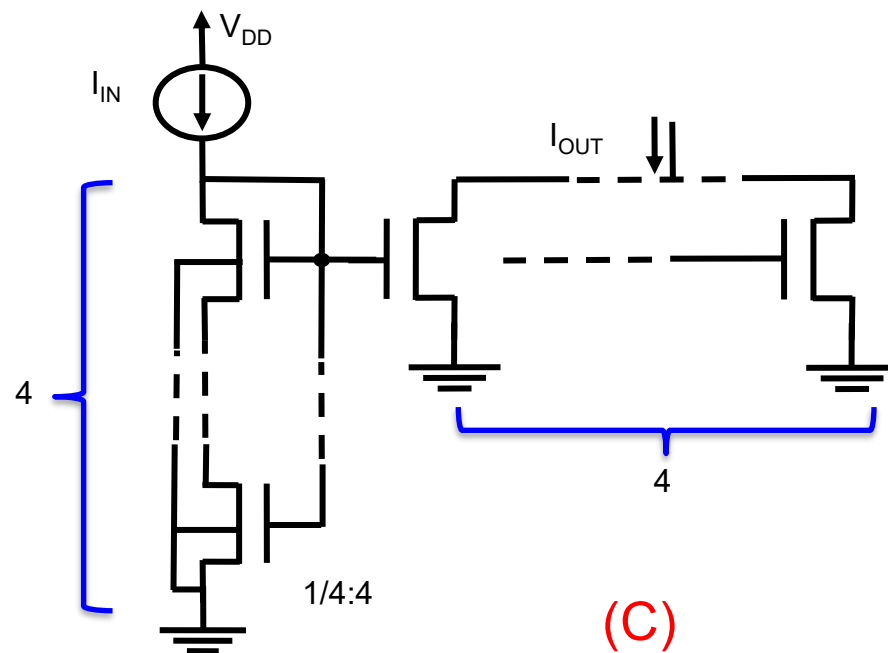
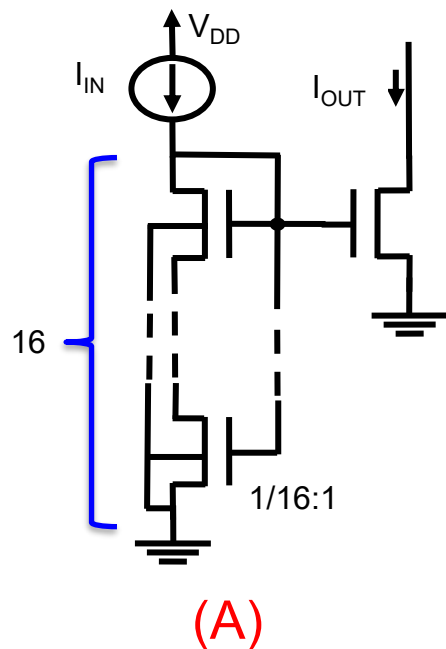
$$g(V_G, V_x) = \frac{\left(\frac{W}{L}\right)_S g(V_G, V_S) + \left(\frac{W}{L}\right)_D g(V_G, V_D)}{\left(\frac{W}{L}\right)_S + \left(\frac{W}{L}\right)_D}$$

$$I_D = \left(\frac{W}{L}\right)_{eq} [g(V_G, V_S) - g(V_G, V_D)]$$

$$\left(\frac{W}{L}\right)_{eq} = \frac{\left(\frac{W}{L}\right)_S \left(\frac{W}{L}\right)_D}{\left(\frac{W}{L}\right)_S + \left(\frac{W}{L}\right)_D}$$

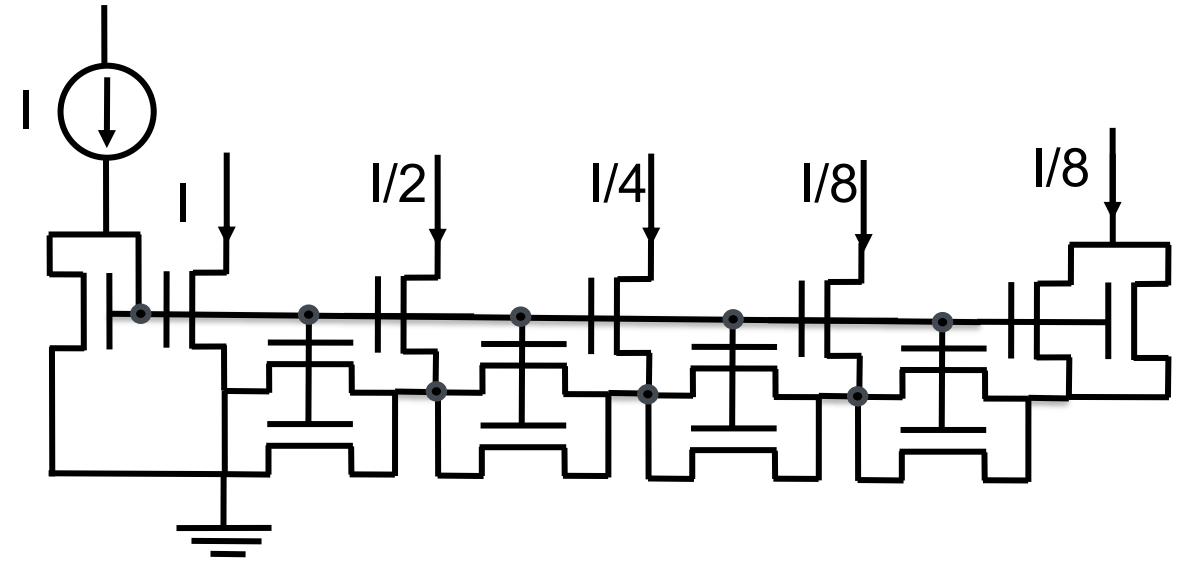
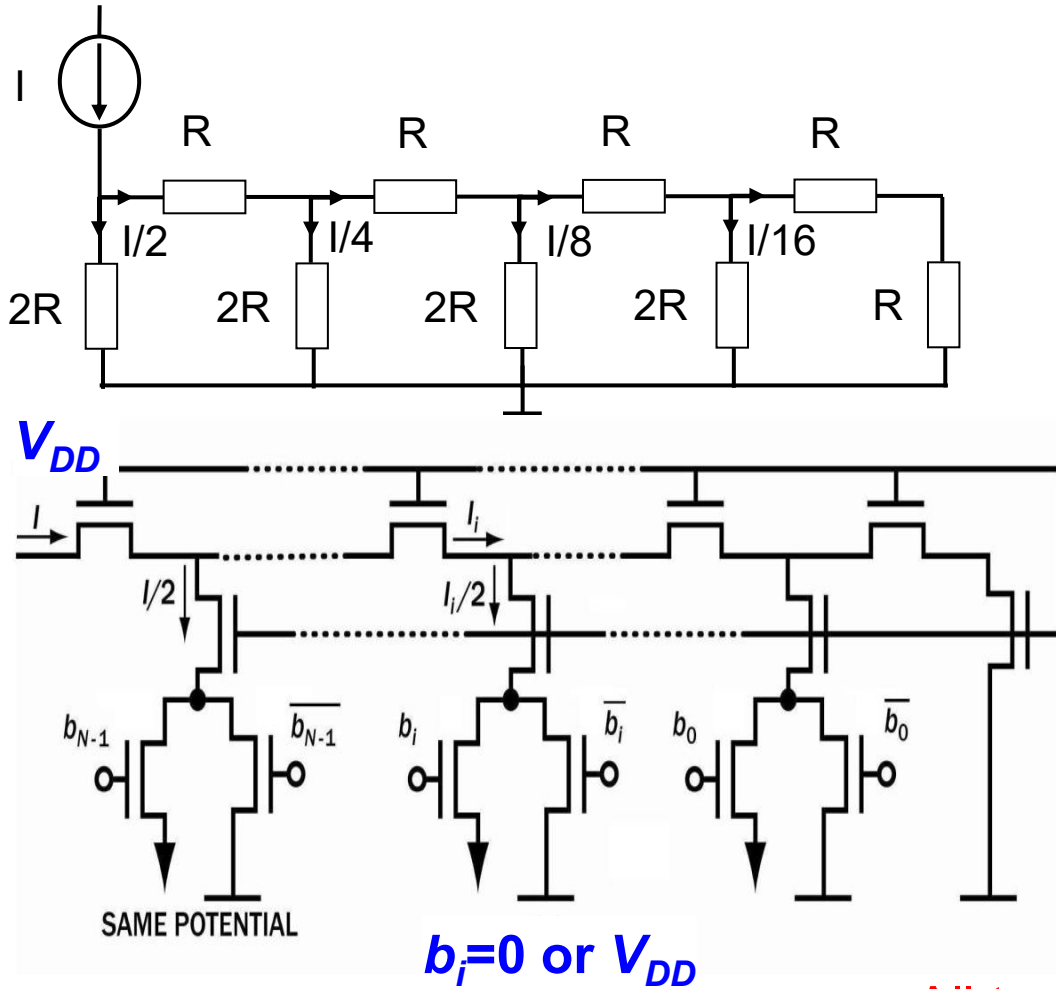
Current gain schemes

Example: Gain-of-16 current mirror



Current gain schemes

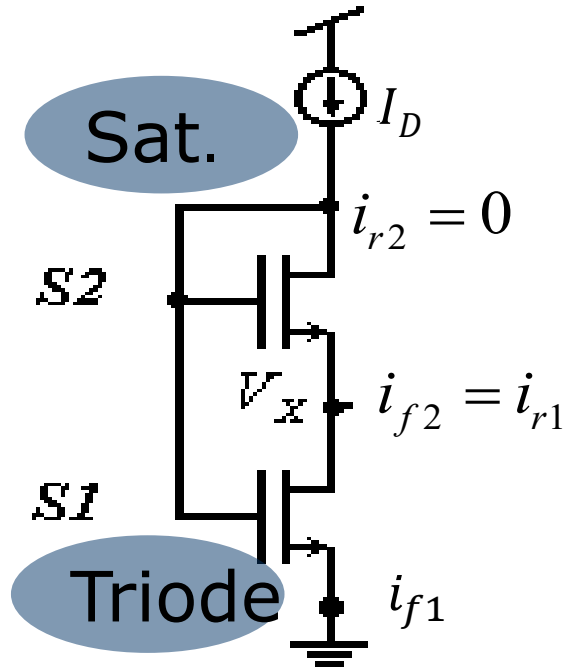
Binary division of currents



All transistors: same W and L

Current reference

The self-cascode MOSFET (SCM)



$$I_D = I_S (i_f - i_r)$$

$$I_S = (W / L) I_{SH} = S I_{SH}$$

$$I_D = I_{S2} i_{f2} = I_{S1} (i_{f1} - i_{f2}) \quad \Rightarrow \quad i_{f1} = \left(1 + \frac{I_{S2}}{I_{S1}}\right) i_{f2} = \left(1 + \frac{S_2}{S_1}\right) i_{f2} = \alpha i_{f2}$$

$$I_D = \frac{I_{S2} i_{f1}}{1 + \frac{S_2}{S_1}} = \frac{S_1 S_2}{S_1 + S_2} I_{SH} i_{f1}$$

In weak inversion, $i_{f1} \ll 1$

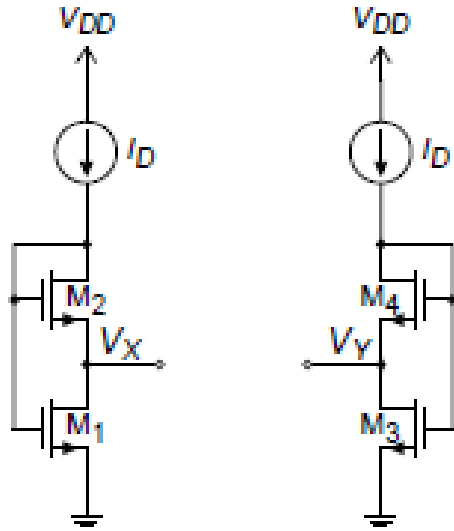
$$I_{D1} = I_{D2} \Rightarrow S_1 \left(1 - e^{-\frac{V_x}{\phi_t}}\right) = S_2 e^{-\frac{V_x}{\phi_t}} \Rightarrow$$

$$\frac{V_x}{\phi_t} = \ln \left(1 + \frac{S_2}{S_1}\right) = \ln \alpha$$

Series
association of
conductances

Current reference

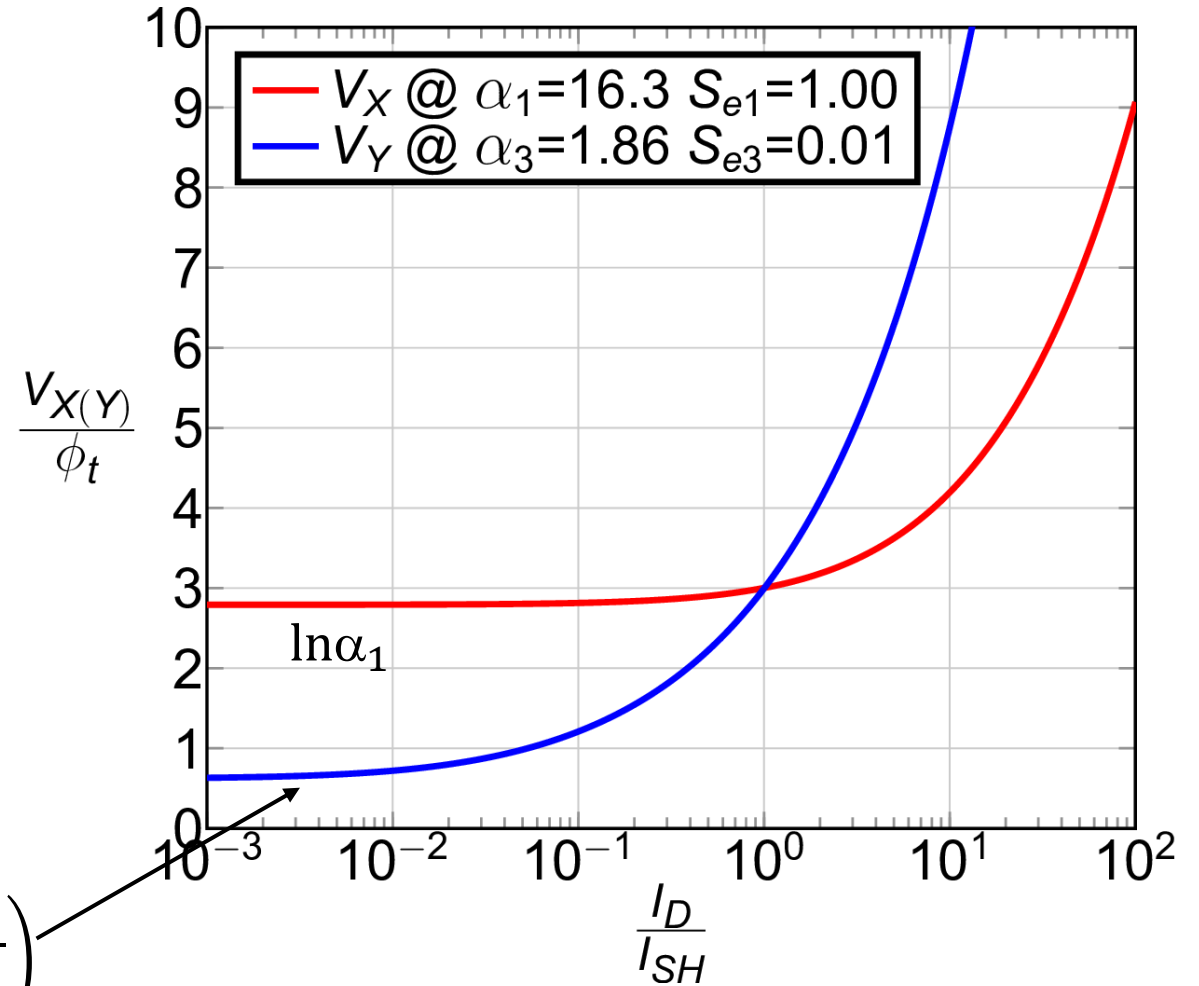
A self-biased current source (SBCS)



$$\frac{S_1 S_2}{S_1 + S_2} = 1 \quad \frac{S_3 S_4}{S_3 + S_4} = 0.01$$

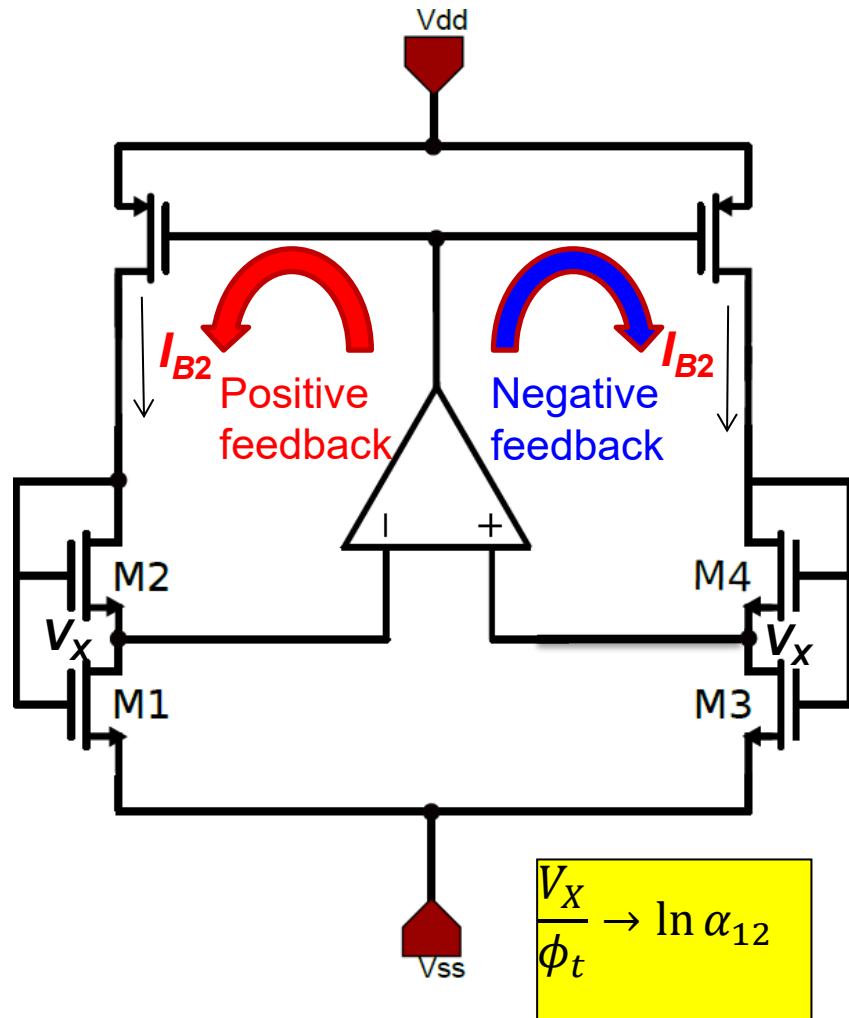
$$\alpha_1 = 1 + \frac{S_2}{S_1} \quad \alpha_3 = 1 + \frac{S_4}{S_3}$$

$$\frac{V_Y}{\phi_t} = \sqrt{1 + i_{f3}} - \sqrt{1 + i_{f3}/\alpha_3} + \ln \left(\frac{\sqrt{1 + i_{f3}} - 1}{\sqrt{1 + i_{f3}/\alpha_3} - 1} \right)$$



Current reference

A self-biased current source (SBCS)



If M_1 and M_2 operate in WI :

$$\ln \alpha_{12} = \sqrt{1 + \alpha_{34} i_{f4}} - \sqrt{1 + i_{f4}} + \ln \left(\frac{\sqrt{1 + \alpha_{34} i_{f4}} - 1}{\sqrt{1 + i_{f4}} - 1} \right);$$

$$\alpha_{12} = 1 + \frac{S_2}{S_1} \quad \alpha_{34} = 1 + \frac{S_4}{S_3}$$

Thus, i_{f4} (the inversion level of M_4) is constant (it depends only on the geometrical ratios α_{12} and α_{34})

The reference current $I_{B2} = I_{S2} i_{f2}$ is proportional to the specific current of M_2 . $I_{S2} = I_{SH} (W/L)_2 = I_{SH} S_2$

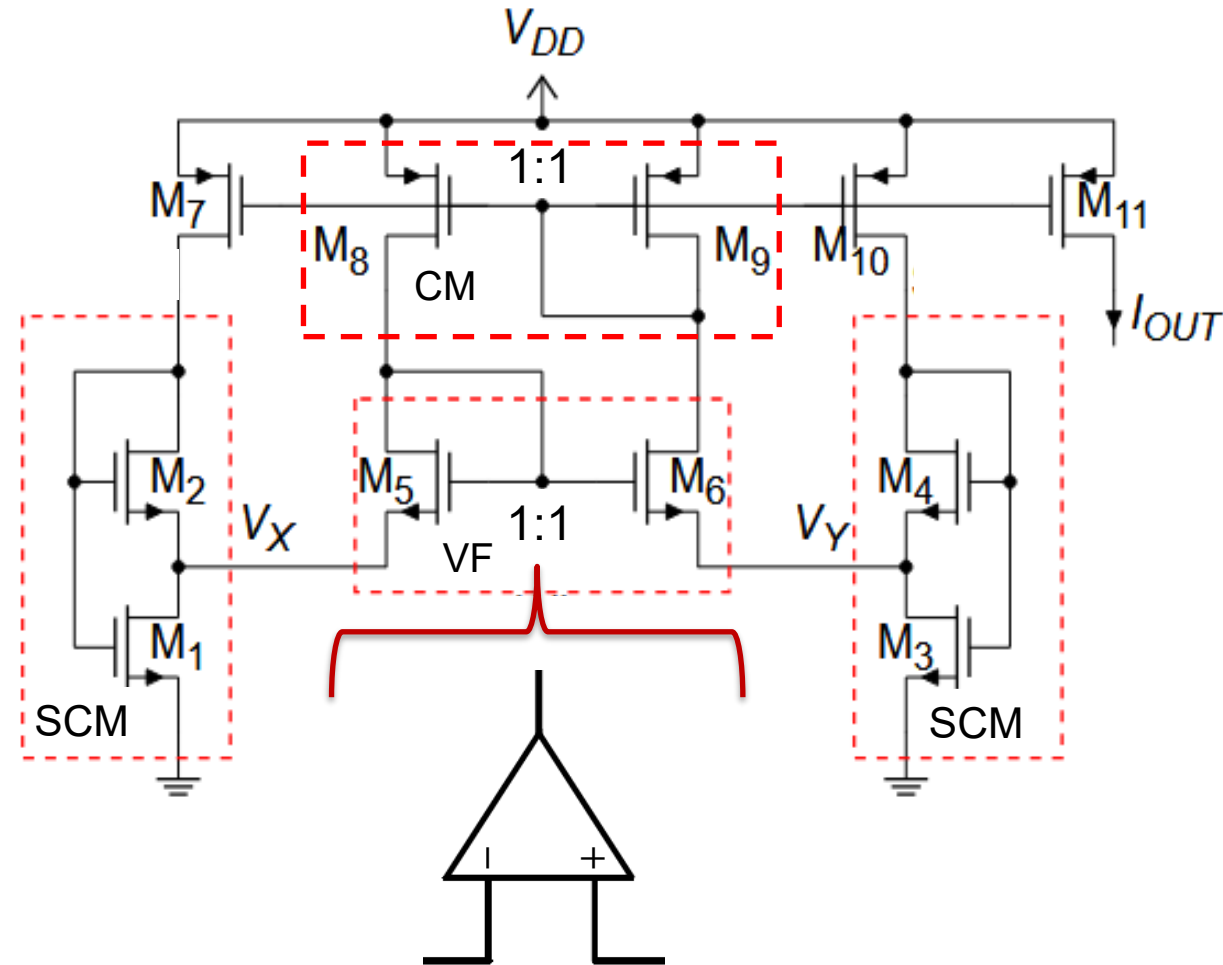
• Stability at the operating point:

Negative feedback > positive feedback

Current reference

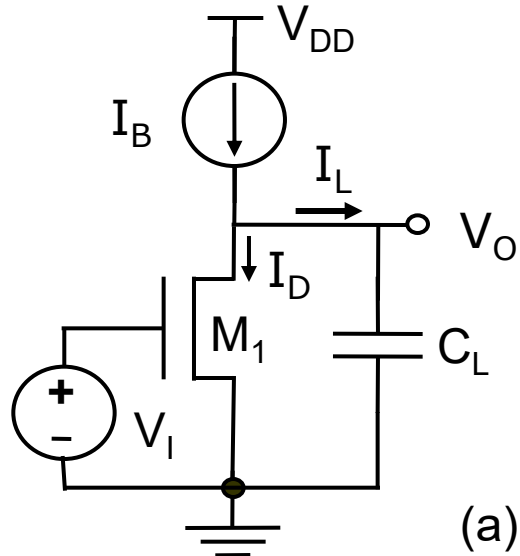
A self-biased current source (SBCS)

The voltage following current mirror – VFCM



Common-source amplifier

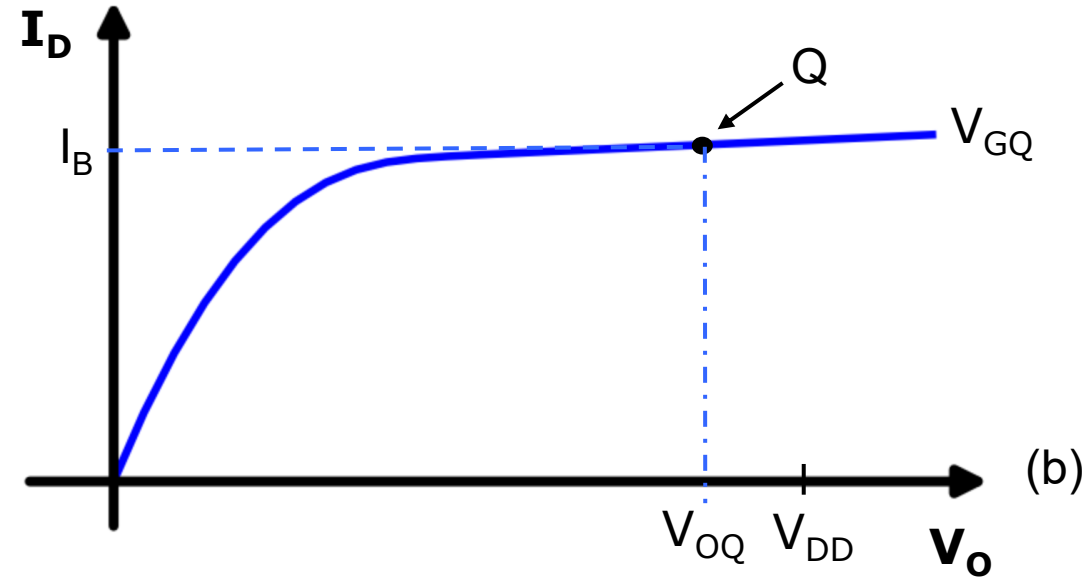
Intrinsic gain



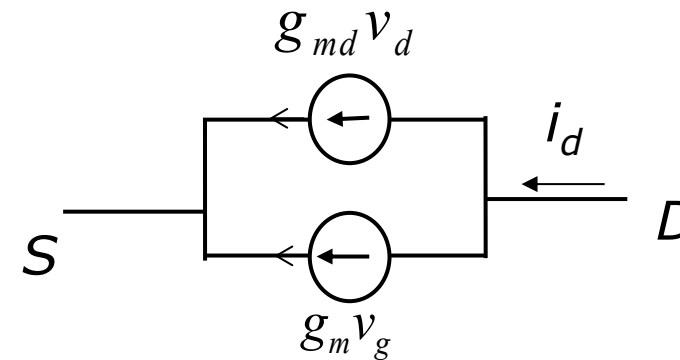
(a)

$$A_V = \frac{-g_m}{g_{md}}$$

The gain is a function of technology, transistor channel length, bias current, and output voltage



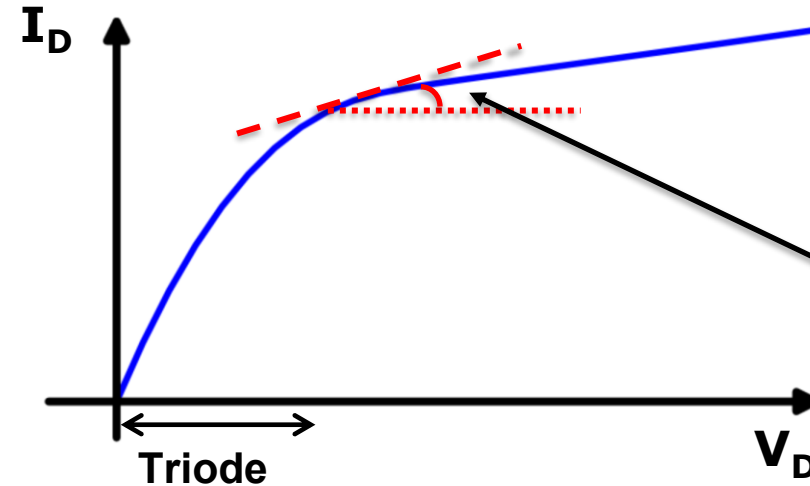
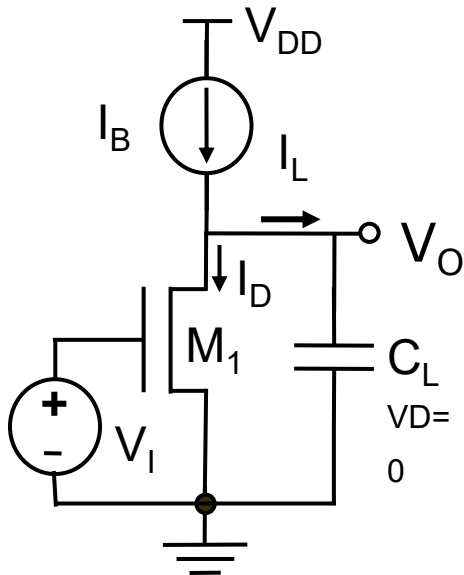
(b)



Common-source amplifier

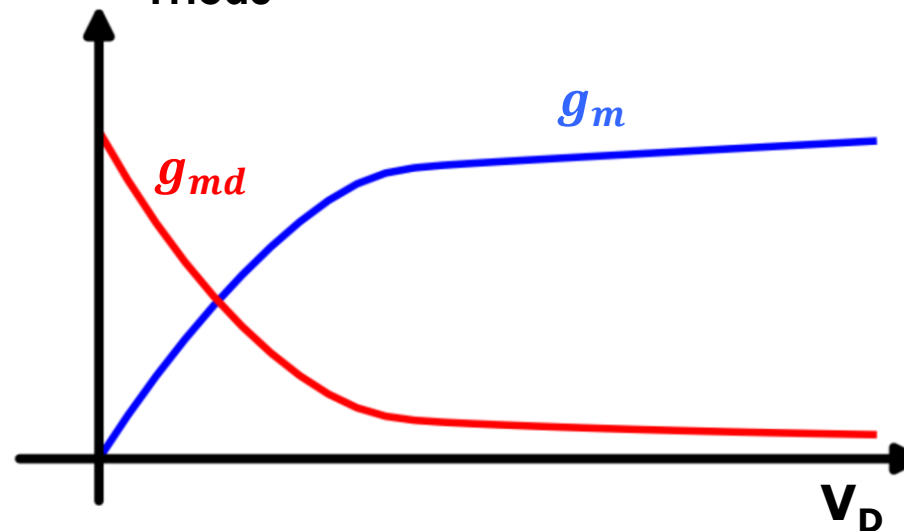
Intrinsic gain

Velocity saturation not accounted for :



$$g_m = \frac{2I_S}{n\phi_t} (q_s - q_d)$$

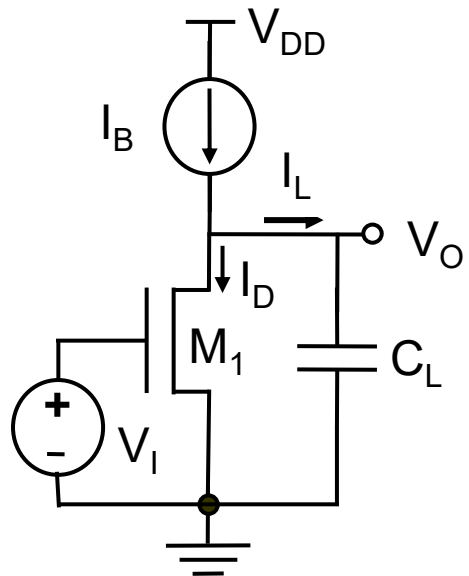
$$g_{md} = \frac{2I_S}{n\phi_t} (q_s - q_d)\sigma + \frac{2I_S}{\phi_t} q_d$$



$$A_V = -\frac{g_m}{g_{md}} = -\frac{\frac{q_s}{q_d} - 1}{\sigma \left(\frac{q_s}{q_d} - 1 \right) + n}$$

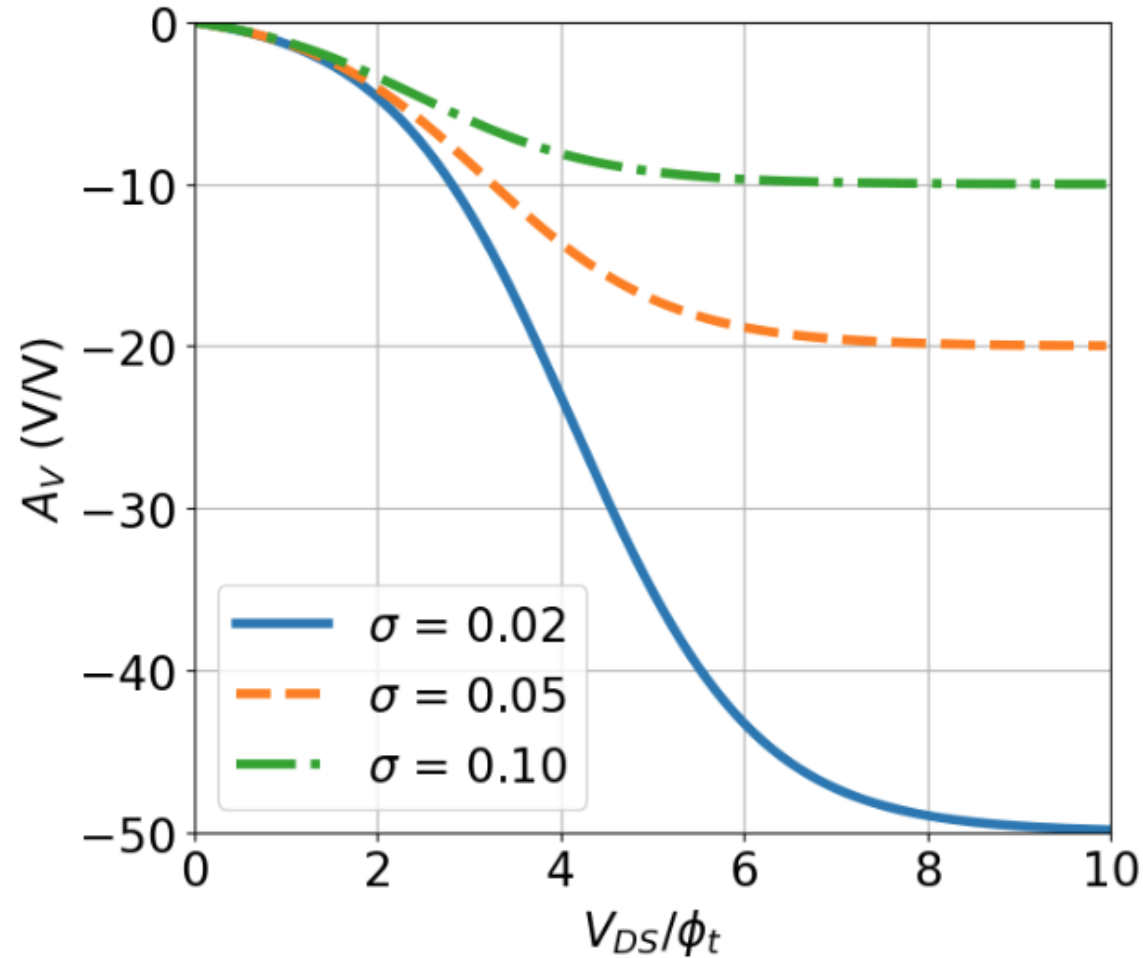
Common-source amplifier

Intrinsic gain



$$A_V > 1$$

$$V_{Dmin} = \phi_t \ln \left(1 + \frac{n}{1 - \sigma} \right)$$



$$A_V = -\frac{g_m}{g_{md}} = -\frac{\frac{q_s}{q_d} - 1}{\sigma \left(\frac{q_s}{q_d} - 1 \right) + n}$$

$$\frac{q_s}{q_d} \gg \frac{1}{\sigma}$$

$$A_V \cong -\frac{1}{\sigma}$$

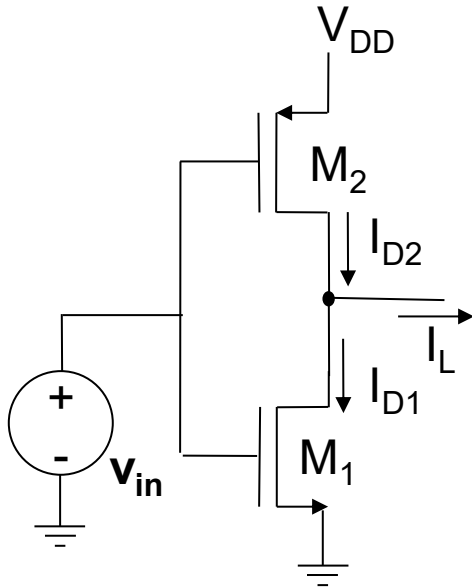
Weak inversion



$$A_V = -\frac{e^{\frac{V_{DS}}{\phi_t}} - 1}{\sigma \left(e^{\frac{V_{DS}}{\phi_t}} - 1 \right) + n}$$

Common-source amplifier

CMOS inverter

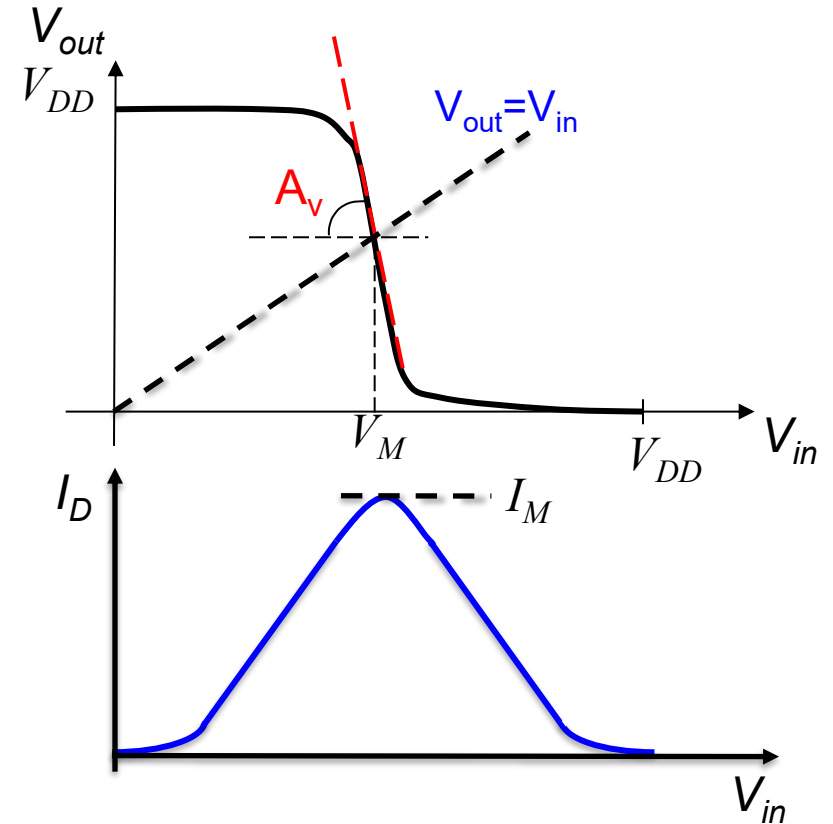


Voltage gain (saturation):

$$A_V = - \frac{g_{m1} + g_{m2}}{g_{md1} + g_{md2}}$$

$$g_{m1(2)} = \frac{I_M}{n_{1(2)} \phi_t} \frac{2}{1 + \sqrt{1 + i_{f1(2)}}}$$

$$g_{md1(2)} = \sigma_{1(2)} g_{m1(2)}$$



For “deep” saturation:

$$\frac{1}{A_V} = - \left(\frac{g_{m1} \sigma_1}{g_{m1} + g_{m2}} + \frac{g_{m2} \sigma_2}{g_{m1} + g_{m2}} \right)$$

$$A_V = - \frac{2}{\sigma_1 + \sigma_2} \text{ for } g_{m1} = g_{m2}$$

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